NISM V-D Complete Summary

Every chapter of the September 2026 workbook (MF Ch 1–12, equity derivatives Ch 13–17, interest rate derivatives Ch 18–22, appendices) — 468 subtopics in tables, keywords and formulas. Single source: the workbook. Each subtopic rated 1–5★ for exam importance; read in any order you like.

★★★★★ must know, asked often ★★★★☆ high ★★★☆☆ medium ★★☆☆☆ low ★☆☆☆☆ skim

‡ rule or number that changes often; check workbook edition

Ch	Chapter	Marks	Priority	Subtopics	5★
1	Investment Landscape	5	★★★☆☆	17	8
2	Concept & Role of a Mutual Fund	4	★★★☆☆	24	11
3	Legal Structure of Mutual Funds in India	3	★★★☆☆	11	6
4	Legal & Regulatory Framework	7	★★★★☆	22	9
5	Scheme Related Information	7	★★★★☆	17	9
6	Fund Distribution & Channel Management Practices	4	★★★☆☆	19	9
7	NAV, Total Expense Ratio & Pricing of Units	5	★★★★★	15	9
8	Taxation	3	★★★★☆	14	12
9	Investor Services	10	★★★★★	53	28
10	Risk, Return & Performance of Funds	5	★★★★☆	29	15
11	Mutual Fund Scheme Performance	5	★★★★☆	15	8
12	Mutual Fund Scheme Selection	10	★★★★★	17	11
App	Appendices 1–4: Codes of Conduct & Resources	in Ch 4/6	★★★☆☆	7	2
13	Basics of Derivatives	10	★★★★★	8	7
14	Understanding the Index	5	★★★☆☆	12	4
15	Introduction to Forwards & Futures	15	★★★★★	31	22
16	Introduction to Options	13	★★★★★	21	16
17	Strategies using Equity Futures & Equity Options	9	★★★★★	26	23
18	Interest Rate, Interest Rate Instruments & Fixed Income Market	6	★★★★☆	48	31
19	Interest Rate Derivatives	2	★★★☆☆	14	7
20	Exchange Traded Interest Rate Futures	10	★★★★★	19	13
21	Exchange Traded Interest Rate Options	6	★★★★☆	15	11
22	Strategies using Exchange Traded Interest Rate Derivatives	6	★★★★☆	14	11

5★ list (282) — read these first

Ch 1 · Factors to evaluate investments · Four asset classes · Investment risks · Systematic vs unsystematic classification · Behavioural biases · Risk profiling · Asset allocation: SAA, TAA · Rebalancing

Ch 2 · Key concepts: units, face value, unit capital, expenses, NAV, AUM, MTM · Advantages of mutual funds · Classification by structure · SEBI categorisation 2026 — five broad categories · Market-cap definitions (Box 2.2) · A. Equity schemes (13) · B. Debt schemes (17) · C. Hybrid schemes (7) · D. Life Cycle Funds (new) · E. Other schemes; one scheme per category rule · Specialized Investment Fund (SIF) — strategies

Ch 3 · Legal structure — checks and balances · Trustees (Regulation 11) · Asset Management Company · Custodian · Service providers · AMFI — role and functions

Ch 4 · Permitted investments — Regulation 39 · Restrictions — Regulation 41 · Sixth Schedule restrictions · Advertisement code — Fifth Schedule · Performance advertisements · Celebrity endorsement at industry level · SIF branding & advertisement · Investors' rights · Grievance redress mechanism

Ch 5 · Mandatory scheme documents; caveat emptor · Fundamental attributes (Box 5.1) · Key Information Memorandum (KIM) · Updating SID, SAI, KIM · Current value & daily NAV disclosure · TER / Base Expense Ratio disclosure · Risk-o-meter · Risk-Band for SIFs · Portfolio disclosure

Ch 6 · Stock exchange platforms · MF Utilities (MFU) & MFCentral · Prerequisites to become a distributor · Distributor revenue: trail commission only · Trail commission calculation · Additional commission: B-30 & women investors · Distributor vs investment adviser; advisory vs execution only · Mis-selling (Box 6.8.3) · Change of distributor

Ch 7 · 10 principles of fair valuation · Net assets — worked example · NAV drivers & profitability metric · NAV worked examples (workbook) · Mark to market — why · TER limits (2026) · Charges over and above base TER · IDCW & distributable reserves · Entry & exit load

Ch 8 · Two levels: fund vs investor · What decides the tax rate · Capital gains: holding periods · Grandfathering (Budget 2018) · ₹1.25 lakh LTCG exemption · Tax table for unitholders · IDCW taxation · Stamp duty · Set-off & carry forward · Securities Transaction Tax (STT) · Section 80C — ELSS Tax Saver · Tax Deducted at Source (TDS)

Ch 9 · Direct and regular plans · Growth, IDCW payout, IDCW reinvestment · Illustration 9.1 — three investors · Allotment of units · Account statements · Eligible individual investors · Application form — key fields · Financial transactions: purchase, additional purchase, repurchase · Third-party payments (Box 9.1) · Cash payments · Repurchase payments & multiple bank accounts · Instant Access Facility (IAF) · Cut-off timings & applicable NAV · Cut-off worked examples · KYC — who and when · KYC documents & PAN exemptions · PAN-exempt investments & PEKRN (Box 9.2) · KYC Registration Agencies & CKYC · Institutional investors & UBO · SIP & rupee cost averaging · Systematic Withdrawal Plan (SWP) · Systematic Transfer Plan (STP) · Nomination — rules · Change in folio details · Transfer and gifting of MF units · Transmission of units · Minor turned major · Turnaround times (SEBI)

Ch 10 · General risk factors · Derivative risks · Debt fund risks · Systematic vs unsystematic risk · Fundamental ratios · Growth vs value; top-down vs bottom-up · Drivers of debt scheme returns · Debt fund management: duration, floaters, credit spreads · Simple, annualised and compounded return · CAGR with IDCW reinvested (SEBI method) · Risks from the investor's view · Variance & standard deviation · Beta, modified duration, WAM, credit rating · Gating — restriction on redemption · Segregated portfolio (side pocketing)

Ch 11 · Benchmarks — requirements & choice · Price Return Index vs Total Return Index · Two-tier benchmarks · Sharpe ratio · Treynor ratio · Alpha · Information ratio · Tracking error

Ch 12 · Selection based on investor need, risk profile, allocation · Core and satellite portfolio · Risk-return hierarchies (Illustrations 12.1.1–12.1.6) · Active vs passive funds · Open-ended vs close-ended · Diversified vs focused vs sector vs thematic · Debt fund selection: FMP, TMF, short duration, liquid, ultra-short, floater · Selecting among AMCs / within a category · Portfolio turnover ratio · Selecting options · Do's and don'ts

Ch App · Appendix 3 — AMFI Code of Conduct for MFDs: scope & fiduciary duty · Appendix 3 — client obligations

Ch 13 · Meaning of a derivative & underlying assets · Indian derivatives market history · Four generic derivatives · Market participants · OTC vs exchange-traded derivatives · Significance of derivatives · Risks in derivatives trading

Ch 14 · Market-cap weighted index (worked example) · Free-float market-cap index · Price-weighted index · Impact cost & bid-ask spread

Ch 15 · Forward contracts · Limitations of forwards · Futures contracts: features & limitations · Lot size rules (SEBI Master Circular 30 Dec 2024) · Contract cycle, expiry day & rollover · Basis · Cost of carry · Margins & MTM · Open interest vs volume · Forwards vs futures · Futures payoffs · Cost of carry (no-arbitrage) model · Cash-and-carry vs reverse cash-and-carry · Cost of carry with income (dividend yield) · Pricing with known dividends · Expectations model · Price discovery & convergence · Hedgers, traders, arbitrageurs in futures · Systematic vs unsystematic risk & beta · Hedge ratio · Long hedge, short hedge, cross hedge, hedge month · Arbitrage types & formulas

Ch 16 · What an option is · Option contract specifications · Moneyness · Intrinsic value & time value · Long call (worked) · Short call (worked) · Long put (worked) · Short put (worked) · Options vs futures; leverage · Futures vs options — full comparison · Five factors that determine option price · Option Greeks · Option pricing models · Implied volatility · Call buyer & seller analysis (index 17,562) · Put buyer & seller analysis (index 17,562)

Ch 17 · Long hedge with stock futures · Short hedge with stock futures · Hedging a portfolio with index futures · Futures for trading: bullish & bearish · Cash-and-carry arbitrage (worked) · Reverse cash-and-carry arbitrage (worked) · Calendar spread arbitrage (futures) · Option spreads: vertical, horizontal, diagonal · Bull call spread (worked) · Bull put spread (worked) · Bear call spread (worked) · Bear put spread (worked) · Long & short straddle (worked) · Long strangle (worked) · Covered call · Collar · Butterfly spread · Hedging with options; protective put · Put-call parity arbitrage · Delta hedging · Open interest & delta-adjusted OI · Put-call ratio (PCR) · Volatility index (India VIX)

Ch 18 · Macro & micro factors affecting interest rates · Effective rate; nominal vs real rate · Fixed income securities & key components · By issuer · By coupon · Risk-free rate & risks in bonds · Term structure of interest rates · What sets short-term & long-term rates · Yield curve shapes · Yield curve shifts · Simple vs compound interest; FV & PV · Day count conventions · Accrued interest; clean & dirty price · Sources of bond return · Price-yield relation; current yield · YTM, BEY, discount yield, effective yield · Bond pricing (worked) · Spot rates vs YTM; YTM drawbacks · Price-yield relationship rules · Interest rate risk; price volatility properties · Macaulay duration · Modified duration · PV01 (price value of a basis point) · Convexity · G-sec primary market: auctions · G-sec secondary market · G-sec settlement & market operations · Corporate bond primary market · Corporate bond secondary market · Money market instruments · Credit rating agencies

Ch 19 · Approaches to risk · Economic functions of derivatives · Forwards & FRA · Futures, options, caps, floors, collars · Swaps & swaptions · Underlyings of IRDs · OTC vs exchange-traded derivatives

Ch 20 · History of IRF in India · Futures terminology · Futures payoffs · Cash-settled GOI bond futures (6/10/13-yr) specs · 91-day T-bill futures specs · Overnight MIBOR futures specs · Corporate bond index futures (CBIF) · Lot size, tick value & contract value · FRA vs interest rate futures · Forward rates · Cost of carry, income & basis · Bond futures pricing (worked) · Expectancy model & convergence

Ch 21 · Option styles · Moneyness; intrinsic & time value · Five option price factors · Option Greeks (bond examples) · Put-call parity · Option pricing models · Implied volatility · Long & short call on G-sec (worked) · Long & short put on G-sec (worked) · Exchange-traded IRO specs · Open interest (worked) & PCR

Ch 22 · Choosing instrument, side, contract month · Short hedge (worked) · Portfolio (duration-based) hedge · Hedging future borrowing & long hedge · Vertical spreads on G-sec options · Straddles & strangles on G-sec options · Covered call & protective put on bonds · Speculation with IRF · Arbitrage with single-bond IRF (worked) · Spread trading with IRF · Limits of IRDs for hedgers

4★ list (140)

Ch 1 · Goal-setting steps · Goals, horizon & inflation; the pool approach · Saving vs investing · Table 1.1 — avenues by asset category · Risk management strategies & risk measures

Ch 2 · Concept of a mutual fund · Investment objectives (Table 2.1) · Limitations of mutual funds · Exchange Traded Funds (Box 2.1) · Classification by portfolio management; MF Lite · Other categories: FMP, TMF, capital protection, IDF, REMF · REITs & InvITs (box)

Ch 3 · Definition of 'mutual fund' (SEBI MF Regulations 2026) · Sponsor eligibility · AMC organisation structure · Stock exchanges & transaction platforms · Execution Only Platforms (EOP)

Ch 4 · Four financial regulators · Role of SEBI · 22 areas covered by regulations · IDCW / bonus advertisement rules · Standard additional benchmarks · Unclaimed amounts & illiquid security proceeds · AMFI Code of Ethics & AGNI

Ch 5 · SID — nature & cover page · Using the SID — where to find what · Statement of Additional Information (SAI) · Addendum · Financial results & annual reports · Fund factsheet (non-mandatory)

Ch 6 · Role of the MF distributor vs fund manager · Kinds of distributors · Transaction charges — withdrawn · Commission disclosure by AMCs · Due diligence of distributors by AMCs · Nomination for distributors & commission to nominee

Ch 7 · Expenses: rules · Recurring expense heads (a)–(o) · Perpetual (AT-1 / Tier 2) bonds · Accounting & reporting requirements · Segregated portfolio: NAV, TER, pricing

Ch 8 · Bonus stripping · GST in MF schemes

Ch 9 · NFO process · NFO & ongoing prices · Digital payment modes · One-Time Mandate (OTM) · Cheque / DD rules · Time stamping · e-KYC via UIDAI & KYC through intermediaries · KYC process & online KYC · FATCA & CRS · Switch & Transfer of IDCW Plan (DTP) · Operational aspects — SIP registration · SIP top-up & SIP pause · Nominee for incapacitated investors; joint holder death · Pledge / lien of units · Demat account for MF units · Change of bank account · NRI to resident Indian · Unclaimed assets & MITRA · Voluntary lock-in / debit freeze

Ch 10 · Standard vs specific risk factors (SID) · Specific risk factors — equity, short selling, mid/small cap, dividend · Securitised assets · 'Subject to market risks' — meaning · Drivers of equity scheme returns · Gold fund performance · Scheme returns vs investor returns · Credit events & the redemption spiral · Segregated portfolio: TER, NAV, risks · Segregated portfolio illustration · Risk mitigation for investors

Ch 11 · Equity scheme benchmarks · Debt scheme benchmarks · Standard additional benchmarks · Relative returns vs risk-adjusted returns

Ch 12 · Selection by strategy: objective → strategy → portfolio · Large vs mid vs small cap; multi/flexi · Growth vs value funds · International equity funds · Hybrid and gold fund selection · Research agency rankings & ratings (Box 12.1)

Ch App · Appendix 1 — Fourth Schedule: Code of Conduct for AMC & trustees · Appendix 2 — AMFI Code of Ethics (ACE) · Appendix 3 — compliance, infrastructure, records · Appendix 3 — other obligations & integrity

Ch 14 · What an index is & why it matters · Equal-weighted & capped indices · Attributes of a good index · Index management · Derivatives on non-benchmark indices (SEBI 30 Oct 2025) · Applications & types of indices

Ch 15 · Contract specifications — Nifty futures example · Other contract terms · Price band & base price · Positions: long, short, open, naked, calendar spread · Transaction costs & no-arbitrage bounds · Assumptions of the cost of carry model · Convenience yield · Trading: naked & spread positions · Workbook sample numericals

Ch 16 · Option quote example & terminology · Opening & closing option positions · Long & short options — general · Choosing ITM / ATM / OTM · Workbook sample answers

Ch 17 · Short strangle · Rollover

Ch 18 · Interest rate concept: loan vs debt instrument · By maturity · By embedded options · By security & seniority · Other instruments · Equity vs debt; pros & cons of debt · Spot (zero) rate & holding period return · Valuing bonds between coupon dates (worked) · Price time path; pull to par · Portfolio duration · Role of the debt market · Primary vs secondary debt market; participants · Holding G-secs · Call money market, MIBOR & repo utility · Role of regulators

Ch 19 · Derivative definition & accounting criteria · Legal definition (SCRA 1956) · Bond forwards (RBI 2025) · Four generic derivatives compared · Growth drivers; IRD market size · Participants

Ch 20 · Futures basics · Positions & rollover · Other / earlier IRF designs · Why exchange-traded IRDs in India · 91-day T-bill futures pricing

Ch 21 · Option basics (land example) · Squaring off an option (worked) · Exchange-traded vs OTC IRO

Ch 22 · Hedgers, speculators, arbitrageurs · Long call butterfly on bonds

Source: NISM-Series-V-D MF-SIF Distributors Workbook, version September 2026 (the only source used). Chapter marks are the official weightages from the workbook. TRAP boxes flag common exam confusions and places where the workbook itself is inconsistent.

01 Investment Landscape

≈5 marks · 17 subtopics · chapter priority ★★★☆☆

Why invest: examples & financial goals ★★★☆☆

Workbook example	Goal type
Shalini (8 yrs) — wants to be a space scientist	Child's higher education (~10 yrs away)
Rabindra (45, engineer, no pension)	Retirement (works ~15 more yrs)
Surinder Singh (transferred to big city)	Buy a house (next couple of years)
Mrs. D'Souza (just retired, lump sum)	Regular income for life (immediate + long, uncertain period)

Financial objective	A need (education, house, retirement…)
Financial goal	Objective + **amount** + **timeline**
Common goals	Child's education, child's marriage, retirement lifestyle, vehicle, buy/renovate house, big vacation
Not-so-common	Start own business, sabbatical to fund higher education
Why invest	Goal needs money that current income (salary/fees) can't fund at that time

Start with WHY (Simon Sinek)

Questions about 'best fund' focus on investments, not the investor

Goal-setting steps ★★★★☆

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1. Identify life events (desirable/planned vs undesirable surprises)
2. Assign priorities (responsibilities vs good-to-have)
3. Assign timeline + amount needed at that time
4. Classify goals by horizon and importance
```

Undesirable events (death, hospitalisation, accident, theft, fire) → can't plan via investments → emergency fund + insurance

Retirement / children's education = responsibilities

Grand vacation = good-to-have

Only the individual/family decides priorities; MFD/adviser only guides

Short-term needs vs long-term goals; goal matrix ★★★☆☆

Horizon ↓ / Importance →	Critically important (responsibilities/needs)	Dreams	Good-to-have
Immediate term			
Near term			
Medium term			
Long term			

Retirement goal = 2 parts: accumulate corpus + draw income from it

Mrs. D'Souza: income needed now AND for life

Covey's time management matrix ★★☆☆☆

	Urgent	Not urgent
Important	I	II
Not important	III	IV

Source: 'The Seven Habits of Highly Effective People' — Stephen R. Covey

Focus only on Quadrant I (urgent + important) → it grows and dominates

Plan for important-but-not-urgent goals early, else they become urgent

Many struggle financially because they don't plan the important, not-urgent events

Goals, horizon & inflation; the pool approach ★★★★☆

Inflation (goal value)	Rise in the cost of a goal over time; must adjust goal values
Fast-rising costs	Education and healthcare (impact retirement years)
Pool approach	Meet all needs from one pool of savings — drawback: horizon of investments unknown
Long-term goals	Inflation impact big → must factor in
Immediate / near-term	Little impact of price change

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Case: Shalini's education ₹50 lakh today, 10 yrs, 8%
inflation → ≈ ₹1,07,94,625
Household ₹30,000/month at 6% → ₹31,800 next year ·
₹53,725 after 10 yrs · ₹96,214 after 20 yrs
Common to all goals: mismatch of cash flows (expenses >
income, or no income)
```

Saving vs investing ★★★★☆

'Saving'	Same root as 'safe' — safety of money; reduce consumption so money is saved
'Investing'	Primary objective: earn profits; risk–return trade-off
Relationship	Two steps of the same process: **saving precedes investing**

Factors to evaluate investments ★★★★★

Three most important: **safety, liquidity, returns**. Plus convenience, ticket size, taxability, tax deduction.

Factor	Workbook points
Safety	Safety of capital + degree of surety of income; understand the risks
Liquidity	Ease of converting to cash; product nature or features (lock-in, exit penalty). Penalty doesn't hamper liquidity — only lowers return. **Divisibility**: can you sell part?
Returns	Current income (periodic, without selling) + capital appreciation (only on sale). Exit charges = liquidity vs return trade-off
Convenience	Ease of investing, withdrawing (full/part), checking value, receiving income
Ticket size	Min amount: from ₹50/₹100 to >₹1 lakh or >₹1 crore; don't choose just because affordable
Taxability of income	Post-tax return matters; don't view tax in isolation; some products tax-efficient only if held to maturity/min period
Tax deduction	Raises effective return (on net amount invested); often comes with lock-in → liquidity vs deduction trade-off

TRAP No factor in isolation; consider the investor's situation.

Four asset classes ★★★★★

Asset class	Key traits (workbook)
Real estate	Location = most important factor · illiquid · not divisible · physical or financial form · rent (current income) + appreciation · high transaction costs (brokerage, registration) · maintenance & taxes reduce return · self-occupied house = expense, not investment · sub-types: residential, land, commercial
Commodities	Most can't be invested (perishable/storage). Commodity derivatives not 'investments': leveraged + short-term. Gold & silver = investment commodities; prices in sync globally (adjust for FX, costs); gold = safe haven; gold standard abandoned decades ago but central banks still hold gold; **no current income** — only appreciation; purity hard to judge
Fixed income	Borrowing via bonds/debentures (used interchangeably); issuers: companies, Union/State govts, municipal corps, banks, FIs, PSUs; regular interest; held from issue to maturity → usually no capital gain; secondary trades → gains/losses; safer than equity, not risk-free; classify by issuer (govt/corporate) or maturity (short: liquidity; long: income)
Equity	Owner's capital / risk capital; returns beat inflation historically; Sensex 100 (base 1979) → 82,269 (30 Jan 2026) ≈ 15.4% p.a. before dividends; dividends from profits; prices fluctuate often without regard to fundamentals but follow profits long-term

Comparison	Equity	Bonds	Real estate	Commodities
Form	Financial only	Financial only	Physical or financial	Physical or financial
Consumption use?	No	No	Yes (self-use)	Yes (jewellery)
Intermittent cash flow	Dividends (no guarantee)	Interest	Rent	**None**
Investor is	Owner	Lender	Owner	Owner
Cash flows known?	No	Agreed at issue	No	No

Ownership assets = uncertain returns

Lending assets (bonds, FDs) = agreed receipts

Tangible feels safer to many

International assets add currency exposure (e.g. LSE share → GBP/INR risk)

Table 1.1 — avenues by asset category ★★★★☆

Equity	Fixed income	Real estate / infra	Commodities	Hybrid / others
Blue-chip, mid-sized, small-sized, unlisted, foreign stocks; equity MFs; ETFs; index funds	Bank FD & RD; endowment & money-back policies; PPF; SSY; SCSS; PO MIS; PO RD; company FD; debentures/bonds; debt MFs	Physical: residential/commercial. Financial: REMF, REIT, InvIT	Gold, silver, gold funds, commodity ETFs	Hybrid/multi-asset MFs · Others: rare coins, art, rare stamps

TRAP Endowment and money-back policies are listed under FIXED INCOME. InvIT is under real estate/infrastructure.

Investment risks ★★★★★

Risk	Workbook points
Inflation risk	Erodes purchasing power; hits hard over long periods. Nominal rate = return without inflation; **real rate** = inflation-adjusted. Safety + anytime liquidity usually earns below inflation (e.g. FD 7% vs inflation 8%)
Liquidity risk	Promised return only if held to maturity (FD premature penalty; PPF no/partial liquidity); real estate takes weeks/months; listed equity liquid but volatile → unsuitable for short-term needs
Credit risk	Delay or default of interest/principal. 3 outcomes: pays on time / pays late / doesn't pay. Causes: **ability** (business stability, profitability) or **intention**. Own-country government bonds safest → lowest rates
Market & price risk	Volatility from opinion changes. Market-wide (e.g. war fear) vs company-specific (product sales fall) vs industry-specific (policy, technology — pager industry vanished). Specific risk reduced by diversification; market-wide not
Interest rate risk	Value changes with rates; affects bonds more than stocks; **inverse** relation; longer maturity → bigger price swings; market-wide

Inflation 8% p.a. — money needed to buy ₹10,000 of goods today	Amount
After 5 years	₹14,693
After 10 years	₹21,589
After 20 years	₹46,610
After 30 years	₹1,00,627

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FV = P × (1 + r)^n
100 units for ₹10,000 today → only 68 units after 5 yrs,
46 units after 10 yrs (8% inflation)
Bond ₹1,000 at 8%; new bonds offer 8.5% → old bond sells
at a discount
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Risk management strategies & risk measures ★★★★☆

Strategy	Meaning
Avoid	Skip products whose risk you don't want; experts: avoid what you don't understand (but lose the opportunity)
Take a position	Act on an expected event (e.g. expect rates ↓ → sell short-maturity, buy long-maturity bonds); needs superior knowledge; not for most investors
Diversify	Prudent for lay investors; spreads risk of loss

Measure	For
Credit rating, credit spreads	Credit risk
Variance, standard deviation, beta, modified duration	Volatility / return fluctuation (Ch 10)

Systematic vs unsystematic classification ★★★★★

Risk	Type
Credit risk	Mostly unsystematic (issuer-specific)
Liquidity risk	Unsystematic most of the time (except market-wide crisis, e.g. GFC)
Inflation risk	Systematic
Interest rate risk	Systematic (lower for short maturity, but exists)
Market risk (market-wide volatility)	Systematic
Price risk of individual security	Unsystematic

Unsystematic = specific risk → reduced by diversification

Systematic = market-wide risk → cannot be diversified

Behavioural biases ★★★★★

Dominant emotions: fear, greed, hope

Bias	Meaning
Availability heuristic	Rely on examples that come to mind immediately → too little research, miss risks
Confirmation bias	Seek info confirming existing views; decide first, then find data
Familiarity bias	Prefer familiar ('known devil better than unknown angel') → concentration, poor diversification
Herd mentality	Follow the group ('man is a social animal'); going against the herd often most profitable
Loss aversion	Avoiding ₹5,000 loss preferred over ₹5,000 gain; identified by Kahneman & Tversky (Kahneman: Nobel in Economics)
Overconfidence	Overrate own ability → lower guard, take unassessed risks
Recency bias	Extrapolate recent events; bear market → prefer safe assets, bull market → over-allocate risky assets
Behaviour patterns	Personality (spender/saver/investor) drives saving; behavioural tests help
Interest of the investors	Choose by interest, not suitability (IT employee holding only tech stocks) → concentration
Ethical standards & boundaries	Ethical people more disciplined; shortcuts derail long-term wealth

List not exhaustive

Remedy: detailed analysis; take a third person's opinion — RIA or MFD

Risk profiling ★★★★★

Purpose	Don't sell schemes riskier than the investor can handle
Need to take risk	Need higher returns to reach goals
Ability to take risk	Financial ability + investment horizon
Willingness to take risk	Psychological capacity
Distributor's job	Evaluate all three; strike a balance when they conflict
Method	Distributor free to choose/design own tools

Asset allocation: SAA, TAA ★★★★★

Definition	A **process** of allocating money across asset categories in line with a stated **objective**
Strategic (SAA)	Aligned to goals; uses required return, horizon, risk profile; maintains a **target** % allocation
Tactical (TAA)	Dynamically changes allocation to exploit opportunities; primary aim: improve **risk-adjusted return**; also called **dynamic asset allocation**; suits seasoned investors with large investible surplus
Workbook example	Two-asset portfolio (equity + debt)

Rebalancing ★★★★★

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Target 50:50 in A and B; ₹1,000 each
A +20% → ₹1,200 ; B -5% → ₹950 ; total ₹2,150 → 55.81% : 44.19%
Restore 50:50 → sell ₹125 of A, buy ₹125 of B  (buy low, sell high automatically)
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Needed for both SAA and TAA

Also when investor's situation / risk appetite changes

SEBI passive-breach rebalancing	Deviation from SID asset allocation due to passive breach (not AMC's act/omission): **30 business days** for all schemes except index funds & ETFs
Not applicable	Overnight funds

Do-it-yourself vs professional help ★★★☆☆

Question	Point
Can one do it?	Ability (skills/knowledge) + time (history teacher can't teach higher maths)
Does one want to?	May not enjoy research/admin/accounting; prefer profession, family, hobbies
Can one afford to outsource?	MF costs capped by SEBI; compare with hidden costs of DIY: value of time + emotional mistakes

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Example: DIY 12% vs fund 10% net (2% cost) on ₹10 lakh → ~₹20,000/yr saved — worth the time?
Value of time may be higher for wealthier people
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MF = outsourcing the investing job to an AMC, not a different asset

For most investors MF is the better option

02 Concept & Role of a Mutual Fund

≈4 marks · 24 subtopics · chapter priority ★★★☆☆

Concept of a mutual fund ★★★★☆

Definition	Professionally managed investment vehicle (a **trust**) that mobilises money from investors and invests per stated objectives
Correct phrase	Invest **through** mutual funds (not 'in') — a different **way** of investing, not a different product
Gives access to	Equities, bonds, money market and other securities otherwise unavailable + professional management by AMC
Difference	Professional investing, diversification, regulated vehicle

Role of mutual funds ★★★☆☆

- Primary: help investors earn income / build wealth via securities markets
- 'Fund' and 'scheme' used interchangeably; the offering entity = 'mutual fund' / 'the fund'
- Money raised funds govts & companies → projects → jobs → demand → economic development
- As large investors, check investee companies' operations, corporate governance, ethics
- Livelihood for MF employees, distributors, registrars, service providers
- Higher output → more tax revenue → nation-building
- **Market stabiliser**: in 2022 and 2025 FPIs sold heavily; MF inflows provided counterforce

Investment objectives (Table 2.1) ★★★★☆

Objective (from SIDs)	Category
Reasonable income + high liquidity; only debt & money market securities up to 91 days	Liquid fund
Capital appreciation/income from portfolio predominantly in equity	Equity fund
Long-term capital appreciation via mix: equity, debt, InvITs, commodities	Hybrid fund
Goal-based investing, reduce risk over time via pre-determined maturity + glide path	Life Cycle Fund

Objectives stem from safety, liquidity, returns

Each pool = a scheme with pre-announced objective

Classification by objective ≈ by investment universe

Profits/losses belong only to unitholders; all others earn fees/commission

Investment policy ★★★☆☆

Investment policy	Asset allocation + investment style (to achieve the objective)
Asset allocation	Disclosed in SID (e.g. liquidity → money market; long-term growth → equity)
Style	Growth or value; concentration: focused vs diversified

Key concepts: units, face value, unit capital, expenses, NAV, AUM, MTM ★★★★★

Term	Meaning
Units	Investment translated into units of the scheme
Face value	Typically ₹10 (can be ₹100); relevant for accounting
Unit capital	Units issued × face value
Recurring expenses	Fees/commissions to constituents; % of AUM; deducted in NAV → higher expense = lower NAV; SEBI limits
NAV	True worth of a unit; net realisable value per unit if scheme liquidated
AUM	Scheme size = NAV × units outstanding; AMC size judged by AUM; rises with performance/inflows, falls with IDCW/redemptions
IDCW	Dividend option now called Income Distribution cum Capital Withdrawal
Mark to market (MTM)	Valuing each security at current market value, daily, for NAV → daily NAV fluctuation

Advantages of mutual funds ★★★★★

Advantage	Workbook points
Professional management	Objective-based, research-based, prudent processes; avoids multiple accounts (broking, demat); handles corporate benefits (interest, dividends, rights, buybacks); daily NAV = daily accounting; low cost, even smallest investments
Affordable diversification	Proportionate ownership of diversified portfolio for as little as ₹500; 'eggs not in one basket'; same diversification directly needs several lakhs
Economies of scale	Afford professional managers; research/office costs spread; better terms with brokers, bankers; IDCW/growth options to structure returns
Transparency	3 sources: scheme documents (SID, SAI, KIM), portfolio disclosures (factsheet), NAV — available even to prospective investors
Liquidity	Recover value from the fund (anytime / intervals / closure); bond market wholesale — MF gives small-ticket access; close-ended compulsorily listed; timely 'material' info + liquidity enhances safety
Tax deferral	MFs don't pay tax on income; growth options defer investor's tax
Tax benefits	ELSS: deduction up to ₹1,50,000 per FY u/s 80C (Old Tax Regime)
Convenient options	Options suit liquidity/tax; partial withdrawal, additional investment, systematic transactions
Investment comfort	Further purchases need very little documentation
Regulatory comfort	SEBI checks and balances
Systematic approach	SIP, SWP, STP → investment discipline

Limitations of mutual funds ★★★★☆

Limitation	Points
Lack of portfolio customisation	PMS gives large investors control/customised portfolio; MF unitholder is one of thousands, can't influence securities
Choice overload	Many schemes/options; handled by information, advisers/MFDs; SEBI categorisation brings uniformity
No control over costs	Costs shared in proportion; SEBI limits; market forces push costs below limits (advantage)
No guaranteed returns	Pass-through vehicle; deciding factors: market movement, individual securities, team's skill — only skill is in manager's control

Classification by structure ★★★★★

Type	Features
Open-ended	Perpetual purchase & repurchase after NFO; no closing time-frame; unit capital keeps changing (units created on purchase, cancelled on redemption)
Close-ended	Fixed maturity; buy from fund only in NFO; wound up at maturity; **listing compulsory** (every close-ended scheme other than ELSS); unit capital stable; trades usually at **discount** to NAV (buyer has better bargaining position)
Interval	Largely close-ended, open at pre-specified intervals (e.g. Jan 1–5, Jul 1–5); compulsorily listed; **transaction period ≥ 2 working days**; **interval period ≥ 15 days**; redemption only in transaction period

SIF close-ended/interval strategies must also be listed

Listed close-ended/interval units: poor liquidity, discount to NAV

NFO = New Fund Offer (first offer to public)

Exchange Traded Funds (Box 2.1) ★★★★☆

- Traded on exchange like a stock; usually track an index → passive
- Open-ended fund: single end-of-day NAV; ETF: multiple prices through the day
- Extra liquidity, benefit from intraday price changes; downside: price may fluctuate and gap from NAV
- Buy like a stock, small minimum; needs demat; wide variety of indices
- Market price tracks NAV closely (solves close-ended liquidity issue)
- Silver ETF introduced by SEBI (invests primarily in silver/silver-related instruments), like Gold ETF

Classification by portfolio management; MF Lite ★★★★☆

Active funds	Manager chooses portfolio within objective; higher expenses; expected to beat market
Passive / index funds	Track an index (e.g. S&P BSE Sensex) in same weights; mirror index; not designed to beat market; manager has no role; low running costs
MF Lite (from 31 Dec 2024)	Framework for passively managed schemes: a mutual fund having only index funds, ETFs, FoFs or other specified schemes; **Chapter X of SEBI MF Regulations 2026**
MF Lite scheme	Any scheme launched by an MF Lite or other eligible passive scheme specified by SEBI

Classification by investment universe ★★☆☆☆

Equity, fixed income, money market, gold, international funds

Narrower: large cap, mid cap; G-sec funds, corporate debt funds

SEBI categorisation 2026 — five broad categories ★★★★★

Circular	Categorisation and Rationalisation of MF Schemes, 2026 (SEBI circular dated 26 Feb 2026)
Objective	Uniformity across fund houses; objective evaluation
A. Equity	13 sub-categories
B. Debt	17 sub-categories
C. Hybrid	7 sub-categories
D. Life Cycle Funds	(new)
E. Other	2 sub-categories

Market-cap definitions (Box 2.2) ★★★★★

Category	Rank by full market capitalisation
Large cap	1st – 100th
Mid cap	101st – 250th
Small cap	251st onwards

A. Equity schemes (13) ★★★★★

#	Category	Rule
1	Multi Cap	Min **75%** equity: min 25% each in large, mid, small
2	Large Cap	Min 80% in large cap
3	Large & Mid Cap	Min 35% large + min 35% mid
4	Mid Cap	Min 65% mid cap
5	Small Cap	Min 65% small cap
6	Dividend Yield	Predominantly dividend-yielding stocks; min **80%** equity
7	Value	Value strategy; min 80% equity
8	Contra	Contrarian strategy; min 80% equity
9	Focused	Max 30 stocks; must state focus (multi/large/mid/small cap); min 80% equity
10	Sectoral	Specific sector (bank, power…); min 80% in sector
11	Thematic	Theme (e.g. infrastructure: construction, cement, steel, telecom, power); min 80%
12	ELSS – Tax Saver	Statutory 3-yr lock-in + tax benefit; min 80% equity (ELSS 2005 notified by MoF)
13	Flexi Cap	Min 65% equity; dynamic across large/mid/small

Value AND Contra both allowed if portfolio overlap ≤ 50%

Sectoral/thematic: overlap ≤ 50% with other sectoral/thematic and other equity schemes except large cap

Existing scheme can convert to Flexi Cap (fundamental attribute change rules)

Residual portion: equity, money market, liquid, gold & silver instruments, InvITs (within limits)

ESG = sub-category under thematic

B. Debt schemes (17) ★★★★★

#	Category	Rule
1	Overnight	Overnight securities, 1-day maturity; may put ≤5% in G-secs/T-bills ≤30 days residual maturity as margin/collateral
2	Liquid	Debt & money market up to 91 days maturity
3	Ultra-Short-Term	Macaulay duration 3–6 months
4	Ultra Short to Short Term	Macaulay duration 6–12 months (low duration)
5	Money Market	Money market instruments, maturity up to 1 yr
6	Short Term	Macaulay duration 1–3 yrs
7	Medium Term	3–4 yrs; under adverse situation 1–4 yrs
8	Medium to Long Term	4–7 yrs; under adverse situation 1–7 yrs
9	Long Term	> 7 yrs
10	Dynamic Term	Across duration
11	Corporate Bond	Min 80% in AA+ and above corporate bonds
12	Credit Risk	Min 65% in AA (excl. AA+) and below corporate bonds
13	Banking & PSU Debt	Min 80% in banks, PSUs, PFIs, municipal bonds
14	Gilt	Min 80% G-secs across maturity
15	10-yr Constant Maturity Gilt	Min 80% G-secs; Macaulay duration = 10 yrs
16	Floating Interest Rates	Min 65% floating (incl. fixed converted via swaps/derivatives)
17	Sectoral (debt)	Min 80% in one sector, AA+ and above, across duration; sectors: Financial Services, Energy, Infrastructure, Housing, Real Estate

TRAP Scheme names can't stress only returns (e.g. 'credit opportunities', 'high yield', 'credit advantage') — SEBI circular 6 Oct 2017.

C. Hybrid schemes (7) ★★★★★

#	Category	Rule
1	Conservative Hybrid	Debt 75–90%, equity 10–25%
2	Balanced Hybrid	Equity 40–60%, debt 40–60%; **no arbitrage**
3	Aggressive Hybrid	Equity 65–80%, debt 20–35%
4	Dynamic Asset Allocation / Balanced Advantage	Equity/debt managed dynamically
5	Multi Asset Allocation	≥3 asset classes, ≥10% each; foreign securities NOT a separate asset class
6	Arbitrage	Min 65% equity (arbitrage); debt limited to G-secs <1 yr maturity + repo of govt bonds
7	Equity Savings	Min 65% equity, min 10% debt; **net equity 15–40%**; min hedged/unhedged stated in SID

Residual in hybrid: InvITs (not arbitrage funds), ETCDs, Gold ETFs, Silver ETFs

Balanced and Aggressive listed together (one AMC offers either)

D. Life Cycle Funds (new) ★★★★★

Strategy	Glide path across equity, debt, InvITs, ETCDs, gold & silver ETFs; risk falls as maturity nears (equity ↓, debt ↑)
Tenure	Min 5 yrs, max 30 yrs; multiples of 5
Limit	Max **6** life cycle funds open for subscription per MF at a time
Name	Must include maturity date (e.g. Life Cycle Fund 2045, 2055)

If AMC keeps...	It may not launch	May launch
Children's Fund	20-yr life cycle fund	Any of remaining 5
Retirement Fund	30-yr life cycle fund	Any of remaining 5
Both	20 & 30-yr	Only 5, 10, 15, 25-yr (4 funds)
Neither (discontinues both; merged per Board approval)	—	All 6

E. Other schemes; one scheme per category rule ★★★★★

Category	Rule
Index Funds / ETFs	Min 95% in index securities
Fund of Funds (overseas/domestic)	Min 95% in underlying fund

Only ONE scheme per category, except: (1) index funds/ETFs on different indices (2) FoFs with different underlying (3) sector/thematic funds in different sectors/themes

Other categories: FMP, TMF, capital protection, IDF, REMF ★★★★☆

Type	Points
Fixed Maturity Plan (FMP)	Close-ended debt; portfolio duration aligned to scheme maturity; pre-identified investments; no money post-NFO; clarity on likely returns (no guarantee)
Target Maturity Fund (TMF)	Debt fund maturing on a specific date (typically 2–14 yrs); bonds with matching maturity → low reinvestment risk, predictable returns; listed for exit; low credit risk, tax efficiency, liquidity
Capital Protection Oriented	Close-ended hybrid; equity via derivatives; e.g. ₹90 in debt for 3 yrs grows to ₹100; rest in equity derivatives
Infrastructure Debt Fund (IDF)	Sponsored by banks/NBFCs; trust-based IDF = MF (SEBI); company-based = NBFC (RBI). IDF-MF: close-ended, min **90%** in debt/securitised debt of infra companies/projects/SPVs or bank loans of completed revenue-generating projects. IDF-NBFC sponsors: only banks & Infrastructure Finance Companies
Real Estate MF (REMF)	Real estate assets in cities per SEBI circular 2 May 2008; close-ended; must be listed

ESG investing ★★★☆☆

Separate sub-category under thematic equity

Strategies: Exclusion · Integration · Best-in-class & Positive Screening · Impact investing · Sustainable objectives · Transition / transition-related

REITs & InvITs (box) ★★★★☆

	REIT	InvIT
Registered with	SEBI (trust)	SEBI (trust)
Invests in	Commercial real estate	Infrastructure
Asset value (initial offer)	≥ ₹500 crore	≥ ₹500 crore
Min offer size	≥ ₹250 crore	≥ ₹250 crore
Min subscription	₹10,000–15,000 (earlier ₹50,000)	₹10,000–15,000 (earlier ₹1 lakh)
Listing / lot	Listed; trading lot 1 unit	Listed; trading lot 1 unit

SM REITs (small & medium): asset value ≥ ₹50 crore, capped at ₹500 crore

REITs/InvITs are NOT mutual fund schemes

REIT raises later via follow-on, rights, institutional placements

New types of funds ★★★☆☆

Fund	Idea
Factor-based	Factors: value (inexpensive), size (smaller cos), momentum (recent winners), low volatility, quality (strong fundamentals), dividend yield; factors rotate; often via ETFs
Smart beta	Extension of index/ETF; less market-cap weighting (e.g. equal weight); aim: better returns, diversification, lower risk
Quant	Machine/data-driven models from past data; removes human element
International REITs	Exposure to overseas REITs: international + commercial real estate

Specialized Investment Fund (SIF) — strategies ★★★★★

Why	Gap between MFs and PMS in portfolio flexibility; regulation gets more flexible MF → PMS → AIF
What	New product line under MF regulations
Minimum	₹10 lakh per investor across all strategies of the same AMC
Proliferation	Only ONE strategy per category

Strategy	Rules	Structure / redemption
Equity Long-Short	Min 80% equity; unhedged short ≤ 25%	Open-ended/interval; daily or less frequent
Equity Ex-Top 100 Long-Short	Min 65% of NAV in stocks excluding top 100; short (non-large cap) ≤ 25%	Open-ended/interval; daily or less
Sector Rotation Long-Short	Min 80% equity; max 4 sectors; short at sector level (all stocks of sector); short ≤ 25%	Open-ended/interval; daily or less
Debt Long-Short	Debt across duration; unhedged short via exchange-traded debt derivatives	Interval; redemption once a week or less
Sectoral Debt Long-Short	Min 2 sectors; max 75% per sector; short across all instruments of a sector; short ≤ 25%	Interval; once a week or less
Active Asset Allocator Long-Short	Equity, debt, equity/debt/commodity derivatives, REITs, InvITs; short ≤ 25%	Interval; twice a week or less
Hybrid Long-Short	Min 25% equity + min 25% debt; short ≤ 25%	Interval; twice a week or less

Growth of the MF industry in India ★★★☆☆

Metric	Data
AUM	₹12.33 lakh crore (Mar 2016) → ₹79.46 lakh crore (Mar 2026); 10-yr CAGR 20.48%
Folios	4.8 crore (Mar 2010) → 27.39 crore (Mar 2026)
MF share in financial investments	10% (Mar 2016) → 12% (2017) → 14% (2018); bank deposits 71% → 69% → 65%
India's share of world MF industry	0.33% (2008) → 0.60% (2018)
Monthly SIP contribution	₹8,055 crore (Mar 2019) → ₹32,087 crore (Mar 2026)
SIP AUM	₹15.11 lakh crore (Mar 2026)

03 Legal Structure of Mutual Funds in India

≈3 marks · 11 subtopics · chapter priority ★★★☆☆

Definition of 'mutual fund' (SEBI MF Regulations 2026) ★★★★☆

Definition	Fund established as a **trust**, registered under regulation 3, raising money by sale of units to the public under one or more schemes, investing in securities, money market instruments, gold/gold-related, silver/silver-related and other SEBI-specified assets
Must have	A separate Asset Management Company (AMC)

Established as a trust · Sells units to public / section of public · Under one or more schemes · Invests in securities incl. money market, gold, silver, real estate assets

Legal structure — checks and balances ★★★★★

Element	Rule
Governing law	Indian Trusts Act, 1882 (MFs are trusts)
Creator	One or more **sponsors**
Beneficiaries	Unitholders (investors)
Trust deed	Executed between sponsor and trustees; SEBI prescribes clauses; names first trustees + procedure to change them; registered under **Indian Registration Act, 1908**
Trustees	Protect beneficiaries; individuals (= Board of Trustees) or Trustee Company (acts via Board of Directors)
AMC	Day-to-day management; appointed by sponsor or trustees (with SEBI approval)
IMA	Investment Management Agreement: trustees ↔ AMC
Custodian	Holds assets (securities, gold, real estate, silver); appointed by **trustees**
RTA	Investor records — by AMC itself or an appointed RTA

Sponsor eligibility ★★★★☆

- Applies to SEBI for MF registration; then invests in AMC capital
- Sound track record, reputation of fairness and integrity
- Business in financial services for **≥ 5 years**
- Net worth (share capital + reserves − accumulated losses) **positive in all 5 preceding years**
- Positive **liquid net worth** > proposed capital contribution to AMC (on change of control via share acquisition: ≥ higher of par or market value of shares acquired)
- **Net profit** (after depreciation, interest, tax) in **each** of preceding 5 years
- **Average net annual profit ≥ ₹10 crore** over preceding 5 years
- Fit and proper person

Liquid net worth = net worth in unencumbered liquid assets: cash, money market, G-secs, T-bills, repo on G-secs etc.

AMFI website lists AMCs by sponsor category: Banks, Institutions, Private sector (Indian, foreign, predominantly Indian JV)

Trustees (Regulation 11) ★★★★★

Eligibility	Fit & proper (Schedule II, SEBI Intermediaries Regs 2008) · NOT a director (incl. independent), officer or employee of an AMC · NOT trustee of any other MF · furnish info SEBI seeks
SEBI approval	Prior approval needed to appoint a trustee
Independence	≥ **two-thirds** independent (not associates of sponsor, AMC or trustee company)
Rights & responsibilities	1. Appoint/terminate AMC 2. Conflict of interest & broker empanelment 3. Regulatory compliance 4. Oversight for investor interests 5. Additional oversight 6. Due diligence as specified by SEBI

Asset Management Company ★★★★★

Rule	Detail
Appointment	Sponsor/trustees appoint AMC with SEBI approval
Directors	Adequate professional experience in finance; directors & key personnel fit and proper
Board independence	≥ **50%** directors not associates of sponsor, its subsidiaries or trustees
Cross-holding	AMC directors (incl. Chairman) can't sit on any MF trustee board; trustees' prior approval to appoint AMC director; a person can't be director of more than one AMC
Net worth	≥ **₹50 crore** (deployed in specified assets, effective 1 Jan 2024); maintained continuously — sponsor's responsibility
Change of control	Prior approval of trustees + SEBI; letter/email to each unitholder; ad in one English national daily + one regional-language paper (HO region); exit at NAV **without exit load** within ≥ **30 calendar days**
Duties	Offices, staff, software, advertising, regulator interaction; due diligence so investments comply with regulations and trust deed
Termination	By **majority of trustees** or **75% of unitholders**; change needs prior approval of SEBI and unitholders
SIF	AMCs launching SIF must meet additional eligibility

Custodian ★★★★★

Role	Custody of assets; accepts/gives delivery; settles all transactions; tracks corporate actions (dividends, bonuses, rights)
Registration	SEBI (Custodian) Regulations, 1996
Appointed by	Trustees (custodial agreement trustees ↔ custodian)
Independence	If sponsor/associates control ≥ 50% voting rights, or ≥ 50% of custodian's directors represent sponsor → can't be appointed (unless specific conditions met)
Why	Ensures securities are really held for investors — control aspect

AMC organisation structure ★★★★☆

Function	Role
Compliance	Compliance Officer signs due-diligence certificate in new scheme documents; reports **directly to head of AMC**; works with trustees; reports non-compliance directly & immediately to trustees
Fund management (most critical)	Analysts (identify opportunities: equity, credit, interest rates) · Fund managers (decide; responsible for performance) · Dealers (place orders with brokers)
Operations & customer service	Front office + back office; call centres, chat bots; RTA unit (records, allot/redeem, account statements); Custody team (settlement); Fund accounting (books per scheme, daily NAV)
Risk management	Independent specialised function (SEBI RMF); dedicated risk officers for investment, compliance, operational and cyber security risk
Sales & marketing	Branding, ads, events, distribution; interacts with MFDs; training
Support	Accounts (AMC's own finances — different from fund accounting) · Administration (often reports to finance) · HRD · IT (website, investor/distributor tech)

Service providers ★★★★★

Provider	Key points
Fund accountant	Calculates NAV; in-house or outsourced; **no SEBI registration** needed
RTA	Investor records; offices = Investor Service Centres (ISCs); process purchases/redemptions, update unit capital & folios; appointed by **AMC**; not compulsory (AMC may do in-house); must register with SEBI
Auditors	Scheme accounts separate from AMC; scheme auditor ≠ AMC auditor; scheme auditor appointed by **trustees**, AMC auditor by **AMC**
Distributors	Individuals or institutions; empanel with many AMCs; NISM V-A + AMFI registration; new cadre: NISM V-B (limited schemes; SEBI circular 13 Sep 2012)
Collecting bankers / payment aggregators	Appointed by AMC; scheme bank accounts; payment gateways, digital wallets, payment banks
KRAs	KYC mandatory under PMLA; SEBI KRA regulations 2011 → common KYC; registered with SEBI; issue letter/number usable with any intermediary; confidentiality; computerised connectivity among KRAs
Valuation agencies	For non-traded/thinly traded debt; ≥ 2 agencies; AMFI appointed **CRISIL Ltd. and ICRA Ltd.**
Credit rating agencies	Ratings = initial input; corporate bond & credit risk funds defined by ratings; capital protection schemes rated; FMPs may disclose proposed rating profile and must invest within it or better
Depositories & DPs	Hold demat units; NSDL and CDSL; DP ≈ bank branch, depository ≈ head office; ETFs compulsorily demat

Stock exchanges & transaction platforms ★★★★☆

Platform	Point
Exchange listing	Close-ended funds & ETFs compulsorily listed on ≥ 1 exchange
BSE StAR MF	BSE's MF segment
NSE NMF-II	NSE Mutual Fund II platform
MF Utilities India	AMFI industry initiative; one platform for transactions across many MFs
MF Central	Investor services hub by **KFintech & CAMS**; all folios in one view, service requests

Execution Only Platforms (EOP) ★★★★☆

Definition	Digital/online platform for subscription, redemption, switch in **direct plans**
Excluded	Platforms of SEBI RIAs and stock brokers for their own clients
Registration	From SEBI or AMFI — none may operate without it
Category 1	Registered with **AMFI**; body corporate; agent of **AMCs**; integrates with AMCs/RTAs; may aggregate direct-plan transactions
Category 2	Registered as **stock broker** (SEBI Stock Brokers Regs 2026) in EOP segment of exchanges; body corporate; agent of **investor**; only via exchange platforms

AMFI — role and functions ★★★★★

- Association of all registered AMCs
- Define and maintain high professional and ethical standards
- Recommend best practices and code of conduct
- Interact with and represent to SEBI; represent to Government, RBI and others
- Nationwide investor awareness programme
- Disseminate information; studies & research
- Regulate conduct of distributors incl. disciplinary action (cancellation of ARN)
- Protect interest of investors
- Register distributors — allots **ARN** (mandatory); issues best-practice circulars

> **TRAP** AMFI is NEITHER a regulator NOR a Self-Regulatory Organisation (SRO).

04 Legal & Regulatory Framework

≈7 marks · 22 subtopics · chapter priority ★★★★☆

Four financial regulators ★★★★☆

Regulator	Regulates
RBI	Banking system + money markets
SEBI	Securities markets
IRDAI	Insurance
PFRDA	Pension

All under the Ministry of Finance

Purpose: protect consumers + regulated development of markets

Role of SEBI ★★★★☆

Regulates	MFs, depositories, custodians, RTAs, credit rating agencies (among others)
Preamble	'to protect the interests of investors in securities and to promote the development of, and to regulate the securities market...'
'Securities'	Transferable, tradeable instruments — Sec 2(h) of SCRA, 1956
Three aspects	Disclosures by issuers · efficiency of transactions · low transaction costs
Also checks	Deliberate speculation · insider trading · excessive risk by MFs · inadequate collateral by debt issuers

Unchecked → loss of investor trust → household savings stop flowing to capital markets

Regulatory reforms: MF Regulations 1996 → 2026 ★★★☆☆

MF Regulations first issued 1996; replaced by **SEBI MF Regulations 2026**

Covers investor services, NAV accounting, valuation, disclosures, investment norms

Master circulars on sebi.gov.in

22 areas covered by regulations ★★★★☆

#	Area	Key point
1	Scheme documents (offer document = SID + SAI)	Content, update frequency; performance vs **Total Return Index (TRI)**, not Price Return Index
2	Registration	AMC, sponsor, trustees; mergers, change of management
3	Conversion & consolidation	Scheme mergers; one scheme per category; market-cap definitions; PRC matrix for debt
4	New products	e.g. Infrastructure Debt Funds, ESG
5	Risk management	Operational risk, exposure limits (true-to-label), segregated portfolios (after 2018 credit crisis); 2008 crisis: no premature redemption of FMPs, no 'liquid plus' names
6	Disclosures & reporting	Formats, frequency
7	Governance	Audit & valuation committees, independent directors, trustees, min investors, benchmarking, systems audit, role in corporate governance
8	Secondary market activities	Equity, debt, G-sec, derivatives
9	NAV	Disclosure, rounding, cut-off, time stamping, sale/purchase price uniformity
10	Valuation	Of all securities
11	Loads, fees, expenses	Limits; commission disclosure
12	IDCW distribution	Procedure; distributable surplus
13	Investment by schemes	Restrictions and limits
14	Advertisements	Content, statutory ads, disclaimers
15	Investor rights & obligations	Statements, payout timelines, delay penalty, **instant access facility** (liquid funds)
16	Certification & registration of intermediaries	Exam + CPE
17	Transactions in MF units	Exchanges, record-keeping, KYC
18	CDMDF	Corporate Debt Market Development Fund — **backstop facility** to buy investment-grade corporate debt in stress
19	Execution Only Platforms	Scope, categories, eligibility, fees, grievance, conflicts, tech, disclosures
20	MF Lite Framework	Light-touch regime only for passive schemes
21	Specialized Investment Fund (SIF)	Bridge MF–PMS gap
22	Miscellaneous	Everything else

TRI: dividends/interest notionally reinvested on ex-date + capital gains

Segregated portfolio = separate distressed/illiquid assets after a credit event

Other regulators' rules MFs must follow ★★☆☆☆

RBI: money market, overseas investment, non-resident investment, foreign remittances

Stock exchanges (SEBI-regulated): listing, trading, margining rules

Permitted investments — Regulation 39 ★★★★★

- (a) Securities
- (b) Money market instruments
- (c) Privately placed debentures
- (d) Securitised debt (asset-backed or mortgage-backed)
- (e) Gold / gold-related
- (f) Silver / silver-related
- (g) Others specified by SEBI

MF = pass-through; investors have no control over investment management → SEBI limits risk

Restrictions apply at the time of investment

Restrictions — Regulation 41 ★★★★★

- All investments (except gold/silver ETFs) subject to **Sixth Schedule** + other SEBI restrictions
- Gold/silver ETFs may invest in exchange-traded commodity derivatives
- Settle only in **dematerialised** securities (except specified instruments)
- Gold/silver ETF funds only in gold/silver (related) instruments, except for redemption liquidity; pending deployment → short-term deposits of scheduled commercial banks
- No **carry forward** transactions (except as SEBI specifies)
- Derivatives only on a recognised stock exchange (SEBI framework)
- Short selling only on recognised exchange (short selling & SLB framework)
- Inter-scheme transfers within same MF allowed under SEBI conditions

Sixth Schedule restrictions ★★★★★

- Equity, equity-related and debt only in **listed or to-be-listed** securities — except unlisted G-secs, money market instruments other than **CPs**, and unlisted NCDs as SEBI specifies
- Unrated debt subject to SEBI conditions
- Scheme may invest in another scheme (same or other MF) **without fees**, within limits (not applicable to FoFs)
- **No scheme may invest in a FoF scheme**
- A FoF may not invest in another FoF; only in MF schemes (except liquidity for redemptions)
- Buy/sell on **delivery** basis — take/give delivery (exceptions: short selling framework, exchange derivatives, sale of contracted G-sec per RBI)
- Securities held in the name of the MF on account of the scheme
- Pending deployment → short-term deposits of scheduled commercial banks
- Prudential limits as SEBI prescribes

SIF investment restrictions — Regulation 51 ★★★☆☆

SIF may invest in instruments permitted under Regulation 39

SEBI may specify manner of investment

Subject to Sixth Schedule restrictions

Advertisement code — Fifth Schedule ★★★★★

Clause	Rule
(a)	Accurate, true, fair, clear, complete, unambiguous, concise; not misleading
(b)	No false/biased/deceptive statements, no assumptions/projections, **no testimonials, no ranking**
(c)	Not designed to be misunderstood or disguise significance; no misleading by implication/omission
(d)	No exaggerated/unwarranted slogans or slogans inconsistent with risk-return
(e)	**No celebrities** in advertisements
(f)	Don't exploit inexperience; avoid technical/legal jargon and excessive detail
(g)	Timely; consistent with SID, SAI, KIM
(h)	Don't discredit other ads or make unfair comparisons
(i)	Standard warning in legible font: 'Mutual Fund investments are subject to market risks, read all scheme related documents carefully.' — no words added/deleted
(j)	Vernacular ads: warning in the vernacular language
(k)	Audio-visual: visual + voice-over clear; e.g. **14 words for at least 5 seconds**

IDCW / bonus advertisement rules ★★★★☆

- Disclose IDCW in ₹ per unit + face value + prevailing NAV at declaration
- Close-ended maturity / interval completion: 'the entire distributable surplus at the time of maturity or at the completion of the interval period shall be distributed'
- Immediately below payout figure: NAV will fall to the extent of payout and statutory levy (if applicable)

Performance advertisements ★★★★★

Scheme age	Rule
> 3 years	CAGR for 1, 3, 5 yrs and since inception (since date of allotment) + point-to-point value of ₹10,000
< 6 months	**No past performance**
6 months – 1 year	Simple annualised growth for past 6 months
Overnight, liquid, money market	Simple annualisation of yields if data for ≥ 7, 15, 30 days (not misleading)

- Period computed to last day of month-end preceding the ad
- State regular or direct plan; footnote: plans have different expense structures
- Footnote if same fund manager hasn't managed for the full period
- Include performance of **all other schemes** of the fund manager (CAGR 1, 3, 5 yrs with benchmark); if > 6 schemes → total number + top 3 and bottom 3
- Internet ads may give exact website link to fund manager's other schemes' data

Standard additional benchmarks ★★★★☆

Scheme type	Additional benchmark
Equity	Sensex / Nifty
Debt ≤ 1 yr duration/maturity, Arbitrage	1-year T-Bill
Other debt	10-yr dated GoI security
Conservative Hybrid	10-yr dated GoI security
Balanced/Aggressive Hybrid, BAF, Multi Asset	Sensex or Nifty
Equity Savings	10-yr dated GoI security
Retirement / Children's	Sensex or Nifty
Index funds/ETFs/FoFs	Based on underlying asset allocation

Shown in INR and CAGR | Part of SAI and all advertisements

Celebrity endorsement at industry level ★★★★★

- Allowed only to increase awareness of MFs as a product category
- Must not promote a particular MF's scheme or brand an AMC
- Expenses limited to industry-level pooled investor education & awareness amounts
- **Prior SEBI approval** required

SIF branding & advertisement ★★★★★

- Distinct identity, separate from the MF: distinct **brand name and logo**
- May use sponsor's/MF's brand in SIF documents and ads for **5 years** from SEBI approval for SIF launch
- Use phrases like 'brought to you by', 'offered by'
- Sponsor/MF brand font size ≤ SIF brand font size
- MF advertisement guidelines apply to all SIF strategies
- Separate website or dedicated web page for SIF

Unauthenticated news guidelines ★★★☆☆

- Internal code of conduct; staff must not circulate rumours/unverified info
- Access to blogs/chat forums/messenger sites restricted, supervised or blocked
- Logs of such usage kept as records
- Market news forwarded only after Compliance Officer's approval; violations → action; Compliance Officer also liable

Investors' rights ★★★★★

Right	Detail
Beneficial ownership	Proportionate; can ask for unit certificate; demat allotment (open & close-ended); demat units freely transferable (ELSS restricted during lock-in); demat statement = SOA
Change distributor	Written request; no NOC from existing distributor
Inspect documents	Trust Deed, IMA, Custodial Services Agreement, RTA agreement, AMC's Memorandum & Articles
Nominees	Up to **3** nominees with % split; if not specified → equal
Pledge	Units pledged as security to financier
Grievance redressal	Complaint status in annual report, MF and AMFI websites; pending complaints can stop new scheme approval
Fundamental attribute change	Exit at NAV without load for ≥ **30 days**; written communication (email/SMS), website display
Terminate AMC / wind up	**75% of unitholders** can terminate AMC or wind up scheme
Unclaimed amounts	Claim within 3 yrs → prevailing NAV; after 3 yrs → NAV at end of 3 yrs
Other services	Account statements, portfolio statements, half-yearly disclosures

Trustees must get unitholders' consent: when SEBI requires; when 75% of unitholders require; when majority of trustees decide to wind up / prematurely redeem close-ended

Trust is a notional entity → investors can't sue the trust, but can sue **trustees** for breach of trust

Caveat emptor applies; prospective investors don't have unitholders' rights

Daily disclosure of debt & money market trades (incl. inter-scheme) with **15-day lag**

Unclaimed amounts & illiquid security proceeds ★★★★☆

Deployment	Money market + separate plan of a liquid scheme for unclaimed amounts
AMC fee	Max **0.50% p.a.**; **no exit load** on this plan
Reminders	AMC reminds investors; annual report shows unclaimed amount & number of investors per scheme
Written-off security later recovered	Substantial & recovered within **2 years** → paid to old investors; else → Investor Education Fund

Due diligence of distributors ★★☆☆☆

AMCs regulate distributor practices via due diligence

AMFI coordinates common unified due diligence

Details in Ch 6

Grievance redressal mechanism ★★★★★

```
Investor service centre → senior levels at AMC → SEBI (SCORES) → SMART ODR
AMC must redress within 21 calendar days of receipt
```

SCORES	SEBI's web-based centralised system (scores.sebi.gov.in): lodge, follow up, track online; physical complaints at SEBI offices are scanned & uploaded
Entities covered	Listed cos/RTAs · brokers/exchanges · DPs/depositories · MFs · portfolio managers · others (KRAs, CIS, merchant bankers, CRAs, FPIs)
SMART ODR	Online dispute resolution: formal online conciliation and arbitration

Cyber security & resilience (CSCRF) ★★★☆☆

Framework	Cyber Security and Cyber Resilience Framework for SEBI Regulated Entities
5 goals (from CERT-In CCMP)	Anticipate · Withstand · Contain · Recover · Evolve
AI threats	Task force **cyber-suraksha.ai**
Incident reporting	Email mkt_incidents@sebi.gov.in within **6 hours**; SEBI Incident Reporting Portal within **24 hours**; aligned with FSB's FIRE format

AMFI Code of Ethics & AGNI ★★★★☆

Fourth Schedule	SEBI Code of Conduct for AMCs & trustees (Appendix 1)
ACE	AMFI Code of Ethics — higher, more explicit standards for AMCs & trustees (Appendix 2)
AGNI	AMFI Guidelines & Norms for Intermediaries — agents, brokers, distribution houses, banks (Appendix 3)

```
Breach by intermediary:
1. AMFI writes, asks explanation within 3 weeks
2. No/unsatisfactory reply → warning letter (next violation = cancellation)
3. Proven second violation → registration cancelled, all AMCs informed
Intermediary may appeal to AMFI
```

05 Scheme Related Information

≈7 marks · 17 subtopics · chapter priority ★★★★☆

Mandatory scheme documents; caveat emptor ★★★★★

Scheme related documents	SID, SAI and KIM
Caveat emptor	'Let the buyer beware' — investor presumed to have read them; signing the form = legal acceptance; can't later claim unawareness
Two key documents	**SID** (scheme details) + **SAI** (statutory info on MF/AMC; one SAI for all schemes)
Legal technicality	SAI is part of the SID (in practice separate documents)
Format	Prescribed by SEBI, same sequence; MF may add any 'material' disclosure; simple language
SEBI role	Does NOT approve; gives observations → documents are **'vetted'**, not approved
Where	Draft SID/SAI on SEBI website (public); final documents on AMFI website; every MF's site

Contract: offer via documents → investor accepts by signing & paying → MF completes by issuing units

SID — nature & cover page ★★★★☆

Purpose	Info a prospective investor ought to know before investing
Validity	Effective until a 'material change' (other than fundamental attribute) → filed with SEBI and communicated / newspaper ads
Cover page	Scheme name, type (open/close/interval; equity/debt/liquid/hybrid), face value, NFO dates (open, close, re-open), SID date, MF name, AMC & trustee contacts
Standard clause	Prepared per SEBI MF Regulations 2026, filed with SEBI with AMC's **Due Diligence Certificate**; units 'not approved or recommended by SEBI nor has SEBI certified the accuracy or adequacy' of the SID

SID structure ★★★☆☆

Table of contents · Highlights · Introduction

Risk factors (standard + scheme-specific) · Minimum number of investors

Special considerations · Definitions · Due Diligence Certificate (AMC)

Information about the scheme: name & type, objectives & policies, asset allocation, where it invests, strategy, fundamental attributes, benchmark, managers, restrictions, performance

Units and offer · Fees & expenses · Rights of unitholders

General unitholder info · Penalties, litigation

Using the SID — where to find what ★★★★☆

Question	Where
Eligible / ineligible investors	NFO & Ongoing Offer Details
Risk-o-meter + suitability statement	Front page of SID
Risks	Risk section: standard risks (no guarantee; sponsor not liable beyond contribution; past performance not indicative) + specific (credit, interest rate, liquidity, pre-payment, reinvestment; equity volatility; unlisted liquidity; foreign, securitised, derivatives) + mitigation
Type, objective (primary/secondary), asset allocation, instruments, restrictions, strategy, investment management structure	'Information about the Scheme' (objective also in Highlights)
When to buy	Cover page: open-ended (NFO + continuous at NAV), close-ended (only NFO), interval (NFO + transaction periods)
How to apply	SAI + SID together
Purchase/redemption price	NFO & Ongoing Offer (applicable NAV, cut-off, amount)
Loads	Load structure segment
Min purchase, min balance, min redemption	Highlights + Ongoing Offer
Plans (regular/direct) & options	Highlights + Ongoing Offer
Modes of transaction	SAI 'How to Apply' (also SID)
Liquidity & redemption	Highlights
Fees, expenses (with examples)	SID
SIP, STP, triggers, switch, IDCW transfer	SID special products
Periodic info, performance vs benchmark	SID
NAV computation, tax (brief)	SID

TRAP SEBI has WITHDRAWN the transaction charges framework (Aug 2025 circular).

Fundamental attributes (Box 5.1) ★★★★★

Attribute	Includes
(i) Type of scheme	Open/close/interval; sectoral, equity, balanced, income, index, other
(ii) Investment objective	Main objective (growth/income/both); tentative portfolio break-up with min–max allocation (may alter short-term on defensive considerations)
(iii) Terms of issue	Liquidity (listing, repurchase, redemption); aggregate fees & expenses; any safety net or guarantee

Regulation 22(9)(c)	Trustees ensure no change in fundamental attributes / trust / fees & expenses / other modification unless: (a) written communication (email/SMS ok) to each unitholder (b) details on AMC website (c) exit option at prevailing NAV without exit load

Statement of Additional Information (SAI) ★★★★☆

Content	Description
Constituents	Sponsors, AMC, trustee company, service providers (custodian, RTA, statutory auditor, fund accountant if outsourced, collecting bankers): experience, condensed financials (schemes of last 3 FYs), key personnel, rights & obligations
How to apply	Guidance
Rights of unitholders	Beneficial ownership, service timelines, information, exit on changes
Investment valuation norms	SEBI-standardised valuation
Tax, legal & general	Nominations, transfer/transmission, minors, pledge/lien, taxation
Investor grievance	How to register

Downloadable on MF website; right to a printed copy; all SAIs on AMFI website

Key Information Memorandum (KIM) ★★★★★

Why	Circulating SID/SAI with forms is costly → KIM mandatorily circulated with application form
What	Summary of SID and SAI
Rule	Every application form accompanied by KIM

AMC, MF, trustee, fund manager & scheme names
Issue open/close/re-opening dates · Objective · Asset allocation
Risk profile (risk to principal, horizon, securities) · Plans & options
Benchmark · Dividend policy
Performance vs benchmark: 1, 3, 5 yrs, since inception · Expenses
Grievance info

Addendum ★★★★☆

Interim changes between updates
Part of scheme documents; must accompany KIM

Updating SID, SAI, KIM ★★★★★

Document	Rule
SID (open-ended & interval)	First: within 6 months from end of the half-year (H1/H2) of launch, data as of previous month-end. Then: within **1 month** from end of each half-year, data as of end-September / end-March
SAI	Regular update by end of **3 months** of every FY; material changes ongoing, uploaded on MF & AMFI websites
KIM	At least once every **half-year**, within 1 month of half-year end (Sept/March data); filed with SEBI electronically

Change	Procedure
Fundamental attribute (Reg 22(9)(c))	Addendum on website immediately · SID revised after exit window (≥ 30 days from notice) · public notice in one English national daily + one regional-language paper (HO region)
Other changes	Addendum on website immediately · circulated to distributors/brokers/ISCs to attach to KIM & SID in stock · latest addendum part of KIM/SID (e.g. load change) · account statements continue to show load structure

Current value & daily NAV disclosure ★★★★★

```
Current value of investment = Unit balance × Current NAV
```

Daily NAV	End of each business day on AMC website + AMFI website (amfiindia.com/net-asset-value)
Open-ended	NAV every business day, released to press
Liquid & overnight	NAV declared on **Sundays/holidays** too (accrual)
Close-ended	NAV at least **once a week**
Overseas schemes	Revised time limit (SEBI, Mar 2023)

TER / Base Expense Ratio disclosure ★★★★★

```
TER = Base Expense Ratio (BER) + Brokerage cost +
Transaction cost + Statutory levies (incl. GST)
```

Requirement	Rule
Daily disclosure	Scheme-wise, date-wise TER (all schemes except IDFs) in section 'Total Expense Ratio of Mutual Fund Schemes' on AMC + AMFI websites; downloadable, machine-readable; TER broken into components
BER change notice	Email/SMS at least **3 working days** before; website section updated 3 working days ahead
No prior notice	If BER change due only to AUM fluctuation, or a decrease due to regulation
Governance	BER changes intimated to AMC Board (with rationale) and to trustees quarterly
SID/KIM	All expenses identified; web link to TER of last 6 months, daily TER, factsheet; numerical illustration of TER impact for direct & regular; separate regular/direct disclosures
Half-yearly CAS	Average TER % with breakup: investment & advisory fees · distributor commission · other expenses
Annual accounts	Any single expense item > **10%** of total expenditure disclosed separately
Abridged annual report	TER as % of average AUM, plan-wise

Scheme dashboard ★★★☆☆

Scheme name · Classification/category · Open/close-ended · Objective
Fund manager · Benchmark (+ additional) · Exit load & lock-in
Plans & options · Min investment · Month-end AUM · TER
Inception date · Month-end portfolio · Performance

Risk-o-meter ★★★★★

Level	Colour
Low	Irish Green #08A04B
Low to Moderate	Chartreuse #7FFF00
Moderate	Neon Yellow #FFFF33
Moderately High	Caramel #C68E17
High	Dark Orange #FF8C00
Very High	Red #F70D1A

- Pictorial risk to principal + written statement below
- Risk value per SEBI parameters; evaluated **monthly**
- Change → notice-cum-addendum + email/SMS to unitholders
- Show risk-o-meter of scheme AND benchmark when disclosing performance
- First step only — not the only factor; consult MFD/RIA (product labelling circular 18 Mar 2013)

Commodity annualised volatility (quarterly, 15 yrs of benchmark prices)	Risk value
< 10%	3 (Moderate)
10–15%	4 (Moderately High)
15–20%	5 (High)
> 20%	6 (Very High)

Risk-Band for SIFs ★★★★★

Why	SIFs riskier than all MF schemes → separate meter to avoid confusion
Levels	5: level 1 (lowest) → level 5 (highest)
Assigned	At NFO; change → notice-cum-addendum + email/SMS
Evaluation	Monthly; disclosed on website within 10 days of month close
Annual	Risk level as on 31 March + number of changes in the year, on AMC & AMFI websites

Portfolio disclosure ★★★★★

Item	Rule
All schemes	As on last day of month / half-year, with ISIN, on AMC + AMFI websites within **10 days**; downloadable spreadsheet
Debt schemes	**Fortnightly**, within **5 days** of every fortnight; plus yield of each instrument
Email	Monthly & half-yearly portfolio statements emailed within 10 days to registered emails
SIFs	Last day of every **alternate month**: May, Jul, Sep, Nov, Jan, Mar; within 10 days; all securities with ISIN
SIF scenario analysis	In Investment Strategy Information Document (**ISID**) — expected loss from market moves

Financial results & annual reports ★★★★☆

Item	Rule
Half-yearly results	Unaudited, on website within **1 month** of half-year end (31 Mar, 30 Sep); SEBI format; email/SMS to unitholders on availability
Annual report	Full or abridged on AMC + AMFI websites; emailed within **4 months** of FY end (31 Mar); physical copy on request if no email; ad in all-India editions of ≥ 2 dailies (one English, one Hindi); functional link to full scheme-wise annual reports
Change in control of AMC	Draft letter/email + draft newspaper ad filed with SEBI while seeking approval

Fund factsheet (non-mandatory) ★★★★☆

Status	Not a regulatory requirement; voluntary monthly market practice; part of AMFI best practices; SEBI disclosure rules apply (marketing document)
Contents	Inception date, AUM, NAV, benchmark, style box, performance vs benchmark, SIP returns, sector & security allocation (some only top 10), derivatives exposure
Equity measures	PE, standard deviation, beta, Sharpe ratio, **information ratio** (mandatory — SEBI circular 17 Jan 2025)
Debt measures	Rating profile, average maturity, duration
Investment details	Min amount, plans/options, loads, expenses, systematic facilities

> **TRAP** MFs may use derivatives for hedging but cannot take open (naked) derivative positions — as stated in the factsheet section.

06 Fund Distribution & Channel Management Practices

≈4 marks · 19 subtopics · chapter priority ★★★☆☆

Role of the MF distributor vs fund manager ★★★★☆

```
Investor → Mutual fund distributor → Fund manager → Portfolio   (Chart 6.1)
```

Professional	Analyses
Fund manager	Market factors: company, industry, economy, liquidity, mood → portfolio per scheme objective
MF distributor	Investor's needs, limitations, resources, goals → asset allocation → suitable schemes

Kinds of distributors ★★★★☆

Type	Points
Individual	Long history (LIC, UTI agents; small savings); single-handed; = 'individual' channel
Non-individual	Partnerships, regional/national distributors, NBFCs, banks, stock brokers; distribution cos & banks = 'institutional'; models: own branches & employees (fixed cost, salary) or sub-agents (variable cost, commission)
Banks	MNC banks entered first, then private & PSU, now some co-operative banks; segments: retail, wealth/priority/private banking; PSU banks for non-urban reach
Sub-agents (agency channel)	Not employees; extended distribution arm of NBFCs/brokers
E-commerce / online	No physical office; EOP guidelines apply

Channels: individual MFDs, bank branches, national distributors, post offices, AMCs direct

Modes of distribution ★★★☆☆

Physical (paper forms → Official Points of Acceptance, OPOAs) · Digital · Hybrid · Online channel partners: own websites, beyond geography

Stock exchange platforms ★★★★★

Point	Detail
Platforms	NSE **NMF II** (two versions: trading members, distributors) · BSE **StAR MF**
Who can use	Trading members registered with AMFI as MFD and signed with the MF; clearing members; **DPs — redemptions only**; AMFI-registered distributors via limited-purpose membership
Eligible schemes	Notified by MFs; these open-ended schemes are **not listed**
Settlement	Exchange = order routing only; does **not** replace RTA; **no settlement guarantee** — AMC responsible; exchange grievance mechanism available
Direct access	SEBI circular 26 Feb 2020: investors can directly use exchange infrastructure to buy/redeem from AMCs
Facilities	Units in demat; pay-in/pay-out via exchange & clearing corporation; SIP, STP, SWP, switch possible
Listing	Close-ended & interval schemes must be listed; ETFs traded on exchange

MF Utilities (MFU) & MFCentral ★★★★★

MFU	Detail
What	Transaction aggregation platform: investors, RTAs, distributors, banks, AMCs
CAN	Common Account Number — consolidates holdings; KYC compliance needed (MFU facilitates KYC); multiple CANs possible by holding pattern/mode
Common Transaction Form	One form, many schemes across MFs; submit via MFD or neutral Points of Service (PoS); single payment for multiple subscriptions
Override	Bank & nomination details given to MFU override folio details
Online access	Transact, view holdings, lodge & track complaints

MFCentral	Detail
What	Unified single-window portal by **CAMS and KFintech** under SEBI mandate (RTA interoperable platform, operational since 31 Dec 2021)
Functions	Consolidated portfolio view (demat + non-demat) · service requests (bank mandate, email/mobile, KYC, nominee) · transactions (lump sum/SIP, redemption, switch) · folio management (consolidation, minor→major, FATCA/CRS) · CAS, unclaimed dividends · family portfolio tracking

Apps, AMC platforms, new-age platforms ★★☆☆☆

Distributor apps (smartphones, feature phones, tablets) · AMC web/mobile apps, SMS, WhatsApp = dealing directly with MF · New-age platforms (Groww, Kuvera, Paytm Money, Coin): low cost, direct plans, little paperwork

Prerequisites to become a distributor ★★★★★

Step	Rule
Who	AMC may appoint individual, bank, NBFC, distribution company
Certification	NISM Series V-A mandatory for all in sale/distribution; SIF needs additional eligibility
Exemption (grandfathered)	Age ≥ 50 or ≥ 10 yrs experience in MF distribution as of **31 May 2010** → pass exam OR CPE classroom credits
KYD (Know Your Distributor)	Self-attested PAN + address proof at **CAMS-PoS** with originals; biometric: **right-hand index finger** (new & renewal; authorised persons for non-individuals); acknowledgement sent to all empanelled AMCs. New cadre: no KYD/biometrics, only ID & address proof
ARN	Allotted by AMFI after certification + KYD; minimum age **18**
Empanelment	With any number of AMCs, or work as agent of an empanelled distributor — compulsory to earn commission
Employees of institutions	Clear NISM V-A + get **EUIN**; quote EUIN on forms; no separate ARN (operate under institution's ARN)
Online only	From **1 Oct 2024**, ARN & EUIN applications only online on AMFI website

AMC empanelment form & declaration ★★★☆☆

Personal info (name, age, trade name, contact, ARN, PAN, tax category)

Key sales/ops people

Business details (branches, employees, area, experience, investors, agents, AMCs, AUM)

Bank details (direct credit) | Info preferences | Nominee

- Info correct; keep transactions confidential; follow instructions/codes
- No ads/publicity except AMC-provided or pre-approved
- Risk factors with performance info
- Give documents AMC asks for
- All selling staff have EUIN
- No rebating, gifts or pass-back
- AMC may terminate empanelment any time

No standard format — each AMC has its own

Some AMCs empanel only high-business distributors; commission may be linked to volume

Distributor revenue: trail commission only ★★★★★

- Upfront commission and transaction charges **withdrawn** by SEBI
- Commission on Regular Plan investments, paid by AMCs per their commission structure (typically monthly)
- SEBI MF Master Circular **20 March 2026**: full trail model; no upfront, no upfronting of trail, directly or indirectly, cash or kind, sponsorships or other route
- Trail paid only in monetary terms; no other incentives
- Reg 66(6): AMC pays distribution commission as SEBI specifies (to curb mis-selling)
- Exception: **upfronting of trail allowed only for SIP inflows**

Trail commission calculation ★★★★★

```
Trail for the day = AUM × trail rate p.a. ÷ 365
Ex: ₹10 lakh at NAV 10 → 1,00,000 units; 1% p.a.
Day 1: 10,00,000 × 1% ÷ 365 = ₹27.40 ; August 2019 total
= ₹853.13
Bought 1,000 units at ₹10; NAV rises to ₹15 → trail on
₹15,000, not ₹10,000
```

% of net assets; daily balances; paid monthly/quarterly

Rises with market value

Paid as long as money stays invested (unlike insurance)

No commission on distributor's own investments (self-business)

Must disclose commissions from competing schemes considered

Change of distributor → no trail to either for 12 months

Additional commission: B-30 & women investors ★★★★★

Eligible inflows	(a) New individual investors (new PAN) from **B-30 cities** (industry level) (b) New **women** individual investors (new PAN) from Top-30 and B-30 cities
Lump sum	1% of first application, max **₹2,000**, if investor stays ≥ 1 year
SIP	1% of total investment in first year, max **₹2,000**
Source	2 bps of daily net assets set aside annually for investor education, awareness & financial inclusion; claw-back provisions
Over and above	Existing trail commission
No dual incentive	Woman from B-30 paid once; Top-30 woman only if B-30 commission not claimed
Reference	Reg 52(4A); SEBI circular 27 Nov 2025

Transaction charges — withdrawn ★★★★☆

- Earlier: AMCs could pay transaction charges on subscriptions > ₹10,000
- SEBI circular **Aug 2025**: framework withdrawn → no investor-borne transaction charges; no upfront commission by investors
- Distributors paid directly by AMCs (agents of AMCs); nothing deducted from investor contributions

GST on distributor commission ★★★☆☆

From	July 2017
Threshold	Annual turnover > **₹20 lakh**
Registered MFD	Raises invoice, pays GST
Reverse charge	Sec 9(4) CGST Act & Sec 5(4) IGST Act — registered recipient pays tax on procurement from unregistered supplier

Commission disclosure by AMCs ★★★★☆

Disclose total commission & expenses for distributors meeting ANY (non-institutional incl. HNI)	
Multiple points of presence	> 20 locations
AUM raised across industry	> ₹100 crore
Commission across industry	> ₹1 crore p.a.
Commission from single MF/AMC	> ₹50 lakh

Data to AMFI; AMFI publishes consolidated data

Yearly: distributor-wise gross inflows (flag associate/group of sponsor), net inflows, average AUM, AUM/gross inflow ratio

Portfolio turnover > 2× industry average → additional due diligence

Due diligence of distributors by AMCs ★★★★☆

Applies to	Same 4 criteria (any one)
Reference	SEBI circular CIR/IMD/DF/13/2011, 22 Aug 2011
Fit & proper factors	(a) Business model, experience, proficiency (b) Record of levies, fines, penalties, suits, compensation + corrective actions (c) Associates & subsidiaries (d) Controls delinking customer risk evaluation and scheme appropriateness from sales/RM

Distributor vs investment adviser; advisory vs execution only ★★★★★

Investment adviser	Paid to give investment advice, or holds out as adviser; **excludes** AMFI-registered MFD giving advice incidental to primary activity
Rules	Distributor cannot call itself investment adviser; adviser can't earn both advisory fee and commission
Advisory	Subject to **appropriateness**: sell only products best suited within defined upper ceiling of risk appetite — no exception
Execution only	If inappropriate: written communication of unsuitability acknowledged by investor + confirmation of 'execution only' before execution; investor pays nothing extra
No third category	Only advisory or execution only
Group/associate products	Disclose conflict of interest

Distributor compliance reviews: product review criteria & periodicity; customer risk appetite factors & review

Other due diligence: internal audit of transactions/exceptions; recruitment, training, certification; onboarding, servicing, grievances; audits; investor sample surveys

Distributors can't project as investment advisers or financial planners

Mis-selling (Box 6.8.3) ★★★★★

Exemption	SEBI Investment Adviser Regulations 2013 exempt MFDs giving incidental advice
Suitability	Required in both advisory and execution-only
Regulation	SEBI (Prohibition of Fraudulent and Unfair Trade Practices) 2003 — includes MF mis-selling

- False or misleading statement
- Concealing/omitting material facts
- Concealing risk factors
- Not taking reasonable care for suitability

Nomination for distributors & commission to nominee ★★★★☆

- AMFI advised AMCs to offer nomination to MFDs at empanelment
- Commission paid to nominee, or legal heirs if no nominee (heirship documents)
- Paid until investor changes the deceased's ARN code / AUM becomes nil; no new business or new/changed systematic transactions under deceased's ARN
- Nominee: death certificate only, no succession certificate; need not be an ARN holder
- Only for assets procured while ARN valid; ARN valid & not suspended on date of death; later expiry ignored
- Nominee receives as **trustee** for legal heirs
- Assets transferred to nominee's account only on investor's request and if nominee holds valid ARN
- Nominee/heir informs AMFI with Gazette-officer-attested death certificate; AMFI informs AMCs (not for overseas distributors)

AUM transfer to nominee	Deceased's ARN valid/not suspended at death; nominee has valid ARN, KYD compliant, annual self-certification done; inform all clients — investors get **15 days** to object to AMC

Change of distributor ★★★★★

- Investor may change distributor without reason, by written request, without NOC
- No commission to old or new distributor; AMCs may pay new distributor after **12-month** cooling-off
- Exception: change due to distributor's **voluntary cessation** → new distributor gets trail per SEBI/AMFI rules

Distributor-initiated change allowed only for	
i	Change in name/legal status (individual → partnership → LLP; private ↔ public ltd)
ii	Merger/acquisition/consolidation/transfer within same group (non-individuals)
iii	AUM consolidation within family/close relatives (individuals)
iv	Transfer of business by individual distributors

Entire AUM must transfer, to a valid KYD-compliant ARN holder

Old ARN surrendered; no further business

07 NAV, Total Expense Ratio & Pricing of Units

≈5 marks · 15 subtopics · chapter priority ★★★★★

Why fair valuation ★★★☆☆

Ownership changes as investors enter/exit → each must get a fair price

Valuation per **Seventh Schedule** of SEBI MF Regulations 2026

Aims: fair treatment of existing, entering and exiting investors; address mispricing; reduce short-term debt schemes' vulnerability to redemption pressure

10 principles of fair valuation ★★★★★

#	Principle
1	Fair valuation = realisable value; good faith, true and fair, via policies & procedures
2	AMC Board-approved policies name methodology per security type; invest in a **new type** of security only after its valuation method is approved by AMC Board
3	Consistent valuation; process for exceptional events when quotes are unreliable
4	Periodic review; update Trustee & AMC Boards; reviewed by **independent auditor at least once a FY**
5	Policies address conflict of interest
6	Disclose valuation policy in **SAI**, AMC/MF website and as SEBI specifies
7	AMC responsible for true & fair valuation and correct NAV; may deviate from policy to reach fair value — with reporting to both Boards and disclosure to investors
8	Policies to detect and prevent incorrect valuation
9	Document rationale (incl. inter-scheme transfers) for audit trail
10	Debt & money market: consider trades of same/similar security on all public platforms

TRAP If Fair Valuation Principles conflict with SEBI Valuation Guidelines, the PRINCIPLES prevail.

Valuation per SEBI & AMFI guidelines for traded, non-traded and thinly traded securities, gold & silver; adjust for illiquidity

Net assets — worked example ★★★★★

Liabilities side	₹ cr
Unit capital (20 cr units × ₹10)	200
Profits: 8 income − 4 paid − 1 payable	3
Capital appreciation (10% of 140)	14
Unitholders' funds (net assets)	**217**
Expenses payable	1
Scheme liabilities	218

Assets side	₹ cr
Investments at market (140 + 10%)	154
Bank & money market (60 + 8 − 4)	64
Scheme assets	218

```
NAV = Net assets ÷ units = 217 ÷ 20 = ₹10.85
Alt: (Total assets - liabilities other than to
unitholders) ÷ units = (218 - 1) ÷ 20 = ₹10.85
```

Net assets = original money + booked profits + appreciation

Rise with market price even if unsold

Accrual principle: expenses and income of the period counted whether paid/received or not

Current liabilities: payables for securities bought; borrowings ≤ 6 months for liquidity; intra-day borrowing returned same day

NAV drivers & profitability metric ★★★★★

```
Profit = Interest + Dividend + Realised gains + Valuation
gains - Realised losses - Valuation losses - Scheme
expenses
Higher income/gains → higher NAV · Lower expenses →
higher NAV
```

NAV worked examples (workbook) ★★★★★

```
Ex 1: (150 stocks + 67 bonds + 2.36 MM + 1.09 dividend
accrued + 2.68 interest accrued - 0.36 fees payable) ÷
1.90 cr units
     = 222.77 ÷ 1.90 = ₹117.25
Ex 2: (230 stocks + 5 MM + 2.39 dividend accrued + 2.34
receivable on sale - 7.5 payable on purchase - 0.41 fees)
÷ 2.65
     = 231.82 ÷ 2.65 = ₹87.48
NAV = (Investments + income accrued + current assets -
current liabilities - accrued expenses) ÷ units
```

Mark to market — why ★★★★★

Meaning	Value each security at current market value
Example	Shares bought at ₹15 now worth ₹50 — valuing at ₹15 meaningless

Case (NAV ₹15, 10,000 units, net assets ₹1,50,000)	Result
Issue 100 units at face value ₹10	Net assets 1,51,000 ÷ 10,100 = NAV **₹14.95** (falls for all)
Redeem 100 units at ₹10	1,49,000 ÷ 9,900 = NAV **₹15.05** (rises for rest)

MTM → fair buy/sell prices; helps judge fund manager performance

Expenses: rules ★★★★☆

- All expenses clearly identified and paid from the scheme, within base expense limits + brokerage + transaction cost + statutory levies
- Excess over base limits borne by AMC, trustees or sponsors
- Investment & advisory fees charged by AMC — disclosed in SID
- AMC pays distribution commission as SEBI specifies

Recurring expense heads (a)–(o) ★★★★☆

(a) Marketing & selling incl. distribution commission · (b) Brokerage · (c) Registrar services · (d) Trustee fees · (e) Audit fees · (f) Custodian fees · (g) Investor communication · (h) Fund transfer between locations · (i) Account statements, IDCW/redemption cheques & warrants · (j) Insurance premium · (k) Winding-up costs · (l) Statutory advertisements · (m) Storage & handling of commodities / gold / silver · (n) Listing fees · (o) Others approved by SEBI

TER limits (2026) ★★★★★

Equity-oriented (for TER)	Scheme (other than index fund/ETF) investing ≥ 65% of net assets in equity per SID
TER covers	Investment & advisory fee; excludes issue/redemption expenses

Open-ended: daily net assets	Equity-oriented	Other than equity
First ₹500 cr	**2.10%**	**1.85%**
Next ₹250 cr	1.90%	1.65%
Next ₹1,250 cr	1.60%	1.40%
Next ₹3,000 cr	1.50%	1.25%
Next ₹5,000 cr	1.40%	1.15%
Next ₹40,000 cr	−0.05% per ₹5,000 cr (or part)	−0.05% per ₹5,000 cr
Balance	0.95%	0.70%

Scheme type	TER cap
Index fund / ETF (incl. advisory fee)	**0.90%**
FoF investing in liquid, index, ETF	0.90% (incl. weighted avg TER of underlying)
FoF ≥ 65% in equity-oriented schemes	2.10%
Other FoF	1.85%
Close-ended / interval — equity-oriented	1.00%
Close-ended / interval — others	0.80%

FoF's own TER over underlying ≤ 2 × weighted average TER of underlying (within ceilings)

Charges over and above base TER ★★★★★

Item	Limit
Brokerage & transaction cost — cash market	Up to **0.06%** of trade value
— derivatives	Up to **0.02%** of trade value
Total charged to investors	Base expense + brokerage + transaction cost + statutory levies
Nothing else	Except base expense ratio, brokerage, transaction cost, statutory levy and exit load
Additional commission	For MFDs onboarding new investors (B-30, women) as SEBI specifies

Daily scheme-wise TER on AMC + AMFI websites under 'Total Expense Ratio of Mutual Fund Schemes'

Perpetual (AT-1 / Tier 2) bonds ★★★★☆

Bank-issued perpetual bonds that absorb losses before equity / convert on trigger

SEBI exposure limits apply

AT-1 bonds valued on **Yield to Call** basis (SEBI Aug 2024)

IDCW & distributable reserves ★★★★★

- Accrued income/expense = only a matter of time; valuation gains may never be realised
- IDCW only from **distributable reserves**: all accrual profits count; **valuation gains ignored**; **valuation losses adjusted**; part of new-unit sale price attributable to valuation gains excluded
- Trustees decide quantum and record date; up to monthly frequency may be delegated to AMC officials, ratified at next Board meeting
- Record date = cut-off for eligibility; NAV adjusted at end of record date
- AMC issues public notice within **1 day** of trustees' decision; record date = **5 calendar days** from notice; notice states NAV will fall by payout + statutory levy
- Delay → interest from record date
- No notice needed for liquid/debt schemes with daily-to-monthly IDCW (disclosed in SID); listed schemes follow listing agreement
- No assurance of rate, quantum or regularity
- Realised gains in sale price → **equalisation reserve**, usable for IDCW; disclose how much is NAV appreciation vs capital distribution

Entry & exit load ★★★★★

```
Old entry load: NAV 11 + 1% → sale price ₹11.11 (banned
now → sale price = NAV, subject to stamp duty)
Exit load: NAV 11 - 1% → repurchase price ₹10.89
₹25,000 at NAV 43.21 → 578.570 units
```

- Graded load (e.g. 3% year 1, 2% year 2) encourages longer holding
- Earlier: different loads by investor class; loads funded AMC selling expenses
- Now: no distinction by subscription amount; new/higher load only for prospective investments; parity at portfolio level
- **No exit load on bonus units and IDCW-reinvestment units**
- Exit load credited back to the scheme immediately

Accounting & reporting requirements ★★★★☆

- Scheme accounts distinct from AMC; different auditors
- Norms on when interest, dividend, bonus, rights are booked
- NAV to **4 decimals**: index funds, liquid funds, other debt funds
- NAV to **at least 2 decimals**: equity and balanced funds
- Units may be held in fractions; exchanges may restrict to whole units

Segregated portfolio: NAV, TER, pricing ★★★★☆

- No investment & advisory fee on segregated portfolio; other TER pro-rata only upon recovery
- NAV of segregated portfolio declared **daily**
- Disclosed in scheme documents, monthly & half-yearly portfolios, annual report

Credit event basis: SEBI circular 28 Dec 2018

Unrated debt of issuer with no rated debt: allowed (7 Nov 2019) only on actual default of interest or principal

Detail in Ch 10

08 Taxation

≈3 marks · 14 subtopics · chapter priority ★★★★☆

Two levels: fund vs investor ★★★★★

Principle	Not what you earn but what you keep after tax; MF is pass-through → tax at fund level and investor level
Fund's income	Interest, dividend, capital gains
Fund taxation	**Exempt** — MFs are trusts for unitholders: **Section 10(23D)** of Income Tax Act
Investor	Growth option → only capital gains on sale; IDCW option → IDCW + capital gains

What decides the tax rate ★★★★★

Factor	Point
Type of income	Capital gains taxed differently from IDCW
Holding period	Short-term vs long-term
Type of scheme	Equity-oriented: **> 65%** of AUM in equity shares listed on recognised Indian exchanges; all others non-equity. FoF investing in equity MFs = **non-equity**
Type of investor	Resident, NRI, non-individual; joint holding → income of **first holder**
Date of purchase	Budget 2023 changes from 1 Apr 2023; earlier investments keep old structure
Date of redemption	Budget July 2024: new rates for redemptions on/after **23 July 2024**; before → old rates

Capital gains: holding periods ★★★★★

Gain = sale price − purchase price (loss if lower)

Only two holding periods: 12 months and 24 months

12 months: all listed securities + units of equity-oriented funds

24 months: all other assets

Grandfathering (Budget 2018) ★★★★★

Why	Equity LTCG was exempt till Budget 2018; taxing past gains = retrospective tax
Rule	Gains up to **31 Jan 2018** not taxed; cost = higher of purchase price or NAV on 31 Jan 2018

```
Invested ₹1,00,000 in 2017; value on 31 Jan 2018 =
₹2,00,000
Scenario 1: sold for ₹1,80,000 → taxable LTCG = 0
Scenario 2: sold for ₹2,20,000 → taxable LTCG = ₹20,000
```

₹1.25 lakh LTCG exemption ★★★★★

```
Equity shares + equity-oriented MFs: first ₹1.25 lakh of
LTCG per year exempt
LTCG ₹90,000 → all exempt ; LTCG ₹1,35,000 → ₹10,000
taxable
```

Tax table for unitholders ★★★★★

Investment	STCG	LTCG	Long-term after
Equity-oriented funds (incl. hybrids > 65% equity)	20%	12.5% (₹1.25 lakh exempt per FY)	12 months
Debt-oriented funds (incl. hybrids > 65% debt)	Marginal slab rate (MSR)	MSR	—
Hybrid 35%–65% domestic equity	MSR	12.5%	24 months
Other MFs (FoFs, international funds/FoFs)	MSR	12.5%	24 months
Equity ETFs	MSR	12.5%	12 months
Gold ETFs	MSR	12.5%	12 months
Debt ETFs	MSR	MSR	—

IDCW taxation ★★★★★

Till 2019-20	IDCW tax-free for investor; Dividend Distribution Tax (DDT) deducted by scheme
Budget Feb 2020	DDT abolished; from **April 2020** IDCW added to income, taxed at slab
Effect	High-income investors pay more than before; tax-exempt investors (charitable trusts, exempt-slab individuals) now pay nothing
DDT vs new	DDT wasn't investor's tax, no set-off; new tax can use exemptions/adjustments
Common	NAV falls more than net-of-tax amount investor receives

```
Old: Drop in NAV = Dividend paid + DDT
New: Post-tax IDCW = IDCW paid - tax at investor's slab (except tax-exempt)
```

Growth option more tax-efficient — tax deferral, compounding pre-tax

From 1 Apr 2021: MF must show how much IDCW is income distribution (appreciation) vs capital distribution; tax on segregated basis

Stamp duty ★★★★★

Transaction	Stamp duty
Issue of units (purchase, additional purchase, IDCW reinvestment, SIP, STP, IDCW Transfer Plan incl. triggers from past registrations, switch-ins)	**0.005%** of amount (SoA & demat)
Transfer of units (e.g. between demat accounts)	**0.015%**

From 1 July 2020 (Indian Stamp Act amended Feb 2019)

Units allotted on amount after deducting stamp duty

Set-off & carry forward ★★★★★

- Capital loss can't be set off against any other head (e.g. salary)
- Short-term capital loss → against STCG or LTCG
- Long-term capital loss → only LTCG
- Unabsorbed capital loss carried forward up to **8 years**

Heads of income: salaries, house property, business/profession, capital gains, other sources

Bonus stripping ★★★★☆

```
Buy at ₹30 → 1:1 bonus → NAV ₹15 → sell original at ₹15 = loss ₹15
Loss NOT allowed if bought within 3 months BEFORE record date and sold within 9 months AFTER
→ loss becomes cost of acquisition of the bonus units
```

Even if allowed, practice is discouraged — invest for fundamentals

Securities Transaction Tax (STT) ★★★★★

Transaction	Rate	Payable by
Purchase of equity-oriented fund units	Nil	Purchaser
Sale of equity-oriented fund units (delivery, on exchange)	0.001%	Seller
Sale of equity shares / business trust units / equity fund units (non-delivery)	0.025%	Seller
Sale of equity-oriented fund units to the mutual fund	0.001%	Seller

Applies on redemption/switch/sale of equity-oriented funds only

Not on purchase; not on debt securities or debt funds

Section 80C — ELSS Tax Saver ★★★★★

- Deduction up to **₹1.50 lakh** per year for individuals & HUFs, **Old Tax Regime** only
- Lock-in **3 years** from date of investment
- Limit shared with other 80C avenues — if exhausted, ELSS still locked in without extra benefit
- SIP: each instalment locked in for 3 years from its own date
- IDCW reinvestment: each reinvestment gets fresh 3-yr lock-in (most AMCs offer only growth/IDCW payout)
- Joint holding: benefit to first holder
- New tax regime (Budget 2020): more slabs, lower rates, no 80C → ELSS = equity fund with lock-in
- Some older retirement funds also have 80C benefit, **5-yr lock-in** — check scheme

MFs = tax-sheltered vehicles → tax deferral → pre-tax compounding in growth option

Tax Deducted at Source (TDS) ★★★★★

Resident — redemption	No TDS
Resident — IDCW	10% if IDCW > **₹10,000**
Non-resident	TDS at rates by investor type, investment type and income type
DTAA	Lower of Income Tax rate or DTAA rate; investor must provide documents

GST in MF schemes ★★★★☆

GST on	Treatment
Investment management & advisory fees	Charged to scheme **over and above** TER limits
All other fees	Within TER limit
Exit load	Deducted from exit load; net credited to scheme
Brokerage & transaction cost	Within TER limit
Distributor commission	May attract GST for ARN holder; cannot be charged to scheme

From 1 Apr 2026	Cost = base expense ratio + execution brokerage (within limits) + statutory levies (e.g. GST)
Base expense ratio	Investment & advisory fees + out-of-pocket expenses + brokerage commission, excluding statutory levies and transaction cost

09 Investor Services

≈10 marks · 53 subtopics · chapter priority ★★★★★

NFO process ★★★★☆

- AMC picks scheme using CIO (investment objective) + CMO (market interest) inputs
- One scheme per category — NFO only in a category where AMC has none, or multi-scheme categories (sector/theme, index, FoF)
- SID prepared → approved by **Trustees and AMC Board** → filed with SEBI → SEBI observations incorporated → issued
- Timetable per market; ads & PR per SEBI code; events for intermediaries and press; documents & forms circulated

Date	Meaning
NFO open date	Investors can start investing
NFO close date	Last day to invest
Scheme re-opening date	Open-ended: ongoing sale & repurchase at NAV-based prices (close-ended has none — trade on exchange)

NFO period	Minimum **3 working days**; maximum **15 days** (except ELSS)
Allotment / refund	Within **5 business days** of NFO close
Re-open (open-ended)	Within **5 business days** of allotment
Deployment	Reg 35(5): realistic deployment timelines in SID

NFO & ongoing prices ★★★★☆

NFO price = price per unit during NFO (usually face value ₹10)

Ongoing sale price = NAV (no entry load)

Redemption/switch-out price = NAV − exit load (if any)

Direct and regular plans ★★★★★

- Every MF must offer both; different TERs → separate NAVs
- Direct: lower expense (no distribution expense/commission); only for investors buying directly from the fund
- Routed through distributor → regular plan
- No other plans based on expense ratio allowed

Growth, IDCW payout, IDCW reinvestment ★★★★★

- Same portfolio → same portfolio return; differ in cash flows, taxability, units, value → different post-tax returns
- Dividend option renamed IDCW from **1 April 2021**
- Some liquid/very short debt funds declare IDCW daily (if profits)
- **Cum-dividend NAV**: after announcement, before payout · **Ex-dividend NAV**: reduced NAV after payout
- Payout: cash to bank, units unchanged, taxed in investor's hands
- Reinvestment: units added at **ex-dividend NAV** (₹2 × 100 units = ₹200 ÷ ₹20 = 10 units); taxed in year of receipt
- Growth: no IDCW; NAV captures full gain; units unchanged

Parameter	IDCW payout	IDCW reinvestment	Growth
IDCW in bank	Yes	No	No
Tax on IDCW	Yes	Yes	N.A.
More units	No	Yes	No
NAV	Falls by IDCW	Falls by IDCW	Captures all valuation changes

Illustration 9.1 — three investors ★★★★★

```
A (growth), B (payout), C (reinvest): 100 units at NAV ₹10; NAV → ₹12; IDCW 10% (of face value ₹10 = ₹1/unit)
B gets ₹100 cash; C reinvests ₹100 at ex-NAV ₹11 → 9.09 units
A: 100 × 12 = ₹1,200 (₹200 capital gain)
B: 100 × 11 = ₹1,100 + ₹100 IDCW = ₹200 total return
C: 109.09 × 11 = ₹1,200 (₹100 gain + ₹100 reinvested IDCW)
Post-tax returns differ (IDCW vs gains taxed differently)
```

TRAP IDCW % is declared on FACE VALUE, not NAV.

Allotment of units ★★★★★

Case	Rule
NFO	Units at face value ₹10; full allotment to valid applications if minimum subscription met; trustee may reject any application without reason; allotment within 5 business days; demat credit within **2 working days**; SOA (non-demat) within 5 business days of NFO close; FPIs subject to RBI approval; ASBA: account debited on allotment intimation by CAMS
Ongoing	Amount ÷ sale price (NAV): ₹12,000 ÷ ₹12 = 1,000 units
Rights issue	Amount ÷ rights price (less meaningful for MFs)
Bonus 1:3	1 free unit per 3 held; net assets same → NAV falls proportionately; value unchanged
Refund delay	Beyond 5 business days → **15% p.a.** interest, charged to AMC

Demat units fully and freely transferable (subject to lock-in and lien)

Transfer on operation of law/pledge enforcement with satisfactory evidence

Account statements ★★★★★

Statement	Rule
Monthly SOA	Every month if any transaction: value, NAV, units, closing balance & value
Annual SOA	For unitholders with no transaction in last 6 months; may be soft copy by email
Consolidated Account Statement (CAS)	Each calendar month (post/email) for investors with a transaction: all transactions & holdings across all schemes of all MFs
Half-yearly CAS	No transaction in 6 months ending September or March
Common investor	Identified by **PAN**

'Transaction' = purchase, redemption, switch, IDCW payout, IDCW reinvestment, SIP, SWP, STP, bonus

Eligible individual investors ★★★★★

Investor	Rule
Resident adults (18+)	Singly or jointly — max 3 names
Minors (<18)	Through guardian; payment from bank account of minor, parent or legal guardian, or joint account of minor with parent/guardian; existing folios: change of payout bank mandate before redemption
HUF	Karta invests; adds 'HUF' after name
NRIs / PIOs	Repatriable or non-repatriable; PIO/OCI card
Foreign investors (QFIs)	Sub-category of FPI; from FATF member (or member of group in FATF) + IOSCO MMOU signatory country; via demat (direct) or Unit Confirmation Receipt (UCR)

Eligible non-individual investors ★★★☆☆

Indian companies/corporate bodies

Registered & co-operative societies

Trustees of religious/charitable and private trusts

Partners of firms

AOP/BOI

Banks (incl. co-op, RRBs), FIs, investment institutions

Other SEBI-registered MFs

SEBI-registered FPIs (direct: demat via DP; indirect: UCR)

Govt-approved international multilateral agencies

Army/Navy/Air Force, para-military

Scientific & industrial research orgs

Universities & educational institutions

Some gilt plans only for PFs, superannuation, gratuity, pension funds, religious/charitable and private trusts

Check 'Who can invest?' in SID

Application form — key fields ★★★★★

Field	Rule
Direct/Regular	Write 'Direct' in ARN box; regular → ARN/RIA number
Holders	Up to 3; folio in name of first holder; all benefits (IDCW, redemption, tax) to **first holder**; overseas address for FPI/NRI/PIO
Minor	Guardian (natural parent or court-appointed) does KYC/PAN; proof of relationship/guardianship; minor's DOB + age proof; no joint holders; at majority folio **frozen** until status changed (KYC, PAN, bank-attested signature, bank details)
PoA	PoA holder and issuer KYC/PAN; certified PoA copy; grantor can still operate; PoA holder can do everything **except make/change nominations**
Mode of holding	Single / joint / either or survivor; default if not chosen; joint holders can't be added or removed except deletion on death
Address	After first holder's PAN validated for KYC, KYC address overrides form
KYC	KYC acknowledgement + occupation, gross annual income/net worth, PEP status
FATCA/CRS	If birth/citizenship/tax residency outside India: city & country of birth, citizenship, tax countries & Tax IDs
Bank details	Mandatory for sole/first holder: bank, branch, city, account no., type (savings, current, NRO, NRE, FCNR), MICR, IFSC; demat holders → bank linked to demat
Payout account different from paying account	Proof: cancelled cheque with name, passbook/statement ≤ 3 months old, or bank letter
Investment details	Scheme mandatory; plan/option defaults apply if blank
Payment	Payable to MF's specified account; application/folio no. on reverse; paying bank need not equal payout bank
Demat option	DP name, ID, beneficiary account; DP KYC accepted; client master/DP statement copy
Nomination	Max **3** nominees; at application or later; change any number of times; single holder should nominate or declare opt-out (SEBI circular effective 1 Mar 2025)
Minimum investment	Higher for initial, lower for additional; signed by all holders whatever the mode

Financial transactions: purchase, additional purchase, repurchase ★★★★★

- Initial purchase: full application form + KIM (common application form) → folio created
- Existing investor: may use same form with folio no.; personal info not repeated
- Additional purchase: transaction slip (often pre-printed with folio) + payment; folio data overrides conflicting form data; name mismatch → may be rejected
- Repurchase: transaction slip with folio, holders, scheme/plan/option; signed per mode of holding; specify amount or units
- Repurchase price = applicable NAV − exit load; units redeemed on **FIFO** basis
- If balance falls below minimum, all units may be redeemed and folio closed
- Demat units: redemption form to DP → AMC/RTA → paid to demat-linked bank
- Pool accounts of brokers/clearing members discontinued; units credited/debited directly (DIS to DP still allowed for demat redemption)
- Switch = redemption + purchase in one transaction
- Subscription and redemption frequency same for all MF schemes — except SIFs (e.g. daily subscription, weekly redemption)

Digital payment modes ★★★★☆

Mode	Key points
Online / internet banking	PIN-based access; NEFT (batched), IMPS (instant, mobile/net/ATM), RTGS (instantaneous), SWIFT (from abroad, slower); bank acknowledgement with first holder as account holder
2FA	SEBI mandates Two-Factor Authentication for subscription and redemption
NACH	By NPCI; replaced ECS; repetitive bulk payments; same-day presentation & settlement; NACH Credit & Debit; ECS mandates migrated automatically
UPI	VPA via mobile app (BHIM, bank apps); multiple accounts, one default; Aadhaar-based transfer version
ASBA	For NFOs: amount blocked, debited only on allotment; earns interest meanwhile; no refunds
AEPS	Aadhaar-linked; fingerprint as password
NUUP (*99#)	USSD mobile banking without smartphone/internet; 11 regional languages
Cards	Debit & prepaid; **credit cards NOT allowed** for MF purchases
E-wallets	Max **₹50,000** per investor per FY (umbrella limit with cash, per MF); only money loaded via cash/debit card/net banking; not credit card, cashback, promos; no third-party; RBI KYC compliant

One-Time Mandate (OTM) ★★★★☆

- Authorises bank to process debits raised by a specified MF via NACH — one-time registration
- Needs bank name, account no., type, IFSC, MICR + cancelled cheque; folio/application no., email, phone
- Frequency: monthly, quarterly, half-yearly or 'as and when presented'; fixed amount or daily maximum (e.g. ₹5,000 as and when presented — unlimited transactions within daily max)
- Validity period specified; signed by bank account holders; registration takes **3–4 weeks**
- Folio-level; MF specifies eligible schemes/transactions (fresh lump sum, additional, SIP); usable via physical, online, SMS; cancel via OTM form

Cheque / DD rules ★★★★☆

- Cheque/DD in favour of the scheme; investor name/PAN and scheme details on instrument; separate instrument per scheme/plan
- DD accepted mainly where no official collection centre; AMC bears outstation DD charges within limits
- NRI: non-repatriable → NRO/NRE/FCNR cheque; repatriable → NRE/FCNR; drafts bought abroad need debit certificate / FIRC
- Local or 'payable at par' cheques; signed by investor; drawn on account where first holder is an account holder
- No post-dated (except SIP), no stale (> 3 months)
- Payment only through modes with independent traceability of end investor (SEBI)

Third-party payments (Box 9.1) ★★★★★

Definition	Payment from a bank account other than the first holder's (joint account OK if first holder is a joint holder)
Exception 1	Parents/related persons for a minor (love & affection/gift) — max **₹50,000** per purchase or per SIP instalment; payer KYC + Third Party Declaration
Exception 2	Employer via payroll deduction; custodian on behalf of FPIs
Documents	Declaration identifying beneficiary; third party KYC + PAN regardless of amount; source proof: pre-printed cheque name / statement / bank letter / DD funding proof / NEFT acknowledgement
Not accepted	DDs issued against cash for ₹50,000 or more

Cash payments ★★★★★

- Up to **₹50,000 per investor, per MF, per FY** (PMLA & AML compliance)
- Only resident individuals, sole proprietorships, minors via guardians — KYC compliant (with or without PAN)
- Physical application; deposit slip at ISC; cash at designated bank branches; bank acknowledgement + form to ISC for time stamping
- Redemptions/IDCW only via banking channel

Not accepted: stock-invests, postal orders, money orders, cash (beyond exception), outstation cheques, post-dated cheques (except systematic)

AMCs may pay via RTGS, NEFT, IMPS, direct credit, cheque, DD, warrants

Repurchase payments & multiple bank accounts ★★★★★

- Cheque (slow: preparation, post, deposit, clearing) or electronic (direct credit, RTGS, NEFT, NACH)
- Paid to sole/first holder; cheques to unitholder's address; demat → DP-registered bank
- NRIs paid in rupees; conversion cost on investor; repatriable → NRE/FCNR; TDS applies
- Register up to **5** bank accounts (individual) or **10** (non-individual); first holder in each; one default; change default any time
- NRI: paid via NRO → register NRO; paid via NRE → NRO or NRE
- Credit to a non-registered account requested with redemption → paid to default account; change not processed

Instant Access Facility (IAF) ★★★★★

What	Same-day credit of redemption
Where	Only **liquid** schemes
Limit	Lower of **₹50,000** or **90%** of latest value — per day, per scheme, per investor

Cut-off timings & applicable NAV ★★★★★

Scheme	Transaction	Cut-off	Applicable NAV
Equity & debt (not liquid/overnight)	Purchase / switch-in (any amount)	3:00 pm	NAV of business day on which funds are available for utilisation before cut-off
Liquid & overnight	Purchase / switch-in	1:30 pm	Application + funds before cut-off → NAV of day **immediately preceding** day of receipt; after cut-off with funds same day → NAV of day preceding **next business day**; funds not available → NAV of day preceding day funds available
Equity & debt (not liquid)	Redemption / switch-out	3:00 pm	Same-day NAV before cut-off; next business day NAV after
Liquid	Redemption / switch-out	3:00 pm	Before cut-off → NAV of day preceding next business day; after → next business day NAV. IAF before cut-off: lower of previous calendar day NAV and NAV of day of receipt; IAF after: lower of day of receipt and next calendar day
Overnight	Redemption / switch-out	3:00 pm	Before → NAV of day preceding next business day; after → next business day NAV; **online cut-off 7:00 pm**

Liquid funds: NAV for every calendar day

Not applicable to NFOs and international schemes

Funds-realisation rule for all purchases since 1 Feb 2021 (SEBI circular 17 Sep 2020; earlier < ₹2 lakh got same-day NAV)

Cut-off worked examples ★★★★★

```
Week: Mon, Wed, Thu, Fri = business days; Sun, Tue, Sat = holidays
1. Equity purchase ₹3 lakh applied Mon, funds Wed by 2 pm → Wed NAV; funds Wed by 5 pm → Thu NAV
2. Liquid purchase Mon, funds by 1 pm Mon → Sun NAV; funds by 3 pm Mon → Tue NAV; funds 10 am Wed → Tue NAV
3. Equity redemption Mon before cut-off → Mon NAV; after → Wed NAV
4. Liquid redemption Mon before cut-off → Tue NAV; after → Wed NAV
```

Time stamping ★★★★☆

- Official Points of Acceptance (OPoAs) disclosed in SID & website; time stamping done at OPoA
- Distributor may collect but must forward quickly — **time at OPoA counts**, not time given to distributor
- Machines with tamper-proof seal; opened only by vendor/nominated persons, documented and reported to trustees
- Sequential numbering through machine's full range — daily stamping does NOT restart at 1
- Purchase application, reverse of payment instrument and acknowledgement: location code, machine ID, serial no., date and time
- Repurchase application + acknowledgement stamped similarly
- Non-financial (e.g. change of address): date stamp matters, not time
- Online: time of the web server receiving the instruction

KYC — who and when ★★★★★

- All investors: individuals, non-individuals, joint holders, NRIs, PoA holders & issuers, guardians of minors — **irrespective of amount**
- Applies to new/additional purchases, switches, new SIP/micro-SIP, STP, DTP registrations
- Purpose: identity and address under anti-money laundering laws; application carries KRA acknowledgement
- Documents given to distributor or other SEBI intermediary (broker, DP); updated in KRA's central system

KYC documents & PAN exemptions ★★★★★

PAN	Mandatory proof of identity (with photo) except exempt; intermediaries may verify PAN online on IT website if another ID proof given
PAN-exempt	Central/State govt transactions & court-appointed officials · **Sikkim** residents · tax-exempt UN entities/multilateral agencies · MF investments up to **₹50,000 per investor per year per MF** → give alternate photo ID
Address proof	Passport, voter ID, ration card, DL, bank statement, utility bill etc.; both addresses if different; spouse's address proof acceptable
Attestation	Self-attested copies + originals for verification; else attested by authorised persons
Formats	Prescribed by SEBI

PAN-exempt investments & PEKRN (Box 9.2) ★★★★★

- Micro-SIP: annual SIP (rolling 12 months or Apr–Mar) ≤ **₹50,000**
- Composite ₹50,000 limit for micro-SIP + lump sum per MF per FY
- For individuals, minors, sole-proprietary firms
- KYC still via KRA → quote **PAN Exempt KYC Reference Number (PEKRN)** + letter copy

Alt photo IDs: voter ID, DL, govt/defence ID, passport, photo ration card, photo **debit** card (not credit card), ROC-registered company employee ID, IDs by bank managers/gazetted officers/MLAs/MPs, bank employee IDs, senior citizen/freedom fighter ID, university / ICAI / ICMAI / ICSI cards, **PRAN** card (NPS, by CRA NSDL), other govt IDs (ESIC, EPFO)

Current & valid; self-attested or attested by ARN holder (with ARN)

KYC Registration Agencies & CKYC ★★★★★

Central KYC	One KYC with one intermediary → valid across capital market
IPV	In-Person Verification by intermediary + upload to KRA → KYC valid market-wide
Uniform form	SEBI circulars 22 Aug 2011, 5 Oct 2011, 26 Dec 2013 → standard Account Opening Form (AOF): Part I basic KYC, Part II exchange-specific
CERSAI	Notified **26 Nov 2015** as Central KYC Records Registry (CKYCR) under PML Rules 2005
CKYC upload	CERSAI template = Part I of AOF for individuals; upload for accounts opened on/after **1 Aug 2016**
CKYCR functions	Store/safeguard/retrieve KYC records online · share updates with reporting entities · services on prepaid fee · de-duplicate and issue unique **KYC Identifier**

With KYC Identifier, entity downloads records — re-submission only if information changed, current address to verify, or entity needs it for verification / enhanced due diligence / risk profile

e-KYC via UIDAI & KYC through intermediaries ★★★★☆

- UIDAI e-KYC accepted: details + photo = proof of identity and address; client authorises access
- Notified entities may do Aadhaar authentication under **Sec 11A of PMLA** as KYC User Agencies (KUAs); intermediaries/MFDs sign with one KUA and register as **sub-KUAs**
- Demat holding or exchange route: DP's KYC accepted
- Mandatory additional fields: occupation, gross annual income/net worth, PEP status (individuals & non-individuals)
- MFs, depositories, RTAs, KYD-compliant MFDs, brokers can facilitate KYC

KYC process & online KYC ★★★★☆

- Form + documents verified with originals, or true copy attested by Notary, Gazetted Officer or Scheduled Commercial Bank manager
- Originals returned; data uploaded to a KRA
- **IPV mandatory**; name, designation, organisation of verifier recorded; IPV by a scheduled commercial bank accepted
- After upload, no further KYC for any securities market segment; change of address etc. done once with any intermediary
- Online KYC: eSign, DigiLocker, e-signatures (IT Act 2000); mobile/email via OTP; Aadhaar via UIDAI; PAN via IT database; bank via **penny drop**/API; OVDs via DigiLocker/eSign; wet signature scan or cropped signature via eSign
- Supreme Court 30 Apr 2025: digital KYC must be accessible to persons with disabilities (SEBI FAQ revised)

Special case	KYC rule
Minor	Guardian's KYC + minor's age proof
PoA	Both investor and PoA holder; PoA holder can't apply for KYC on issuer's behalf
NRI	PAN = sole identification number; passport/PIO/OCI copy + overseas address proof mandatory

Institutional investors & UBO ★★★★★

- Eligibility to invest: incorporation documents (MoA/AoA, trust deed) must permit; some states need Charity Commissioner's permission for religious/charitable trusts
- Authorisation to invest: Board Resolution
- Authorised signatories (in Board Resolution); MF acts only on forms signed by authorised signatories

Ultimate Beneficial Owner (UBO)	Threshold
Company	> **25%** of shares or profits
Partnership / body of persons	> **15%**
Trust	Settlor, trustees, beneficiaries with ≥ 15% of benefits
Exempt	Listed companies and their (majority-owned) subsidiaries; individual investors

Client Due Diligence under PMLA

UBO proof (name, address, PAN/passport) + UBO declaration to AMC/RTA; intimate changes

FATCA & CRS ★★★★☆

- Due diligence to identify foreign reportable accounts; report to US IRS / foreign govt / Indian tax authorities
- Info if citizenship, nationality, birthplace or tax residency outside India; tax countries + taxpayer IDs
- Covered investor → entire value of all folios reported, with direct/indirect beneficiaries and controlling persons
- Change in status reported to MF within **30 days**

Systematic transactions — why ★★★☆☆

Tailor investments/payouts to goals

Align with investible surplus, regular income needs, rebalancing

Execute without repeated instructions → protects from investor inertia

SIP & rupee cost averaging ★★★★★

Month	NAV	Units for ₹1,000	Cumulative
1	10	100.000	100.000
2	12	83.333	183.333
3	9	111.111	294.444
4	11	90.909	385.353
5	10	100.000	485.353
6	12	83.333	568.686

```
Average cost = 6,000 ÷ 568.686 = ₹10.55  <  average NAV ₹10.67  → Rupee Cost Averaging
```

Buy volatile markets over time without predicting

Smaller amounts easier than lump sums

Via post-dated cheques, NACH, Standing Instructions

Can open a folio or add; lower minimum than lump sum; can start during NFO

Systematic Withdrawal Plan (SWP) ★★★★★

- Redeem a constant value periodically — avoid exiting all at a trough
- ₹1,000/month: NAV 10 → 100 units; NAV 12 → 83.333; NAV 9 → 111.111
- Exit load raises units needed: 1% load at NAV 10 → effective ₹9.90 → **101.0101 units** for ₹1,000
- Register amount, periodicity (usually monthly), period; some allow only appreciation or IDCW to be withdrawn (variable SWP — capital not withdrawn)
- Reasons: avoid trough risk; regular expenses (retirement); regularly encash profits
- Exit loads and taxes apply

Systematic Transfer Plan (STP) ★★★★★

- SWP from source scheme + SIP into target scheme of the same MF
- One transfer = switch; multiple pre-defined tranches = STP
- Transfer = redemption → exit loads & taxes apply
- Scheme minimum investment/redemption limits do NOT apply to STP
- MF defines eligible source and target schemes
- Avoids idle funds & double paperwork; used for phased equity entry, rebalancing, profit booking; fixed amount or appreciation
- SIP, SWP, STP may be offered for SIF strategies (with minimum investment threshold)

```
₹10 lakh in short-term debt fund → STP ₹1 lakh × 10 to equity; exit load 0.25% for 3 months
First 3 instalments: load ₹250 → ₹99,750 transferred; thereafter full ₹1,00,000
```

Switch & Transfer of IDCW Plan (DTP) ★★★★☆

Switch	Redemption + purchase in one transaction (e.g. book equity profits → short-term debt)
DTP	Invest IDCW of one scheme into another scheme of the same MF

- Must hold IDCW option of source scheme; DTP overrides payout/reinvestment instructions; covers all units in that option
- Target units in same folio; target scheme minimum doesn't apply; MF may set minimum IDCW to transfer
- Invested on the **next business day after the record date**
- Register a set number of days before record date; ends automatically when source holding is nil or by written request
- Uses: low-risk investor transfers debt IDCW into equity; equity IDCW into debt to rebalance

Operational aspects — SIP registration ★★★★☆

- MF specifies schemes, minimum per tranche, dates, minimum tenure; investor picks amount, frequency (monthly, quarterly, semi-annual, annual), period, dates; defaults if not chosen
- Minor: SIP registered only till majority
- New investor: application form + SIP enrolment form (must tally); existing: only enrolment form with folio no.; signed per mode of holding (all holders if joint)
- Enrolment typically **15–30 days** before first instalment (e.g. submitted 5 Sep, SIP date 15th, 30 days needed → first on 15 Oct); shorter if OTM registered
- Payment: post-dated cheques or NACH, direct debit, Standing Instructions (bank/branch tie-up)
- First instalment: cheque on the registered bank, or any bank + photocopy of registered bank's cheque, or no cheque if form attested by debiting bank

SIP top-up & SIP pause ★★★★☆

```
Top-up fixed: ₹2,500/month, +₹500 a year → Y1 ₹2,500, Y2 ₹3,000, Y3 ₹3,500
Top-up %: +10% a year → Y1 ₹2,500, Y2 ₹2,750, Y3 ₹3,025
```

- Upper limit or end date for top-up; after that SIP stays constant till tenure ends
- At enrolment or later for an existing SIP (give existing SIP details)
- May not apply to micro-SIPs, or SIP + top-up ≤ ₹50,000 in rolling 12 months/FY
- **SIP pause**: 1–6 months (per AMC) without cancelling mandate; auto-resumes; units keep growing; avoids NACH bounce charges/new mandate

Renewal & cancellation; SWP/STP registration ★★★☆☆

- SIP renewal form: scheme, plan, option, amount, date, period
- SIP cancellation: notice to AMC (period set by AMC) + cancel bank mandate; may auto-cancel on insufficient funds; online/offline via AMC site/app, MFDs, platforms
- SWP: register scheme, plan, option, amount, frequency, period; minimum notice; credited to default or registered account; minimum per tranche usually lower than normal redemption; ends when all units redeemed or by written notice
- STP/switch: source & target per MF list; MF sets minimum; notice period to register/cancel; switch via transaction slip in rupees or units; target units in same folio
- Each tranche at applicable NAV of that date (each leg of a transfer at its own NAV); loads & taxes apply; cancel any time by written notice

Triggers ★★★☆☆

- Profit-booking trigger: redeem when market reaches a level (lose further upside)
- Date trigger: redeem specified units on a date
- Stop-loss trigger: redeem at a NAV level or % fall
- Entry trigger: transfer money into equity scheme when market falls to a level

Triggers — more detail ★★★☆☆

- Set on index value or investment value (e.g. switch into equity if market falls 20% or NAV falls x%)
- On trigger: redeem only appreciation, entire value, or a % of value; or switch to another scheme
- Stop-loss usually → redeem entire value; appreciation trigger → redeem appreciation
- MF decides trigger types and schemes; separate request per investment in a folio; notice period may apply
- **One-time** facility — extinguishes once activated and action completed

Nomination — rules ★★★★★

Rule	Detail
Scope	Folio nomination covers all investments in the folio
Number	Up to **3** nominees with % totalling 100; if % not given → equal shares
Single holder	Nomination required unless holder declares intent not to nominate
Joint holding	All holders sign, whatever the mode
Who can nominate	Only individuals; NOT minors' folios; NOT PoA holders
Who can be nominee	Individuals incl. minors (with guardian) and NRIs; Central/State govts; local authorities
Can't be nominee	Trust (except religious/charitable), society, body corporate, partnership, Karta of HUF, PoA holder
Change/cancel	Any time, by all original holders; new nomination supersedes old; death of sole nominee cancels; multiple → remaining nominees
Demat units	Demat account nomination applies (depository bye-laws)
Opt-out	Investors subscribing on/after **1 Aug 2022**: nominate or sign opt-out declaration

TRAP The workbook says 'up to 3 nominees' in Ch 4 and Ch 9, but the new SEBI nomination section (incapacitated investors) says 'Investors can nominate up to 10 persons'. For the exam, 3 is the figure used in the sample question; know both.

Nominee for incapacitated investors; joint holder death ★★★★☆

- Investor may empower any one nominee (not a minor) to operate the folio if physically incapacitated but able to contract
- Specify % or absolute value the nominee can encash; change mandate any number of times
- On death of a joint holder: assets transmitted to surviving holder(s) by name deletion — survivors receive as **owners, not trustees**
- No limit on how often nomination is made/changed/cancelled

Pledge / lien of units ★★★★☆

- Pledge form by unitholder (pledger) in favour of lender (pledgee)
- Units must have completed lock-in; all holders sign whatever the mode; name, scheme, plan, option, number of units
- Non-individuals: board resolution/authorisation
- Pledged units can't be sold/switched until pledgee's written NOC
- Lender can enforce and redeem with documents; scheme minimum redemption doesn't apply
- IDCW on pledged units to unitholder or lender per agreement

Demat account for MF units ★★★★☆

- MFs must offer demat option; ISIN for each option of a scheme, shown on account statements
- Tie-up with NSDL / CDSL; demat account with DP; KYC incl. IPV for demat → no separate KYC by AMC/distributor
- Application: DP name, DP ID, beneficiary account no. (+ DP statement); names, holding mode, PAN matched; units credited after funds realised
- Mismatch → treated invalid for demat; physical units issued if otherwise valid
- Address, bank, nomination per depository records; changes via depository; redemption via depository or exchange
- Existing SOA units can be dematerialised; proceeds to demat-linked bank
- Demat → physical = **re-materialisation**

Benefits: less paperwork · direct credit of bonus & rights units · single change of address with DP · all investments in one account

Change in folio details ★★★★★

Change	Where / how
Name, status/nationality, PAN, address, contact, spouse's/father's name	KRA change form (once) → KRA informs all MFs; give name, DOB, PAN or PERN, Aadhaar as per KYC; self-attested PAN + supporting documents (originals for verification)
Bank account, mode of holding, nomination	With each MF separately
Demat address	Per DP procedure
Change of guardian (minor)	New guardian KYC + PAN, bank details, **NOC from existing guardian**
Minor → major	Investor's PAN/KYC replaces guardian's; signature & new bank updated; guardian stops operating

Change of bank account ★★★★☆

- Register with each MF; up to **5** (individuals) / **10** (non-individuals); one default
- Before deleting the default account, designate another default (Part B of Multiple Bank Registration Form)
- Details: bank, account no. & type, branch, IFSC, MICR + cancelled cheque with first holder's name (else statement/passbook/bank certificate); originals if photocopies
- Signed per mode of holding; without multiple-bank registration use Change of Bank Details form

Transfer and gifting of MF units ★★★★★

- Units now transferable in SOA or demat mode — can be gifted
- Eligible relatives: **spouse, children, parents, siblings, parents-in-law**
- Excluded: **ETFs** and **retirement/children's solution-oriented** funds
- Both giver and receiver KYC-compliant
- Units under lock-in (e.g. ELSS < 3 yrs) or lien can't be transferred

Transmission of units ★★★★★

Situation	Units go to
First holder dies (joint)	Second holder becomes first holder
Single holder with nominee	Nominee
Joint holding with nominee	Joint holder's right takes precedence
No nomination	Legal successors

- Nomination = authorisation to MF to transfer; nominee holds in trust for legal heirs — no title or beneficial interest; inheritance laws apply
- MF needs nominee's KYC, death certificate, indemnity
- Standard Transmission Request Form + common documents (AMFI), on AMC, RTA and AMFI websites; NOC form; documents vary by situation

Minor turned major ★★★★★

- Folio **frozen** for guardian on date of majority; no financial/non-financial transactions till status changed
- IDCW payout/reinvestment continues (net of TDS)
- SIP/SWP/STP registered beyond majority stop from that date (suspended if documents not received)
- Bank accounts & demat of minors-turned-major: no debits until signature attested by banker; demat must be reopened as new account (minor accounts are single-name)
- PAN reissued with same number and new signature
- AMCs usually send reminder letters before majority

Before MAM form	(i) Get PAN (ii) Complete KYC (iii) Change bank account status or open new account with pre-printed cheque book

MAM documents: MAM form · PAN copy · KYC acknowledgement/form · cancelled cheque or statement/passbook · banker's signature attestation · nomination form · fresh SIP/STP/SWP mandate

Signature in MAM form attested by registered parent/guardian, or Notary, or Judicial Magistrate First Class (JMFC)

SEBI circular 24 Dec 2019 (partly modified May 2023)

NRI to resident Indian ★★★★☆

Bank	Can't operate NRO/NRE/FCNR(B); inform bank; open Resident Rupee account; may open Resident Foreign Currency (**RFC**) account for NRE/FCNR balances
Demat	Inform authorised dealer (Portfolio Investment Scheme) and DP; open new 'Resident' demat; transfer holdings; NRI demat closed
Mutual funds	Inform AMCs of status, address, bank; KYC change form to KRA (acknowledgement issued)

Change in Karta of HUF ★★★☆☆

Letter from new Karta with reason (or AMC form): deceased Karta's name, folio, scheme, units

KYC of new Karta and HUF (HUF KYC not needed if already compliant)

Attested death certificate | Bank certificate of new Karta's signature

Indemnity bond by all co-parceners + new Karta

Turnaround times (SEBI) ★★★★★

Service	Turnaround
NAV calculation & disclosure	Daily
NFO open (other than ELSS)	Max 15 days
Allot units / refund	Within 5 business days of NFO close (delay → 15% p.a. interest, not charged to scheme)
Re-open open-ended scheme (other than ELSS)	Within 5 business days of allotment
IDCW & redemption transfer	As specified by SEBI (delay interest borne by AMC)
Annual report / abridged summary	4 months from FY close
Portfolio statement to unitholders	Within 10 days from close of each half-year (31 Mar, 30 Sep)
Half-yearly unaudited results on website	Within 1 month of half-year close (schemes other than debt/money market)
Debt & money market trades (incl. inter-scheme)	Daily, with 15-day lag (from 1 Oct 2020)
CAS monthly / half-yearly	As specified by SEBI
Unit certificate	Within 5 working days of request
Close-ended: demat units on request	Within 2 working days

SOA or demat units: within 5 working days of closure of initial subscription or receipt of application

Demat units on request within 2 working days

SOA also to dormant investors (no transaction for 6 months) — with portfolio statement/annual return; monthly soft copy if requested

Other services: online consolidated family view, valuation, income, returns, capital gains workings

Unclaimed assets & MITRA ★★★★☆

SEBI measures: norms for inactive/dormant folios · mandatory contact & bank details · nominate or opt out · simpler transmission · central reporting of investor's death

MITRA	Mutual Fund Investment Tracing and Retrieval Assistant — RTA platform to trace inactive/unclaimed folios; industry-level search
Hosted by	Two Qualified RTAs (QRTAs): **CAMS and KFintech**, as agents of AMCs; links on AMC, QRTA, AMFI, SEBI websites
Benefits	Find overlooked investments or ones you're legal claimant to · encourage KYC · reduce unclaimed folios · transparent ecosystem · fraud mitigants
DigiLocker	Govt digital wallet (Aadhaar, PAN, DL, death certificate…); bank, insurance, NPS statements already there; adding MF & demat statements; nomination facility — nominee's mobile & email needed

Voluntary lock-in / debit freeze ★★★★☆

Investor can voluntarily lock folio against unauthorised redemption

Demat and non-demat (SOA) folios

No units debited until unlocked

Phase 1: via **MF Central** (RTAs) — SEBI circular Mar 2026

10 Risk, Return & Performance of Funds

≈5 marks · 29 subtopics · chapter priority ★★★★☆

Standard vs specific risk factors (SID) ★★★★☆

Standard / general	Risks all MF investments face
Specific	Risks of particular asset categories (credit & interest rate → debt; currency → foreign securities)
Who manages	Fund manager manages some; MF structure controls some; some remain with investor
SID	Lists risks + risk mitigation strategies

General risk factors ★★★★★

- Trading volumes, settlement, liquidity, default — incl. possible loss of principal
- NAV moves with markets, interest rates, currency, govt policy, tax, politics, economy, volatility
- Past performance of sponsor/AMC/MF doesn't guarantee future
- Scheme **name** doesn't indicate quality or prospects
- Sponsor not liable beyond initial contribution to set up the MF

Risk	Workbook points
Liquidity	Trading volumes, settlement periods, transfer procedures; settlement delays → uninvested cash or missed opportunities; money market lacks developed secondary market; liquidity premium changes; G-secs more liquid but can freeze in extreme volatility; adverse RBI guidelines
Interest rate	Rates ↑ → prices ↓; extent depends on coupon, maturity, yield level; derivatives too
Re-investment	Interim cash flows reinvested at lower rate than assumed ('interest on interest')
Political	Central/state/local politics, unrest, external tensions, policy delays
Economic	Slowdown, fiscal deficits
Foreign currency	INR vs FPI's home currency; AMC does NOT manage FPIs' currency risk; sponsor/fund/trustees/AMC not liable
Settlement (counterparty)	Floating asset created via swap: counterparty may default
Exchange transactions	Allotment/redemption depend on BSE/NSE and clearing corporations

Specific risk factors — equity, short selling, mid/small cap, dividend ★★★★☆

Risk	Points
Equity	Daily volatility; liquidity limited by volume & settlement; affected by rates, FX, law, tax, politics
Securities lending	Lend via approved intermediary; borrower returns equivalent securities + corporate benefits; risk: counterparty failure, loss of collateral rights / corporate benefits
Short selling	Sell borrowed securities, buy back later; counterparty risk + liquidity risk (cover at much higher price)
Mid & small cap	Faster earnings growth expected but may fail; volatility and liquidity risk rise down the cap spectrum (large 1–100, mid 101–250, small 251+)
Dividend	Due only when declared; past record no assurance

Derivative risks ★★★★★

Highly leveraged — small margin, big gain/loss

Need understanding of derivative + underlying

Risk	Meaning
Counterparty	Counterparty fails → renegotiate at unfavourable price; exchange-traded: exchange guarantees but you take exchange's performance risk
Market liquidity	Can't trade due to low volumes or big price impact
Model	Mis-pricing / improper valuation
Basis	Derivative moves differently from hedged security
Execution	Insufficient bids, trading suspension, circuit breakers
Index futures	Same risk as the index portfolio
Strategy	Depends on manager's ability; no assurance
Fixed income derivatives	Interest rate + liquidity risk

Debt fund risks ★★★★★

Risk	Meaning
Reinvestment	Rates on coupon/maturity dates may differ from original coupon
Rating migration	Downgrade (AAA → AA+) hurts price; upgrade helps
Term structure of interest rates	Rates ↓ → value ↑; rates ↑ → value ↓
Credit	Issuer can't pay; managed by in-house credit analysis; corporate > G-sec risk; AAA < AA risk; derivative credit risk small (no principal exchanged)
Floating rate — spread risk	Spread over benchmark widens → value falls even if benchmark unchanged
Floating rate — basis risk	Benchmark becomes inactive/ceases → doesn't capture rate moves
Repo in corporate bonds	Counterparty may not repurchase; collateral sold — loss only if sale price < repo amount; mitigated by **over-collateralisation**
Segregated portfolio	Can't liquidate till recovery; may realise nothing; listing doesn't ensure liquidity; may trade far below NAV

SEBI credit-risk-based single issuer limits for actively managed schemes

Securitised assets ★★★★☆

Structure	Originator (bank, NBFC, HFC, company) sells receivables to SPV (trust) → investors get rated **Pass-Through Certificates (PTCs)** → paid from loan collections; credit enhancement per rating agency

Risk factor	Points
Asset class	CV, auto, credit card, home loans; macro, collateral, documentation, borrower intent
Loan size	Small loans diversify but tiny tickets = costly recovery
Loan to Value (LTV)	Lower is better: truck ₹20 lakh with ₹10 lakh own equity safer than ₹2 lakh own equity
Seasoning	Pool that repaid 80% of instalments beats one that repaid 10%
Default distribution	Current, 0–30, 30–60, 60–90 DPD (days past due); 60–90 riskier
Credit rating & enhancement	Can reach rating above originator's; non-recourse; schemes mostly buy AA and above; no assurance rating stays
Liquidity & price	Thin secondary market; may sell at discount
Limited recourse / delinquency	Enhancement is limited loss cover; no recourse to issuer/originator; repossession delays
Prepayment	Early repayment, repurchase for misrepresentation, repossession → tenor & yield change
Bankruptcy of originator	If not a 'true sale' → losses/delays; legal opinion obtained
Bankruptcy of investor's agent	Assets must be held as agent/trustee
Co-mingling	Servicer's collections mixed with own funds before deposit; servicer with highest rating

REITs/InvITs risks & risk mitigation strategies ★★★☆☆

REIT/InvIT: price, interest rate, credit, liquidity, reinvestment risk + lower-than-expected distributions (depend on net cash flows)

Risk	How managed
Market liquidity	Portfolio with liquidity access; liquid G-secs, high-quality PSU bonds; cash first
Credit	Good credit profiles; in-house analysis; issuer limits on amount & tenor (except GoI); periodic review
Term structure	Active duration management; mostly short duration
Rating migration	High-grade securities; periodic review
Reinvestment	Short duration → small impact
Equity market	Own research, management meetings, conferences; financials, governance, brand, market share etc.
Floating rate	Very low liquidity → limited investment
Equity tax status	Monitor equity exposure of equity-oriented schemes

Systematic vs unsystematic risk ★★★★★

	Systematic	Unsystematic
Also called	Market risk, non-diversifiable	Company-/firm-specific, diversifiable
Example	Rise in inflation	Labour strike at a plant
Reduced by	Only staying out of market (cash, within SEBI limits)	Diversification

Finance theory: rewarded only for non-diversifiable risk

Active managers take unsystematic risk to beat benchmark

Some managers stay fully invested (investor chose the risk)

'Subject to market risks' — meaning ★★★★☆

- MF = pass-through; investor is owner and bears investment risk
- Company FD promises fixed return → depositor bears credit risk only
- Diversification protects against credit/security-specific risk, not market-wide price fluctuation
- Equity growth funds: big NAV swings; overnight funds: stable NAV
- Risk lies in underlying investments; MFs pass on some, reduce others

Drivers of equity scheme returns ★★★★☆

Portfolio = main driver: asset class, sectors, style, strategy

Equity returns linked to business earnings

To beat benchmark, portfolio must differ from it

Security selection (pick winners, avoid losers) vs **market timing** (time entry/exit)

	Fundamental analysis	Technical analysis
Studies	Business, financial statements, management, competition	Price-volume charts ('chartists'): support, resistance, breakouts
Best for	Long-term, security selection	Short-term, intra-day, timing

Fundamental ratios ★★★★★

Ratio	Formula / reading
EPS	Net profit after tax ÷ number of equity shares
P/E	Market price ÷ EPS; forward P/E on projected EPS; low P/E ≠ cheap automatically
PEG	P/E ÷ expected earnings growth: = 1 fair, < 1 undervalued, > 1 overvalued
Book value per share	Net worth ÷ number of shares (historical, accounting-based)
Price to book	Market price ÷ book value per share
Dividend yield	Dividend per share ÷ market price; conservative investors; high yield may mean fewer growth prospects; falls in bull markets, rises in bear markets

Growth vs value; top-down vs bottom-up ★★★★★

Style	Features
Growth	Faster-growing companies; high valuations; fall more in corrections; high P/E & PEG, low dividend yield; do well early in bull runs
Value	Priced below intrinsic value; needs longer horizon; winners offset failures; safer when markets heat up

High valuation ≠ high share price (₹100 share can be cheap; ₹5 share can be over-valued)

Blend possible

Approach	Order
EIC framework	Economy (inflation, rates, GDP, fiscal/monetary policy, BoP) → Industry (regulation, competition, inputs, cycles) → Company (management, financials, products)
Top-down	Economy → industries → companies; avoids big exposure to a poor sector
Bottom-up	Company → industry → economy; sector allocation results from stock picks; good stock even in a cold sector

Drivers of debt scheme returns ★★★★★

Two main risks	Interest rate risk and credit risk — yields rise with maturity and with credit risk
Tenor / redemption	Pre-specified period; repaying at maturity = redemption
Money market securities	Mature within a year
Yield / YTM	Total return; YTM = return if held to maturity
Holding period return (HPR)	Interest + capital gain/loss vs purchase price
Credit spread	Yield on non-govt debt − yield on gilt of same tenor
Rating agencies	CRISIL, ICRA, CARE, Fitch (India Ratings); AAA = highest safety
Floaters	Base + spread, e.g. 5-yr G-sec + 2%

Instrument	Issuer / tenor
G-sec / gilt	Government; no credit risk; lowest yield for a tenor
T-bills	RBI on behalf of GoI, short term
CD	Banks 7 days–1 yr; FIs 1–3 yrs
CP	Companies, up to 1 yr
Bonds / debentures	> 1 yr; govt/PSUs issue bonds, private cos debentures

```
Bought at 8% yield; market yields rise to 9% → value falls (inverse)
Company X 5-yr 9.5% AAA debenture; after 2 yrs, 3-yr AAA rate = 8.5% → trades at a PREMIUM
Longer maturity → bigger price swings
```

Debt fund management: duration, floaters, credit spreads ★★★★★

- Higher **modified duration** → more price volatility for a rate change
- Floaters reset coupon with market → hold value when yields change
- Expect rates ↑ → more floaters / shorter fixed-rate paper (low MD); expect rates ↓ → longer fixed-rate paper (high MD)
- Debt fund returns hinge on interest-rate calls (equity: sector & stock calls)
- Rating upgrade → lower credit spread accepted → price ↑
- Accrual focus: money market, liquid, ultra short, floating rate funds (short maturities, low MD)
- Accrual + gains: mix of short and long; more long → more volatility
- **Duration management**: lengthen if rates expected to fall; risk = view goes wrong
- Credit plays: buy papers expected to be upgraded; risk = default if upgrade doesn't happen
- Tight SEBI mandate (e.g. overnight: only overnight securities) leaves little room; ultra-short may take credit risk (only maturity defined)
- Dynamic bond fund: manager positions portfolio on rate view

Gold fund performance ★★★★☆

- No current income — gain only from price rise; demand-supply drives price
- Truly international asset, objectively measurable quality
- Indian price = international price (foreign currency) × exchange rate + import duties
- Global price: safe haven — rises in political/economic turmoil; central banks & IMF hold gold; their selling weakens price, big countries' buying lifts it
- **Stronger rupee → lower returns**; weaker rupee → higher returns
- Gold funds are passive — no fund-manager decision risk

Real estate fund performance ★★★☆☆

- Local, immovable asset; local factors drive value
- Economy: uncertainty/recession → purchases postponed, prices weaken
- Infrastructure development → values rise
- Interest rates: cheap money → prices up; rate rise softens market
- Varies by type: residential, commercial, industrial, infrastructure, warehouse, hotel, retail
- Returns: rent (regular) + capital appreciation (hard to determine short-term)
- SEBI mandates neutral valuation agencies; currently **no MF schemes invest in real estate**; REITs gaining popularity

Simple, annualised and compounded return ★★★★★

```
Simple return = (Later - Initial) ÷ Initial          NAV 12 → 15 = 25%
Annualised = Simple return × 12 ÷ months (or 365 ÷ days)
   5% in 6 months → 10% ; 3% in 4 months → 9%
Compounded = (Later ÷ Initial)^(1/n) - 1, n in years (days ÷ 365)
   ₹1,000 → ₹4,000 in 2 yrs → 100%
Compounding: ₹10,000 at 10% for 3 yrs → ₹13,310 (simple ₹13,000)
```

These three ignore IDCW — valid only for growth options or IDCW options with no payout in the period

CAGR with IDCW reinvested (SEBI method) ★★★★★

```
₹10,000 at ₹10 on 30 Jun 2019 → 1,000 units
1 Jan 2020: IDCW ₹1 → ₹1,000 ÷ ex-NAV 12.50 = 80 units → 1,080
1 Jan 2021: IDCW ₹1 → ₹1,080 ÷ ex-NAV 15 = 72 units → 1,152 units × ₹15 = ₹17,280
n = 551 days ÷ 365 = 1.51 yrs → CAGR = (17,280 ÷ 10,000)^(1/1.51) - 1 = 43.65% (pre-tax)
```

Assumes IDCW reinvested at ex-dividend NAV

Scheme returns vs investor returns ★★★★☆

- Loads drag investor return below scheme return: exit load 1% → receive ₹14.85 not ₹15; old 2% entry load → paid ₹10.20
- Use amount actually paid (NAV + entry load) and actually received (NAV − exit load); taxes reduce further
- Additional purchases/partial redemptions also change investor return
- Advertised returns ignore loads
- Holding-period returns: CAGR if > 1 year, simple absolute if < 1 year; distorted by unusual start/end values
- **Rolling returns**: average annualised return over many consecutive periods (e.g. all 1-yr returns in 3 yrs, daily/weekly/monthly roll)
- Don't judge on returns alone: consistency, benchmark, peers; active fund should rise well and fall less; consider volatility

Assured returns ★★★☆☆

MFs can't promise returns unless an assured-return scheme

Assured return needs a guarantor named in SID who pays any shortfall

So far no assured/guaranteed product — only capital protection-oriented funds

Risks from the investor's view ★★★★★

Fund	Investor risk
Equity	Price, business and liquidity risk rise large → mid → small; higher in focused funds; goal may fail; invest per risk profile; keep liquid funds to avoid selling equity in downturns
Debt	NAV moves with rates and credit migration; IDCW not guaranteed (schemes have skipped IDCW); liquid/ultra-short/low duration may give partial liquidity if gating or segregation; without segregation, staying investors' bad-paper exposure rises (5% → 10% after 50% redemptions); 2018–19 credit events hit even liquid funds — **'low risk' ≠ 'zero risk'**
Hybrid — arbitrage	Not always pure arbitrage: debt, paired arbitrage, alpha hedging, merger arbitrage = directional views → losses/low returns possible
Gold	Transparent, hard to manipulate; does well in turmoil, war, weak currency; know if Gold Sector Fund or Gold ETF; loses if gold price falls
Real estate	Subjective valuation, illiquid, unorganised agents, high stamp duty/registration, regulatory & litigation risk, low transparency, family-run groups with poor governance — high risk but less than direct real estate

Listing mandatory: every close-ended scheme (except ELSS) and segregated portfolio units (SEBI 28 Dec 2018)

Schemes being wound up (Clause 37(d)) listed for exit

Variance & standard deviation ★★★★★

Month	Scheme 1	Scheme 2
1	5	5
2	4	−5
3	5	10
4	6	5
Average	5	3.75
Variance	0.67	39.58

- Risk = fluctuation of periodic returns vs average (up or down)
- Variance: Excel =VAR(range); relevant for debt and equity
- Standard deviation = √variance = **total risk**; Excel =STDEV(range); compare with benchmark & peers; historic → limited for prediction
- ≥ 30 observations for accuracy
- Annualise: weekly × √52 (0.65 × √52 ≈ 4.70%); monthly × √12; daily × √252 (trading days)

Beta, modified duration, WAM, credit rating ★★★★★

Measure	Points
Beta	From CAPM; systematic risk only; scheme vs diversified index (index β = 1); β 0.8 moves 8% when market moves 10%; β 1.2 moves 12%; **only for equity schemes**
Modified duration	Sensitivity of debt value to interest rates; professionals' preferred measure
Weighted average maturity	70% in 4-yr + 30% in 1-yr = 3.1 yrs; simpler, used with lay investors; longer → more fluctuation
Credit rating	Default risk; G-secs and cash have none; higher rating → lower spread; downgrade (AAA → AA) → yield ↑, value ↓; anticipating upgrades can pay

Credit events & the redemption spiral ★★★★☆

```
Credit risk = default, delayed payment or downgrade (a
'credit event')
₹10,000 cr scheme, 8% (₹800 cr) in debenture M; M
downgraded; 20% redeemed
→ other papers sold → corpus ₹8,000 cr → M = 10%; another
halving → 20%
Regulatory cap: max 10% in a single issuer
```

Market-wide illiquidity can block redemptions even in good portfolios

SEBI's two tools: gating (restriction on redemption) and segregated portfolios

Gating — restriction on redemption ★★★★★

- Only for **systemic** crises affecting almost all securities, not issuer-specific problems
- Triggers: (a) market-wide liquidity issues — not to manage a scheme's own liquidity or a bad investment (b) market failure / exchange closure (political, economic, military, monetary emergencies) (c) operational issues — force majeure, technical failure (e.g. blackout) despite diligence
- Max **10 working days in any 90-day period**
- Needs approval of **AMC Board and Trustees**; SEBI informed immediately
- Requests up to **₹2 lakh** not restricted; above ₹2 lakh → first ₹2 lakh paid, excess restricted
- Possibility disclosed prominently in scheme documents

Segregated portfolio (side pocketing) ★★★★★

Permitted	December 2018 — fair treatment in credit events + liquidity risk
Definitions	Segregated portfolio = debt/money market instrument hit by credit event, separated; main portfolio = rest
Trigger	Credit event at issuer level = downgrade by a SEBI-registered CRA; optional, AMC's discretion
Unrated debt (7 Nov 2019)	For issuer with no rated debt — only on **actual default** of interest or principal; AMC informs AMFI immediately → AMFI informs all AMCs
New schemes	All launched after 7 Nov 2019 must have enabling provisions in SID
Policy	Written policy approved by trustees

- After trustee approval: effective from day of credit event; immediate press release; NAV of both portfolios from that day
- Existing investors get equal units in segregated portfolio
- No subscription/redemption in segregated portfolio; listed within **10 working days**; transfers enabled
- Redeeming investors keep segregated units; new subscribers get only main portfolio units
- If trustees don't approve → transactions at NAV of total portfolio

Segregated portfolio: TER, NAV, risks ★★★★☆

- No investment & advisory fee; other TER pro-rata only upon recovery, capped at simple average of daily non-management expenses of main portfolio while segregated portfolio existed
- Legal recovery costs charged in proportion to recovery, within main portfolio's TER limit; excess borne by AMC
- Costs never charged to main portfolio
- NAV declared daily; disclosed in scheme documents, monthly/half-yearly portfolios, annual report
- Risks: can't liquidate till recovery; may realise nothing; listing ≠ liquidity; may trade well below NAV

Segregated portfolio illustration ★★★★☆

```
30 Sep 2019: 8% XYZ Ltd NCD downgraded A- → C, marked
down 50% (₹99.105 → ₹49.552)
Total net assets ₹1,00,01,227 ; 10,000 units ; NAV
₹1,000.1227
Main portfolio (B+C+D+cash) = ₹87,62,427 → NAV ₹876.2427
Segregated (XYZ) = ₹12,38,800 → NAV ₹123.88
876.2427 + 123.88 = 1,000.1227 per unit (unchanged in
total)
```

Risk mitigation for investors ★★★★☆

Equity: longer holding period narrows return range — 1 yr: extremes; 5 yrs: mostly positive/moderate; 10 yrs: negative returns next to nil

Debt credit risk: check portfolio ratings; G-sec funds, TMFs, corporate bond funds hold better quality

Debt interest rate risk: match holding period to portfolio maturity

11 Mutual Fund Scheme Performance

≈5 marks · 15 subtopics · chapter priority ★★★★☆

Benchmarks — requirements & choice ★★★★★

- Pre-defined comparable to judge performance
- Credible benchmark in sync with (a) investment objective (b) asset allocation (c) investment strategy
- Calculated by independent agency, transparent, published regularly (exchanges, CRAs, research houses, financial publications)
- Index fund: the tracked index; others: subjective — decided by **AMC in consultation with trustees**; stated in SID with benchmark performance alongside
- Benchmark may change (objective change, index construction change, better index) — with trustees, justified and documented; benchmark names disclosed

Price Return Index vs Total Return Index ★★★★★

PRI	Only capital gains of constituents
TRI	Capital gains + all dividends/interest (reinvested)
Mandatory TRI	From **1 February 2018**
Why	Fair comparison — scheme returns include dividends (IDCW option measured by reinvestment method); PRI gave schemes an edge
Effect	Fewer schemes appear to beat benchmark; gap = dividend yield, historically 1.5–2.5% at index level
No TRI for a period	Composite CAGR: PRI till TRI available, TRI thereafter (SEBI circular 4 Jan 2018)

Equity scheme benchmarks ★★★★☆

Driver	Benchmark choice
Scheme type	Diversified: Sensex, Nifty 50, BSE 200, BSE 500, Nifty 100, Nifty 500; sectoral/thematic: Bankex, BSE FMCG, Nifty Infrastructure, Nifty Energy
Investment universe	Large caps: Sensex (30) / Nifty 50; mid caps: Nifty Midcap 50, BSE Midcap
Portfolio concentration	Few stocks → narrow (Sensex, Nifty 50); many → broad (100/200/500 indices)

Debt scheme benchmarks ★★★★☆

- Developed by research/rating agencies recommended by AMFI: CRISIL, ICICI Securities, NSE
- NSE: Nifty Composite G-sec, Nifty 4–8 Year G-sec, Nifty 10-yr Benchmark G-sec; BSE: India Sovereign Bond Index, India Government Bill Index
- ICICI Securities **I-Bex** (sovereign): sub-indices **Si-Bex** 1–3 yrs, **Mi-Bex** 3–7 yrs, **Li-Bex** > 7 yrs
- Liquid schemes (≤ 91 days): NSE Liquid or CRISIL Liquid Fund Index
- Gilt funds: G-sec indices; diversified debt: mixed-debt indices

Other scheme benchmarks ★★★☆☆

Scheme	Benchmark
Hybrid	Blend: e.g. 65% Sensex + 35% I-Bex
Gold ETF	Gold price
Real estate	Indices by real estate services cos — short history, limited acceptance
International	Where it invests: Shanghai Composite (China), S&P 500 (US), synthetic blend for many countries

Standard additional benchmarks ★★★★☆

Scheme type	Benchmark
Equity	Sensex or Nifty
Debt ≤ 1 yr duration/maturity, arbitrage	1-yr T-Bill
Other debt	10-yr dated GoI security
Conservative Hybrid	10-yr GoI
Balanced/Aggressive Hybrid, BAF, Multi Asset	Sensex/Nifty
Equity Savings	10-yr GoI
Retirement/Children's	Sensex/Nifty
Index funds/ETFs/FoFs	Per underlying allocation

Return in INR and CAGR; part of SAI and all ads

Two-tier benchmarks ★★★★★

Circular	SEBI 27 Oct 2021, effective 1 Jan 2022; all benchmarks TRI
Tier 1	Reflects scheme **category** (one broad market index per provider)
Tier 2	Reflects fund manager's **style/strategy** (bespoke)
Debt tier-1	Based on Potential Risk Class, from 1 Apr 2022

Scheme type	Tier 1	Tier 2
Income/debt	e.g. NIFTY Ultra Short Duration Debt Index / CRISIL Ultra Short Term Debt Index	e.g. AAA Bond Index
Growth/equity	e.g. BSE 100 / NSE 100 for large cap	e.g. Nifty 50
Hybrid & solution-oriented	Single broad market (or bespoke) benchmark	
Thematic/sectoral	Single benchmark	
Index funds & ETFs	Single (the replicated index)	
FoFs	Single underlying fund → its benchmark; multiple → broad market index	

Relative returns vs risk-adjusted returns ★★★★☆

Relative return: scheme vs benchmark/peers → initial premise of out/under-performance; AMCs & trustees review periodically

Weakness: ignores different risk levels for same objective

Risk-adjusted: return should be commensurate with risk

Sharpe ratio ★★★★★

```
Sharpe = (Rs - Rf) ÷ Standard deviation
Rf: risk-free (T-Bill index); Rs - Rf = risk premium
Ex: Rf 5%, Rs 7%, SD 0.5% → (7 - 5) ÷ 0.5 = 4
```

Risk premium per unit of total risk | Higher = better

Compare only comparable schemes (not equity vs debt)

Treynor ratio ★★★★★

```
Treynor = (Rs - Rf) ÷ Beta
Ex: Rf 5%, Rs 8%, β 1.2 → 3 ÷ 1.2 = 2.5
```

Higher = better

Beta relevant for diversified equity → restrict comparisons to them

Alpha ★★★★★

Optimal return = return in line with scheme's beta

Alpha = actual return − optimal return

Positive = out-performance; negative = under-performance

Evaluate only for diversified equity schemes

Historical; calculations are arithmetic — evaluation is an art

Information ratio ★★★★★

```
IR = (Portfolio return - Benchmark return) ÷ Standard deviation of excess return
```

- Measures manager's skill and consistency of excess return vs benchmark
- **Only risk-adjusted measure mandatory** to disclose; daily on AMC website; on AMFI website comparably
- Benchmark = tier-I benchmark (per SID); SD of excess returns from daily returns (SEBI circular 17 Jan 2025)

Tracking error ★★★★★

- Index fund β ≈ 1 → should earn market return; gap = tracking error
- Earlier: how closely an index fund tracks (ideally zero); now also: consistency of out-performance
- = **Standard deviation of excess returns**; low for a consistently out-performing fund
- Denominator of information ratio

Performance disclosure & product literature ★★★☆☆

- SID updated twice a year incl. performance; monthly factsheet (not statutory) fills gap; portals; product literature
- Suitability: product label (objective — wealth creation, regular income, liquidity; asset class; holding period) + risk-o-meter
- Returns: cumulative returns for periods vs benchmark; discrete annual returns show consistency
- Portfolio description: equity — diversification, market cap, growth/value; debt — tenor, credit quality; hybrid — allocation

Fund factsheets — macro metrics (Table 11.3) ★★★☆☆

Metric	Purpose
Index levels & yields	Assess scheme performance
Corporate results, earnings revisions, margins	Tone for equity markets
Government spending	Positive for growth; high fiscal deficit → higher borrowing & interest costs, less fiscal room
Inflation	Demand, costs, profits; high inflation → monetary tightening
Interest rates	Consumption, demand, expansion capacity
Domestic & foreign portfolio flows	Expectations from Indian markets
GDP	Economic cycle
Overseas economic & rate data	Export demand, fund flows (low global rates → flows to emerging markets)
Commodity & input prices	Profitability; oil affects currency, exports/imports

Factsheet not mandatory but subject to SEBI ad guidelines; carries fund manager's market views

AMFI website: performance data of all schemes

Data vendors (e.g. Morningstar, Value Research) — not certified by SEBI/NISM

12 Mutual Fund Scheme Selection

≈10 marks · 17 subtopics · chapter priority ★★★★★

Selection based on investor need, risk profile, allocation ★★★★★

- Need: long-term appreciation, periodic income, or parking with high liquidity
- Portfolio = means to an end (goal) — never lose sight of the goal
- Step 1: set financial goals; Step 2: understand situation and risk appetite
- Risk appetite = need + ability + willingness to take risk (SEBI due-diligence circular and AMFI code stress risk profiling)
- Asset allocation decided after risk profile, goals, situation: growth → equity (needs long horizon); income → debt
- Long horizon but unwilling to take risk → may prefer debt
- **Age alone is not a good basis** — goals and situations differ even within an age group; family goals
- Use **time horizon to the goal**: longer horizon → more risk capacity; liquidity may be needed even in long-term portfolios

Core and satellite portfolio ★★★★★

Core	Long-term needs: diversified equity, large cap, mid cap funds etc.
Satellite	Tactical, short-term views: sector funds (cyclical), long-term gilt when rates expected to fall, gold when inflation high or uncertainty
Split	Conservative: very small satellite; moderate: ~80% core / 20% satellite (illustrative); aggressive: higher satellite

Risk-return hierarchies (Illustrations 12.1.1–12.1.6) ★★★★★

Hierarchy (risk and potential return rise left → right unless stated)	
All MFs	Liquid → Debt → Hybrid → Equity
Debt (interest rate risk, similar credit)	Overnight → Liquid → Ultra short → Low duration → Short → Medium → Medium to long → Long duration
Debt (credit risk, similar rate risk)	Gilt → Banking & PSU → Corporate bond → Credit risk
Equity (risk **decreases** left → right)	Small cap → Mid cap → Multi cap → Large & mid cap → Large cap
Concentration	Diversified → Focused → Thematic → Sector
Hybrid	Arbitrage → Conservative hybrid → Equity savings → Dynamic asset allocation → Multi asset → Balanced hybrid → Aggressive hybrid

Large-cap sector fund (concentration risk) not comparable with diversified small-cap fund (size risk)

Different sector funds have different risk

Product labelling (SEBI 2013) against mis-selling: disclaimer to consult financial adviser if unsure; risk-o-meter (Ch 5)

Selection by strategy: objective → strategy → portfolio ★★★★☆

- Same objective can have multiple strategies; same strategy, different portfolios — evaluate each
- Low-risk equity: index fund or value fund; more risk for return: growth fund
- Debt for better returns: regular-income fund, or long-term gilt if rates expected to decline

Active vs passive funds ★★★★★

- Passive: asset-class exposure without manager-selection risk; index funds (not exchange-traded) mimic index constituents and weights; no selection risk; lower cost (no research expenses); ETFs (exchange-traded) also passive
- Active: higher cost and risk to beat benchmark; outperformance not guaranteed
- Want just equity growth, not to beat market → index fund
- **Index fund NAV does fall when the index falls**; long horizon → does well with market

Open-ended vs close-ended ★★★★★

	Open-ended	**Close-ended**
Exit	Repurchase at NAV – exit load	Sell on exchange — needs counterparty
Liquid assets	Kept for redemptions → can dilute equity returns	Not needed — can hold illiquid / long-gestation stocks
Flows risk	Large sales/repurchases pressure manager	No redemption pressure
Price vs NAV	At NAV	May trade at discount (no limit); converges to NAV near maturity
Suits	Those wanting liquidity	Those who can lock money till maturity

Diversified vs focused vs sector vs thematic ★★★★★

- Diversified: multi-sector → less risky; active manager tilts to better sectors
- Focused: concentrated in few stocks → riskier than diversified, less risky than sector
- Sector (pharma, banking): concentration risk; needs skill to time entry before up-cycle and exit before downturn → **satellite** portfolio
- 'Diversification is a protection against ignorance'
- Thematic (MNC, PSU, infrastructure): between diversified and sector

Large vs mid vs small cap; multi/flexi ★★★★☆

- Large caps: established, stable revenues/profits, withstand downturns
- Mid/small caps: early growth; boom in good times, many fall by the wayside in turmoil; attractive when front-line stocks get expensive; winners re-rated as they grow
- Need careful selection — risky to put a large share of portfolio there
- Multi cap: spread across caps, lower risk than mid/small; large & mid: riskier than large, less than mid; flexi cap: manager's freedom
- Choose multi cap (similar allocation), flexi cap (manager decides) or separate large/mid/small funds (investor decides)

Growth vs value funds ★★★★☆

Growth: companies growing above economy; do well in bull phase, fall more in downturns

Value: cheap vs future unlockable value; outperform in falling markets / over long periods

Both can sit in core portfolio; value pays over longer holding

International equity funds ★★★★☆

- Two exposures: foreign equity market + exchange rate (stronger USD helps; stronger rupee hurts)
- Reasons: attractive overall return; diversification per asset allocation; specific opportunity → satellite or small core allocation

Debt fund selection: FMP, TMF, short duration, liquid, ultra-short, floater ★★★★★

Fund	Suitability
FMP	Close-ended debt; horizon = scheme maturity; more predictable return, generally above FD; check credit risk; exit only via exchange
TMF	Gradually replacing FMPs; G-secs/SDLs/AAA PSUs; index TMFs bought/redeemed with AMC like open-ended — **liquidity advantage over FMPs**
Short duration	Maturities 1–3 yrs; market yields; good when rates expected to rise (reinvest at higher rates); low volatility, core for low-risk investors, short-term parking
Liquid	Lowest risk, low returns; comparable to savings account; park up to 91 days; don't use for long periods
Ultra-short	3–6 months; similar role, low volatility, high liquidity
Floater	Floating rate instruments → steady NAV

Accrual strategy → steady, low volatility

Lower-quality paper for yield → be comfortable with it

Total-return/duration strategy → volatile; risk of wrong rate call

Hybrid and gold fund selection ★★★★☆

- Two routes: own mix of equity + debt schemes (more choice, more decisions) or a hybrid scheme (simpler, accept its mix)
- Equity gives appreciation, debt gives stability — for equity exposure with lower risk
- Evaluate equity part (caps, sectors) and debt part (credit, duration) separately; 'conservative' refers only to equity allocation
- Be cautious of dynamic asset allocation variants
- Balanced schemes taxed as equity or non-equity depending on portfolio
- Gold Sector Funds (gold mining/processing shares) vs Gold ETFs (track gold price): miners can rise/fall more than gold and depend on company factors; **ETF = more efficient gold exposure**

Selecting among AMCs / within a category ★★★★★

Factor	What to check
AMC	Comfort with approach, style, values
Portfolio vs objective	Style drift: 'value fund' full of fully-valued front-liners; short-term debt fund with big long-term exposure; hybrid with aggressive small caps/credit; liquid/ultra-short taking high credit risk; non-gilt debt: credit quality and **sector concentration**
Fund manager	Some managers better at spotting trend changes
Performance	Consistent outperformance; protect downside; equity ≥ **5 years**; long-term debt ≥ **3 years**; short-term debt shorter; liquid: 7 days, 15 days, 1 month; vs benchmark, peers, rolling periods, bull & bear cycles; CAGR for 1, 3, 5 yrs and since inception (allotment date)
Portfolio	Equity: diversification, segment, cash, holding length, churn, strategy. Debt: average maturity, duration, credit profile, interest vs gains, liquid holdings
Fund age	Track record; avoid new fund with lacklustre manager; matters more where dispersion is high (multi cap vs large cap)
Fund size	Large-cap fund: size helps (diversification, scale); sector/mid-cap: size can hurt; small funds more flexible
Portfolio turnover	Frequent churn = costs + unsteady management; view with style
Expenses	Drag on returns; especially important for debt and index funds

Portfolio turnover ratio ★★★★★

```
Turnover = lower of (purchases, sales) in period ÷
average net assets
Ex: ₹10,000 cr ÷ ₹5,000 cr = 2 (200%) → average holding =
12 ÷ 2 = 6 months
```

6 months too short for value style, fine for momentum

Short holding may signal tactical rather than long-term investing

Research agency rankings & ratings (Box 12.1) ★★★★☆

- Ranking = order within category; rating = grouping by performance traits; methodologies on agency websites
- Star ratings: 5-star better than 4-star — but almost always based on past performance, no predictive ability
- Best fund this quarter rarely best next quarter; switching costs hurt
- Aim: stay in top 'few' consistently — 3 to 5 in small categories or top 10–15% in large ones
- Taxes and exit loads also matter

Selecting options ★★★★★

- IDCW payout: money flow; growth: grows gross (no annual tax)
- Repurchase = sale → capital gain/loss; may be booked to set off other gains/losses
- Monthly IDCW not assured — depends on distributable surplus → regular income better via **SWP** (STT for equity, capital gains tax applies)
- IDCW taxed at slab → IDCW option not preferable except for tax-exempt investors
- Taxation and liquidity needs decide the option

Do's and don'ts ★★★★★

Rule	Point
Ensure suitability	SEBI Fraudulent & Unfair Trade Practices regulation applies to MF sales
Stick to asset allocation	Keep it at the core of scheme selection
Don't chase past performance	'Past performance may or may not be sustained in future'
Understand objective & strategy	Know what to expect
Watch taxes & loads	Check capital gains and exit loads on redemptions; compare IDCW vs growth tax periodically
Consistent written methodology	Helps stay on course

App Appendices 1–4: Codes of Conduct & Resources

≈in Ch 4/6 marks · 7 subtopics · chapter priority ★★★☆☆

Appendix 1 — Fourth Schedule: Code of Conduct for AMC & trustees ★★★★☆

#	Rule
1	High standards of integrity and fairness
2	Schemes not run or portfolios selected in the interest of sponsors, AMC/trustee directors or their associates
3	Equitable, fair treatment of all classes of unitholders — favouring one set over another is not fair
4	Adequate, accurate, explicit, timely information in simple language to all unitholders
5	Avoid excessive concentration of business with broking firms/associates and excessive holding of units by few investors; avoid conflicts of interest — unitholders' interest paramount
6	Assets, liabilities, bank and securities accounts of each scheme segregated and **ring-fenced**
7	Invest per stated objectives, solely in unitholders' interest
8	No unethical means to sell or induce
9	High service standards, due diligence, proper care, independent professional judgment
10	AMC: no exaggerated statements about qualifications, capability, achievements
11	Sponsor/trustees/AMC/employees giving advice on a security in public media must disclose their interest incl. long/short positions; employees also disclose dependent family members' and employer's interest

Regulated entities and agents (MFDs, PMS distributors) must show registered name and registration number on social-media home page and at the start of each securities-market video/content

Appendix 2 — AMFI Code of Ethics (ACE) ★★★★☆

Section	Key rules
1 Integrity	High integrity & fairness; schemes run for all unitholder classes — NOT for sponsors, member directors, trustees, brokers/intermediaries, associates, or a special class of unitholders
2 Due diligence	High service, due diligence, independent judgment; adequate resources & procedures
3 Disclosures	Timely, simple info; annual disclosure of investment pattern, portfolio, expense ratios, portfolio turnover; trades with associates or significant unitholders → report to trustees with fairness justification + brief disclosure in annual/half-yearly reports; key investment personnel's own trades → compliance officer at least half-yearly, to trustees if conflict
4 Professional selling	No unethical inducement, no exaggerated performance claims; true info; disclose risks before investment; documents on request; fair allotment and timely refunds; handle complaints fairly. Communications: don't present a scheme as a new share issue; no unrealistic expectations; no guaranteed returns except as in SID (with resources to meet them); state market risk clearly; no extraneous benefits; no misrepresentation
5 Investment practices	Manage per fundamental objectives; don't knowingly trade securities with member's or trustee company's directors, officers, employees, or trustees
6 Operations	Avoid conflicts; no **front running** (trading ahead of fund with non-public info); no **self-dealing**; nothing fraudulent/manipulative; no false market, price rigging, passing price-sensitive info; employees can't act as agents/brokers for member's schemes; no fundamental attribute change without unitholder approval (except due to regulation); avoid broker concentration and concentrated unitholding
7 Reporting	Standardised valuation per SEBI; uniform performance reporting on **total return** basis; scheme-wise segregation of cash & securities accounts
8 Unfair competition	Nothing harmful to other members
9 Statutes	Follow letter and spirit of laws
10 Enforcement	Disseminate code; observance a condition of employment; violation = ground for revocation and discipline; employees sign receipt; internal controls; one compliance person with power to investigate; half-yearly & annual reports to trustees; keep records ≥ **3 years**; adequate resources

Significant unitholder	Holds **5% or more** of a scheme's corpus (incl. entities it controls)
Associate	Per Reg 2(d) of SEBI MF Regulations 2026
Fundamental investment policies	Objectives, policies, terms that are fundamental attributes
Trustee	Member of Board of Trustees or director of Trustee Company

Appendix 3 — AMFI Code of Conduct for MFDs: scope & fiduciary duty ★★★★★

- Core values of a fiduciary: transparency, competency, fairness, integrity
- Applies to all ARN holders and binds directors, partners, sub-distributors, employees, representatives, sales personnel
- Investor's interest paramount; due diligence; independent professional judgment
- Avoid conflicts; disclose when unavoidable; disclose conflict when selling group/affiliate/associate schemes
- Financial incentive must not be the basis of recommendations; no aggressive sales tactics for targets
- No rebating / pass-back of commission; no gifts or gift-vouchers as inducement
- No collusion: over-transacting & churning, splitting applications, payment defaults/diversion of funds, false claims for payouts, mis-selling (false statements, concealing facts/risks)

Appendix 3 — compliance, infrastructure, records ★★★★☆

- Follow SEBI MF Regulations 2026 and SEBI/AMFI circulars on selling, distribution, advertising, performance disclosure; segregation of distribution and advisory (IA Regulations 2013)
- KYD compliance; diligent attestation and IPV for KYC
- Know SID, SAI, KIM; explain key features, fundamental attributes, risks
- Suitability: seek client's financial status, investment experience, objectives
- Representatives educated and experienced
- Confidentiality: no sharing/publishing without written consent, no third-party disclosure except by law, **no sharing with group companies for cross-marketing**; AMFI Data Sharing Principles; data protection laws; purge data when no longer needed
- Principal MFD must not engage sub-distributors without valid ARN
- Physical and digital infrastructure incl. cyber security; internal controls to detect/prevent mis-selling and fraud; insurance encouraged
- Records (physical/digital) incl. KYC and suitability correspondence, consent/dissent

Appendix 3 — client obligations ★★★★★

- Give full, updated scheme info (SAI, SID, addenda, performance, factsheets, portfolios, brochures); don't withhold material facts incl. unsuitability
- Highlight risk factors; no misrepresentation; urge reading SAI/SID/KIM
- Disclose all commissions from competing schemes considered
- Disclose list of MFs affiliated with; tell clients other products exist
- If MFD is associate/group/sponsor of an AMC: disclose association and total commission
- **MFDs cannot deal in Direct Plans** (SEBI letter 6 Sep 2021); digital platforms must state the scheme is Regular Plan with commission, hyperlink commission rates and offer documents
- No indicative portfolio, indicative yield or indicative return; never assure returns
- Where assured-return products are also sold, don't mis-sell MFs as assured/regular income; principal at risk
- Forms: investor's own accurate details; never fill MFD's or third-party contacts; changes only on investor's instruction; no tampering; write EUIN
- Help resolve grievances; use only AMC marketing material (no own material or AMC logo without written approval)
- Draw attention to risks: market risk & loss of capital, force majeure, issuer default, segregation; suspension of redemption; NFO risks (price volatility, liquidity, delisting); winding up
- Compare only similar schemes with complete facts

Appendix 3 — other obligations & integrity ★★★★☆

- NISM certification + AMFI ARN/EUIN, renewed on time; non-individuals ensure staff certified; quote valid ARN & EUIN for regular-plan transactions
- Training: fiduciary duty, fraud prevention, responsible social media use
- Cooperate with AMCs, AMFI, SEBI, due-diligence agencies
- Promptly inform AMC & AMFI of changes in status, constitution, address, contacts
- Refund clawback incentives (set-off or payment)
- Notify AMC & AMFI if a representative commits moral turpitude, financial irregularity, is arrested or terminated for such acts
- Reg 3(3) IA Regulations 2013: no 'Adviser', 'Financial Adviser', 'Wealth Manager', 'Consultant' etc. in name unless SEBI-registered IA
- Tagline **'AMFI-registered Mutual Fund Distributor'** with/below name, font size ≥ **12** in printed communication; name + tagline on website, app, business cards, sign boards
- No fraudulent/unfair practices; no false or defamatory statements about AMCs, AMFI, schemes or other MFDs on any forum; no rumours

Appendix 4 — additional knowledge resources ★☆☆☆☆

SID format (SEBI Nov 2023 simplification; Master Circular Mar 2026)

SAI & KIM formats (Master Circular Jun 2024) | KYC info: amfiindia.com/kyc

Industry data (AMFI) | Global data: ici.org

Draft NFOs, SIDs, KIMs, SAIs on SEBI site

Investor education & performance data on AMFI site

13 Basics of Derivatives

≈10 marks · 8 subtopics · chapter priority ★★★★★

Meaning of a derivative & underlying assets ★★★★★

Derivative	Contract/product whose value is derived from an underlying asset
Underlyings	Metals (gold, silver, aluminium, copper, zinc, nickel, tin, lead) · energy (crude & products, cracks, coal, electricity, natural gas) · agri (wheat, sugar, coffee, cotton, pulses) · financial (shares, bonds, forex)

Ind AS 109 criterion	Meaning
1. Value linked to underlying	Price set by underlying (raw material → finished product), not by derivative's demand-supply
2. Settled on a 'future' date	Later than underlying's settlement (underlying T+2 → derivative after T+2)
3. No full cash outlay on trade date	Gives **leverage** — buy without full payment / sell without immediate delivery

World history of derivatives ★★★☆☆

When	Milestone
12th century	European trade fairs: contracts for future delivery
13th century	English Cistercian monasteries sold wool up to 20 yrs in advance
1634–1637	Tulip Mania in Holland (tulip futures bust)
Late 17th century	Dojima (near Osaka), Japan: rice futures
1848	CBOT facilitated forward contracts on commodities
1865	CBOT listed first exchange-traded 'futures contracts' in US
1919	Chicago Butter and Egg Board reorganised → later CME
1972	CME's International Monetary Market (IMM): currency futures
1973	CBOE: first market for listed options
1975	CBOT T-bill futures — first successful pure interest rate futures
1977	CBOT T-bond futures
1982	CME Eurodollar futures; Kansas City Board of Trade: first stock index futures
1983	CBOE options on S&P 100 (OEX) and S&P 500 (SPX)

Growth drivers: more price volatility · global market integration · technology cuts costs · better understanding of risk tools · frequent innovation

Indian derivatives market history ★★★★★

Date	Event
18 Nov 1996	SEBI's 24-member **L.C. Gupta Committee** for regulatory framework
17 Mar 1998	Report: declare derivatives 'securities'
Jun 1998 → Oct 1998	**J.R. Varma** group: risk containment — margining system, initial margin method, membership & net worth, deposits, real-time position monitoring
1999	SCRA amended (Securities Laws (Second Amendment) Act 1999): derivatives = securities
Mar 2000	Govt repealed three-decade-old ban on forward trading in securities
Jun 2000	BSE & NSE equity derivatives; **index futures** on Nifty and Sensex
Jun 2001	Index options
Jul 2001	Stock options
Nov 2001	Stock futures
Feb 2013	MSEI started derivatives

SCRA definition	(a) security derived from debt instrument, share, loan, risk instrument, contract for differences (b) contract deriving value from prices/index of prices of securities (c) commodity derivatives (d) others declared by Central Govt
Section 18A	Derivative contracts legal and valid if traded on a recognised stock exchange, settled on its clearing house per its rules, and between parties/terms notified by Central Govt

Four generic derivatives ★★★★★

Derivative	Key feature	Market
Forward	Buy/sell underlying for cash on a future date; customised; both parties obliged	OTC
Futures	Same but standardised (lot, maturity) — 'exchange-traded forwards'	Exchange
Swap	Exchange cash flows/returns by formula; a series of forwards; manage rate, FX, commodity risk	Mainly OTC
Option	Right, not obligation, to buy/sell on/before a date at a price; buyer pays premium, writer receives it and is obliged	OTC and exchange

Market participants ★★★★★

Participant	Role
Hedgers	Face price risk; use derivatives to reduce it (corporations, institutions, banks — rates, shares, bonds, FX, commodities)
Speculators / traders	Take positions on price views; prefer derivatives for leverage, lower cost, faster execution in size
Arbitrageurs	Profit from price differences across markets — buy cheap, sell dear simultaneously; gaps close quickly

One individual may play different roles at different times

OTC vs exchange-traded derivatives ★★★★★

OTC features	Exchange-traded (ETD)
Tailor-made contracts	Standardised
Counterparty risk managed by each institution	Clearing corporation guarantees settlement
No central limits on positions, leverage, margining	Margins, limits
No formal market-stability rules	Risk management rules
Private, little disclosure	Transparent anonymous auction
Banks, FIs, hedge funds, corporates, HNIs; less regulated	Open access, low cost, credit agnostic

OTC = network of broker-dealers, a way of doing business, not a place

ETD limitations: standardisation → imperfect hedge; cash settlement may not help actual hedgers; daily MTM & margins → operational issues

Convergence: exchanges offer customised contracts via RFQ and Exchange-for-Physical (EFP); OTC offers 'vanilla' and 'exotic'; e-trading platforms; CCP clearing

India: **CCIL** provides CCP with trade guarantee for OTC currency and interest rate derivatives; OTC adopts margining & MTM

Significance of derivatives ★★★★★

Function	Point
Hedging	Contract moves opposite to owned asset → offsets losses
Price discovery	Futures/forward prices show expected future spot prices
Market efficiency	Payoffs replicable → underlying and derivative stay in equilibrium (no arbitrage)
Access to unavailable assets/markets	e.g. interest rate swap to get better rate than direct borrowing
Price stability	Central banks (incl. RBI) use derivatives for currency stability
Higher underlying volumes	More players join due to risk transfer
Speculation	Not the only or most important use; risky if misused; tool for calculated risks
Organised speculation	Shifts speculation from unorganised to organised market with surveillance

Risks in derivatives trading ★★★★★

Risk	Meaning
Counterparty	Default by counterparty
Price	Loss from price move
Liquidity	Can't exit position
Legal / regulatory	Enforceability of contracts
Operational	Fraud, poor documentation, improper execution (e.g. inadequate disaster planning)

Leveraged — may not suit limited resources, limited experience or low risk tolerance

Read the **Model Risk Disclosure Document** given by broker at agreement (issued by exchange members; covers equities and F&O)

14 Understanding the Index

≈5 marks · 12 subtopics · chapter priority ★★★☆☆

What an index is & why it matters ★★★★☆

- Statistical indicator of changes in the economy or specific areas; in markets, a portfolio of securities representing a market or part of it
- Expressed as change from a base value — % change matters more than the number
- Stock index = average share price movement; broad indices should represent typical portfolio returns
- Significance: (1) indicator of market/sector performance (2) **benchmark** for portfolios (3) **underlying for derivatives**

Market-cap weighted index (worked example) ★★★★★

```
Market cap = shares outstanding × price  (5 crore shares × ₹120 = ₹600 crore)
Index value = (Current index market cap ÷ Base market cap) × Base index value
5 stocks, base 100 on 1 Jan 1995: base m-cap ₹18,800 lakh → now ₹42,500 lakh
Index = 42,500 ÷ 18,800 × 100 = 226.06 (+126.06%)
```

Stock	Old price	Shares (lakh)	Old m-cap	Old wt	New price	New m-cap	New wt
AZ	150	20	3,000	0.16	650	13,000	0.31
BY	300	12	3,600	0.19	450	5,400	0.13
CX	450	16	7,200	0.38	600	9,600	0.23
DW	100	30	3,000	0.16	350	10,500	0.25
EU	250	8	2,000	0.11	500	4,000	0.09

Sensex and Nifty were earlier full market-cap weighted

Free-float market-cap index ★★★★★

- Free float = shares readily available for trading (vs promoters, institutions, corporates holdings)
- Weights by free-float market cap = price × shares × free-float factor
- Most global indices incl. **Sensex, Nifty and SX40** moved to free-float basis

```
Same 5 stocks with free-float factors 0.6, 0.5, 0.5, 0.7, 0.8
Base FF m-cap ₹10,900 lakh → now ₹25,850 lakh
Index = 25,850 ÷ 10,900 × 100 = 237.15
```

Price-weighted index ★★★★★

```
Price index = Sum of prices ÷ Number of stocks
Base = (150+300+450+100+250) ÷ 5 = 250 ; Now = (650+450+600+350+500) ÷ 5 = 510 → +104%
Weights by price: AZ 0.12, BY 0.24, CX 0.36, DW 0.08, EU 0.20
```

Higher-priced stocks have more influence

Examples: **Dow Jones Industrial Average, Nikkei 225**

Equal-weighted & capped indices ★★★★☆

```
4 stocks, ₹30,000 each (weight 0.25) on 1 Jan 1995 → total 1,20,000 → now 1,46,000
Index = 146,000 ÷ 120,000 × 100 = 121.67 (+21.67% = Σ price change × old weight)
```

After moves, weights drift (P, R above 25%; Q, S below) → rebalance: sell risers, buy fallers

Capped indices: float-adjusted with a maximum weight per stock

Attributes of a good index ★★★★☆

Reflects market behaviour

Computed by independent third party, free of participant influence

Professionally maintained

Impact cost & bid-ask spread ★★★★★

- Liquidity = large orders executed without moving prices
- Bid-ask spread = best sell − best buy (₹4.50 − ₹4.00 = ₹0.50); buy & sell 100 shares immediately → lose ₹50 — the transaction cost for a small trade
- Bigger order (3,000 shares) hits deeper levels → higher cost
- **Ideal price** = average of best bid and best offer
- **Impact cost** = % degradation of actual price vs ideal price; varies with size; differs for buy and sell side; arises from lack of liquidity — not actually paid as a fee

```
Bids 9.80/9.70/9.60, offers 9.90 (1,000) / 10.00 (1,500) / 10.10
Buy 1,500: ideal = (9.80 + 9.90) ÷ 2 = 9.85
Actual = (1,000 × 9.90 + 500 × 10.00) ÷ 1,500 = 9.9333
Impact cost = (9.9333 - 9.85) ÷ 9.85 × 100 = 0.84%
```

TRAP Impact cost is HIGH when liquidity is poor.

Beta, PRI, TRI ★★★☆☆

Beta	Movement of a stock's/portfolio's returns relative to market returns
Price Return Index	Only price changes; ignores dividends/interest
Total Return Index	Price moves + dividends reinvested in the overall index (not the paying stock)

Index management ★★★★☆

Managers	BSE indices: **Asia Index Pvt Ltd**; NSE indices: **NSE Indices Limited**
Regulation	SEBI (Index Providers) Regulations, 2024 — transparency & accountability
Construction	Choose stocks + calculation method
Maintenance	Adjust for corporate actions: bonus, rights, split, consolidation, mergers — triggered by events
Revision	Change composition (replace stocks) — continuous exercise
Index Committee	Final decision on inclusion/removal

Good index = trade-off between diversification and liquidity

10 → 20 stocks: sharp risk reduction; 50 → 100: very little; beyond 100: almost none

Derivatives on non-benchmark indices (SEBI 30 Oct 2025) ★★★★☆

Prudential norm	Limit
Minimum constituents	14
Top constituent weight	≤ 20%
Top 3 combined weight	≤ 45%
Weights	Descending structure

Index	Deadline	Mode
BANKEX (BSE), FINNIFTY (NSE)	31 Dec 2025	Single tranche
BANKNIFTY (NSE)	31 Mar 2026	Four monthly tranches

```
Tranche adjustment = (actual - desired weight) ÷ remaining tranches
Month 1: 28% → (28 - 20)/4 = 2% → 26%
Month 2: drifted 25.5% → (25.5 - 20)/3 = 1.83% → 23.67%
Month 3: 23% → (23 - 19.5)/2 = 1.75% → 21.25%
Month 4: 20.9% → (20.9 - 18.5)/1 = 2.4% → 18.5%
```

Major Indian indices ★★★☆☆

BSE Sensex · BSE Sensex Next 50 · BSE 100 · BSE 200 · BSE 500 · BSE Bankex · SX 40 · Nifty 50 · Nifty Next 50 · Nifty 100 · Nifty 200 · Nifty 500 · Nifty Bank

Applications & types of indices ★★★★☆

Application	Point
Index funds	Hold index stocks in index proportions (Sensex fund → 30 stocks); small tracking error from expenses & cash for redemptions
Index derivatives	Index futures & options — most popular worldwide; hedge market risk
ETFs	Basket traded like a stock; intraday; small denomination; low cost

Type	Examples
Broad market	BSE Sensex, Nifty 50
Sectoral	BSE Bankex, Nifty IT
Market cap	Large/mid/small cap (e.g. Nifty Midcap 100)
Thematic & factor	ESG/sustainability, dividend yield, volatility (India VIX)

15 Introduction to Forwards & Futures

≈15 marks · 31 subtopics · chapter priority ★★★★★

Forward contracts ★★★★★

Definition	Agreement directly between two parties to buy/sell an asset on a future date at terms fixed today (commodities, FX, equity, interest rates)
Example	Gold spot ₹62,130/10 g on 11 May 2024 = cash/spot transaction; 1-month forward ₹62,337 → you are 'long forward', goldsmith 'short forward'; no money/gold exchanged now
Features	Bilateral; all terms (price, quantity, quality, delivery place, settlement) fixed on contract day; OTC; changes only if both agree
Purpose	Fix the price → avoid price risk

```
After 1 month spot ₹62,700 → buy at 62,337, sell 62,700 → profit ₹363 per 10 g
Spot ₹62,100 → loss ₹237 per 10 g
```

Limitations of forwards ★★★★★

Limitation	Meaning
Liquidity risk	Tailor-made, not exchange-listed → hard for others to access; hard to exit before maturity
Counterparty (default / credit) risk	A buys 100 kg rice at ₹40 from B in 6 months: price ₹50 → B may refuse; price ₹35 → A may buy from market instead
Others	Lack of transparency, settlement complications

Solution: centralised trading platform → futures

Futures contracts: features & limitations ★★★★★

- Agreement through an organised exchange to buy/sell a fixed amount of an asset on a future date at an agreed price — **standardised forwards**
- Clearing corporation guarantees settlement
- Buyer = long, seller = short (figurative — nothing changes hands at origination)
- Exchange fixes all terms except price
- Contract via exchange · centralised platform · price discovery · **margins by both parties** · standardised quality and quantity

Limitations: limited maturities, limited underlyings, no design flexibility, admin costs of MTM

Contract specifications — Nifty futures example ★★★★☆

Field (3 Oct 2025)	Value
Instrument	Index futures
Underlying	Nifty 50
Expiry	28 Oct 2025
Open / High / Low	24,950.40 / 25,020 / 24,878.60
Close (futures)	25,006.60
Underlying (spot)	24,894.25

```
Lot size (example) = 65 → contract value = 65 × 24,894.25 = ₹16,18,126.25
```

Lot size rules (SEBI Master Circular 30 Dec 2024) ★★★★★

Rule	Index derivatives	Stock derivatives
Value at introduction	≥ **₹15 lakh**	≥ **₹5 lakh**
Value at review	₹15–20 lakh	₹5–10 lakh
Lot size	Multiple of 5, not less than 10	Multiple of 25, not less than 50; if > ₹10 lakh at 50 → multiple of 5, min 10; if still > ₹10 lakh at 10 → lot of 5
Review	Every 6 months, average close of last 1 month; 2 weeks' notice	Same; lot same across exchanges
Higher revised lot	Only for new contracts	Only for new contracts

Futures and options on the same underlying have the same contract size

Contract cycle, expiry day & rollover ★★★★★

- Max maturity **12 months** (monthly 1–12); exchanges generally run a **3-month cycle**: near (Oct), mid (Nov), far (Dec)
- Expiry day = last trading day; all open contracts compulsorily settled; to continue, **roll over** (long: sell expiring, buy next) simultaneously
- New contract introduced on the trading day after near-month expiry

SEBI 26 May 2025	All equity derivative expiries of an exchange on either **Tuesday or Thursday**; one weekly benchmark index option per exchange; all other contracts (index futures, non-benchmark index F&O, stock F&O) min 1-month tenor, expiring last Tuesday/Thursday of the month; changing settlement day needs SEBI prior approval
NSE	Nifty 50, Bank Nifty, stock futures & Nifty Financial Services monthly futures: **last Tuesday** (previous day if holiday)
BSE	Index & stock futures: **last Thursday** (previous day if holiday)

Other contract terms ★★★★☆

Term	Rule
Quantity freeze	Orders above freeze quantity rejected
Price quotation	In rupees
Tick size	Minimum price move; NSE index futures by index level, stock futures = cash-market tick; BSE stock futures 5 paise
Daily settlement price	VWAP of last 30 minutes (across exchanges if more than one; else that exchange's); if not traded in last half hour → theoretical price; differs per expiry
Final settlement price	Closing price of the underlying index on expiry (from constituents' closes); stocks: volume-weighted average of closing prices across exchanges (Closing Auction Session SOP)
Pre-open session (SEBI 29 May 2025)	Near-month stock & index futures (also next month in last 5 trading days); call auction 9:00–9:15 am; order entry 9:00–9:10 with random closure in last 2 min
Trading hours	9:15 am – 3:30 pm, Mon–Fri
Holidays	Trading holiday: exchange closed; clearing holiday: trading possible but no clearing/settlement (banks closed)

BSE Sensex futures	Spec
Underlying	BSE Sensex
Contract size	20
Tick size	₹0.05
Cycle	3 serial monthly contracts
Trading hours	9:15 am – 3:40 pm (as stated)
Expiry	Last Thursday
Final settlement	Cash, closing index value on expiry

Basis ★★★★★

```
Basis = Spot - Futures
3 Oct 2025: 24,894.25 - 25,006.60 = -112.35 (negative)
```

- Futures > spot → negative basis → upward expectations → **contango**
- Spot > futures → positive basis → downward expectations → **backwardation**
- Basis differs for 1-, 2-, 3-month contracts; difference between 1- and 2-month basis = cost of carry between months
- Basis becomes **zero at expiry** (final settlement at underlying close)

Cost of carry ★★★★★

```
Cost of carry = Storage (commodities) + Interest - Income
Equity: interest paid - dividend earned
Ex: ABC at ₹100, borrow at 6%, dividend 200% on FV ₹1 =
₹2 → net carry ₹4 → break-even futures ₹104
```

Component	Effect on futures price
Financing cost (interest) — usually biggest	Increases
Storage (warehousing, refrigeration, security; insurance) — not for equities	Increases
Income (dividends, interest)	Decreases

Cost of carry may differ across participants

Margins & MTM ★★★★★

Who	Exchange/CC charges brokers; brokers charge clients — protection against default
Initial margin	Deposit when entering; both buyer and seller; higher for volatile assets
Extreme Loss Margin (ELM)	Covers losses beyond initial margin scenarios; paid upfront
MTM settlement	Profits/losses settled daily; collected from losers, paid to gainers

```
Long Nifty futures (May 2024) at 22,250, lot 25 →
contract value ₹5,56,250
Initial margin 10% = ₹55,625
Close 22,308.70 → MTM gain = 58.70 × 25 = ₹1,468
credited; next day starts from 22,308.70
```

Open interest vs volume ★★★★★

- OI = total contracts outstanding; longs = shorts = OI (count one side); shows market depth
- Volume = contracts traded over a period (day, week, month, life)

Date	Trades	OI	Volume
Mar 1	A shorts 50, B longs 50	50	50
Mar 2	C longs 100, D shorts 100	150	100
Mar 3	A buys back 50, E shorts 50 (replaces A)	150	50
Mar 4	C sells 100, D buys back 100 (both close)	50	100

Price band & base price ★★★★☆

- Price band: daily range around previous close/base (₹100 with 10% → ₹90–110)
- Currently **no daily price range for futures**; a % band only to prevent erroneous orders (dynamic band — next chapter)
- Base price: day 1 = theoretical price; later = close or theoretical price

Positions: long, short, open, naked, calendar spread ★★★★☆

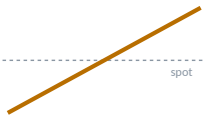
Term	Meaning
Long	Outstanding buy position
Short	Outstanding sell position
Open position	Outstanding long/short (short 5 Infosys + buy 2 later → short 3)
Naked	Futures position without position in the underlying
Calendar spread	Long one maturity + short another, same underlying; computed vs near month; becomes open when near month expires or a leg is closed
Opening / closing	Increases / reduces open position

Forwards vs futures ★★★★★

Feature	Forwards	Futures
Mechanism	Not exchange-traded	Exchange-traded
Specs	Tailor-made	Standardised except price
Counterparty risk	Exists (maybe reduced by guarantor)	Clearing agency is counterparty
Liquidity	Low	High
Price discovery	Not efficient (scattered)	Efficient (common order book)
Information	Poor, slow	Quickly disseminated nationwide

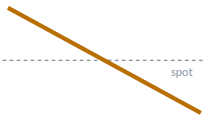
Futures payoffs ★★★★★

Long futures at 100



Expiry 150 → +50; 70 → −30

Short futures at 100



Expiry 60 → +40; 150 → −50

Payoff = profit/loss at expiry vs underlying price (x-axis: price, y-axis: P&L; no negative prices)

Both long and short: unlimited profit/loss → **linear**

Cost of carry (no-arbitrage) model ★★★★★

- Assumes efficient markets → arbitrage can't persist
- Futures position = buy futures OR buy stock in cash and carry it (synthetic futures)
- **Fair futures price = spot + cost of carrying** (transaction, custodial, financing; for commodities warehousing, insurance)

```
Gold spot ₹62,130; carry for 3 months ₹100 → fair ₹62,230
3-month futures at ₹62,280 → buy spot (borrowed), sell
futures, deliver at expiry → riskless ₹50
Arbitrage pushes spot up and futures down until aligned
```

Cash-and-carry vs reverse cash-and-carry ★★★★★

Situation	Arbitrage	Effect
Futures > fair price	**Cash and carry**: buy spot (borrowed money), sell futures, deliver at expiry	Spot ↑, futures ↓ until aligned
Futures < fair price	**Reverse cash and carry**: buy futures, sell spot (borrowed asset), earn interest on sale proceeds, return asset after futures delivery	Spot ↓, futures ↑ until aligned

Transaction costs & no-arbitrage bounds ★★★★☆

- Margins, commissions, taxes create **no-arbitrage bounds** around fair price — no arbitrage inside them
- Every carrying-cost component widens the bounds; wider bounds = further from equilibrium
- Efficient markets need low operating costs → narrow bounds
- Different participants have different costs → different fair values and bounds → continuous trading

Cost of carry with income (dividend yield) ★★★★★

```
Fair futures = Spot + Cost of carry - Inflows
F = S × (1 + r - q)^T        Continuous: F = S × e^((r -
q)T)
Index 17,500, financing 12%, index yield 4%, 90 days:
  17,500 × (1.08)^(90/365) = 17,835.26 ; continuous
17,500 × e^(0.08 × 90/365) = 17,848.63
Futures above 17,848.63 → buy index stocks, sell index
futures
```

Costs, taxes, margins ignored in fair price

Pricing with known dividends ★★★★★

```
F = (S - PV of dividends) × e^(rT)
S ₹200, T 0.5 yr, dividend ₹5 at 0.25 yr, r 6%
PV = 5 × e^(-0.06 × 0.25) = 5 × e^(-0.015) ≈ 4.93
F = (200 - 4.93) × e^(0.03) = 195.07 × e^0.03 ≈ 201.01
```

Assumptions of the cost of carry model ★★★★☆

- Underlying abundant in cash market
- Demand-supply not seasonal
- Easy to hold/store
- Can be sold short
- No transaction costs
- No taxes
- No margins

Fails for seasonal commodities (prices fall when new crop arrives), non-storable assets, assets that can't be shorted

Ignores margins → better suited to **forwards** than futures

Can be adjusted for costs and taxes

Convenience yield ★★★★☆

- Intangible inflow — comfort of holding the asset (e.g. hoarding grains, heating oil during floods)
- Subjective, differs by person and over time, hard to price
- If convenience yield dominates cost of carry → futures at **discount** to spot; reverse arbitrage impossible (no one lends the asset) → model breaks down
- Arises with scarcity / low inventory — **consumption assets (commodities)**, not financial assets

Expectations model ★★★★★

- Relationship of **expected spot** and futures prices moves market (esp. assets that can't be shorted or stored)
- Futures price = expected future spot price; futures may trade at premium or discount
- Futures > spot → expected rise → '**contango** market'; futures < spot → expected fall → '**backwardation** market'

Price discovery & convergence ★★★★★

- May 2024 index futures at 22,308.70 on 14 May → market expects spot to settle there at close of last trading day (30 May 2024)
- All participants predict the spot at one point in time → price discovery
- Futures and spot **converge at maturity** — futures settle at underlying cash price

Hedgers, traders, arbitrageurs in futures ★★★★★

Participant	Futures use
Hedgers	Farmer sells futures to lock crop price; food processor buys; diversified investor sells index futures against correction
Traders (speculators)	Cheaper to create positions; e.g. buy stock futures before good results; leverage is a double-edged sword — riskier than the underlying
Arbitrageurs	Risk-free profit from mispricing; link markets

- Traders take naked positions and give hedgers a counterparty — market depth depends on them
- Farmer fearing wheat price fall sells 3-month futures; buyer may be flour mill/bakery or a trader with the opposite view
- Arbitrageurs lock in profits (unlike traders)

```
Spot ₹100, May futures ₹110, lot 50 → buy spot, sell
futures → lock ₹10/share
Expiry 108: cash +8, futures +2 = ₹10 × 50 = ₹500
Expiry 95: cash -5, futures +15 = ₹10 × 50 = ₹500
(ignores costs)
```

Arbitrage risks: time lag between legs; illiquid leg → only one leg executes → naked exposure; non-cash-settled markets need reversal

Traders & arbitrageurs bring liquidity → better price discovery, lower cost, less manipulation

Systematic vs unsystematic risk & beta ★★★★★

Risk	Examples	Managed by
Unsystematic (specific)	Factory strike, key manager leaves, strong competitor	Diversification; single stock futures hedge total risk of a stock
Systematic (market)	Monetary/fiscal policy change, currency depreciation, war	Index futures

- Total price risk = systematic + unsystematic
- Stock futures vs cash: difference is deferred settlement
- Beta = sensitivity to index (historical): β 2 → stock moved 20% when index moved 10%
- β > 1 = aggressive; β < 1 = conservative
- Beta info from newspapers, Bloomberg, Reuters

```
Portfolio beta = Σ weight × beta
Betas 0.5, 1.1, 1.3, 0.9 with weights 35%, 15%, 20%, 30%
→ β = 0.87
```

Hedge ratio ★★★★★

- Single stock futures: **1:1** hedge (e.g. 550 HDFC Bank shares = 1 futures contract of 550)
- Index futures for a portfolio: not 1:1 (different composition and beta)

```
Contracts = Vp × βp ÷ Vi    (Vi = futures index level × multiplier; futures beta vs spot index = 1)
Portfolio ₹50,00,000, β 1.3 → hedge value ₹65,00,000
Nifty futures 23,900 × 65 = ₹15,53,500 → 65,00,000 ÷ 15,53,500 = 4.18 lots
4 lots → under-hedged; 5 lots → slightly over-hedged → perfect hedge not always possible
```

Hedge ratio changes with the index used; assumes past beta continues

Long hedge, short hedge, cross hedge, hedge month ★★★★★

Term	Meaning
Long hedge	Go long futures to hedge a future purchase (expected inflow; fear of rising market); also lets you invest gradually and unwind futures
Short hedge	Short futures to protect a portfolio you'll sell soon (fear of fall)
Cross hedge	Use futures of a closely related asset (jet fuel hedged with crude/heating oil/gasoline); index futures on a portfolio is broadly a cross hedge
Hedge contract month	Choose contract expiring **just after** the unwind date (sell portfolio on 20 June → short June futures, not May)

Trading: naked & spread positions ★★★★☆

- Naked long when expecting rise (Sensex futures at 61,000, expect settlement higher); naked short when expecting fall; loss if wrong
- Spread = two opposite positions: same maturity, different products = **inter-commodity / inter-product** spread; same product, different maturities = **calendar / time / horizontal** spread
- Equity derivatives: only calendar spreads tradable; inter-commodity spreads (gold–silver, soybean complex) in commodity segment
- Calendar spread counted vs near month: short 3 one-month + long 2 two-month + long 3 three-month = 2 spreads (1st–2nd) + 1 spread (1st–3rd) + 2 naked three-month
- Spread becomes naked when near month expires or a leg is closed
- Spreads = conservatively speculative positions; **lower margin than two independent legs**

Arbitrage types & formulas ★★★★★

Type	Positions
Cash and carry	Long cash + short futures
Reverse cash and carry	Long futures + short cash
Inter-exchange	Same contract on two exchanges (Aug futures of Z at ₹101 NSE, ₹100 BSE → buy BSE, sell NSE; reverse when gap shrinks)

```
F = S + C   ·   F = S(1 + r)^T   ·   F = S × e^(rT)
Futures > fair → cash and carry; futures < fair → reverse cash and carry
Do it only if net inflow after costs, brokerage, margins
```

Futures = quick, cheap tools to add or reduce portfolio risk — risk management and restructuring tools

Workbook sample numericals ★★★★☆

```
Sold 1 XYZ futures at 278, lot 1,200, bought back at 265
→ profit (278 - 265) × 1,200 = +₹15,600
Short 1 June XYZ (multiplier 50) at 3,400; profit ₹10,000
→ closed by BUYING at 3,200
```

Margins in futures: paid by both buyer and seller

When near leg of a calendar spread expires, far leg becomes a regular open position (True)

16 Introduction to Options

≈13 marks · 21 subtopics · chapter priority ★★★★★

What an option is ★★★★★

- Futures = linear payoff, unlimited loss/profit; options let you ride upside while restricting downside
- Option = **right, not obligation**, to buy/sell the underlying on or before a stated date at a predetermined price
- Right bought for the **option premium**, paid upfront by buyer (holder, long) to seller (writer, short)
- Buyer exercises only when favourable; writer legally bound if exercised
- **Call** = right to buy; **Put** = right to sell

Option quote example & terminology ★★★★☆

Field (May 10, 20XX)	Nifty call	Nifty put
Expiry / type	25 May, European call	25 May, European put
Strike	18,400	18,400
Open / High / Low	125 / 130 / 93.65	193.55 / 236 / 171.50
Close (premium)	124.50	177.60
Volume / OI	61,330 / 14,97,650	31,388 / 4,97,950
Underlying	18,315.10	18,315.10

Term	Meaning
Index option / stock option	Underlying = index (Nifty, Sensex) / stock (ONGC, NTPC)
Buyer	Right, pays premium; right to buy (call) or sell (put)
Writer	Receives premium; obligation — sell if call exercised, buy if put exercised
American	Exercise any time on/before expiry
European	Exercise only on expiry; **all Indian index & stock options**
Premium	Per unit; × lot size for contract premium
Spot (S)	Underlying's cash price (18,315.10)
Strike / exercise price (X)	Price at which call buyer may buy / put buyer may sell
Open interest	Outstanding option contracts; SEBI uses **delta-adjusted open position** for options
Exercise	Indian options exercised only at maturity vs settlement/closing price
Assignment	Allocation of exercised options to one or more sellers

Opening & closing option positions ★★★★☆

- Opening purchase = create/increase long on option; opening sale = create/increase short on option
- Long on call = long the underlying; **long on put = short the underlying**
- Closing (squaring off) = offsetting trade in the **same** option (same strike and expiry) — buying a put does not close a long call

Option contract specifications ★★★★★

Spec	Rule
Lot size	Exchange sets; same as futures (e.g. Nifty 50 → call premium 124.50 × 50 = ₹6,225; put 177.60 × 50 = ₹8,880)
Stock options cycle	Same tenor as stock futures; NSE 3-month cycle (near, mid, far)
Index options	Wider range; Nifty 50 & Sensex up to **5 years**; weekly & monthly; one weekly benchmark index option per exchange (Tuesday or Thursday); 3 quarterly (Mar/Jun/Sep/Dec) after 3 serial months; long-term: 8 half-yearly (Jun–Dec cycle) after quarterlies
Weekly options now	NSE: Nifty 50; BSE: Sensex
Expiry rules	SEBI 26 May 2025 (same as futures: Tue/Thu; others last week of month)
Tick size	Set by exchange — see exchange specs
Final settlement	No daily settlement for options; exercise/assignment only on expiry; index = closing price of underlying; stocks = VWAP of closing prices across exchanges (CAS SOP)
Trading hours	9:15 am – 3:40 pm, Mon–Fri (as stated for options)

BSE Sensex options	Spec
Contract size	20
Tick	₹0.05
Cycle	7 serial weekly, 3 monthly, 3 quarterly, 8 semi-annual
Trading hours	9:15 am – 3:30 pm
Style	European
Expiry	Monthly/quarterly/semi-annual: last Thursday; weekly: Thursday
Strikes	Min **101** (50 ITM, 1 ATM, 50 OTM); interval 100 index points
Settlement	Cash, closing index value

> **TRAP** Workbook gives two trading-hour figures for derivatives (futures 9:15–3:30; options 9:15–3:40; Sensex futures 3:40, Sensex options 3:30). Normal market hours are 9:15 am – 3:30 pm.

Moneyness ★★★★★

	Call	Put
In-the-money (ITM)	Strike < Spot	Strike > Spot
At-the-money (ATM)	Strike = Spot (closest strike)	Strike = Spot
Out-of-the-money (OTM)	Strike > Spot	Strike < Spot

ITM: positive cash flow if exercised now; ATM: zero; OTM: negative

Index 18,415 with strikes 18,350/18,400/18,450 → 18,400 is ATM

Quote example: 18,400 put is ITM; 18,400 call is OTM

Intrinsic value & time value ★★★★★

```
Call IV = max(0, S - X)     Put IV = max(0, X - S)     IV never negative
Time value = Premium - Intrinsic value
Put 18,400 with spot 18,315.10: IV = 84.90 ; TV = 177.60 - 84.90 = 92.70
Call 18,400: IV = 0 ; TV = entire 124.50
```

Only ITM options have intrinsic value

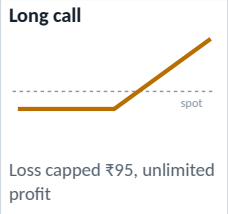
ATM & OTM = pure time value

Long & short options — general ★★★★☆

	Long (buyer)	Short (writer)
Right / obligation	Right to exercise	Obligation; may be assigned (any time if American)
Max loss	Premium paid	Theoretically unlimited
Max profit	Depends on underlying	Premium received
Margin	None	Must pay margin

Long call (worked) ★★★★★

```
Index 17,562; buy call 17,500 at ₹95
Expiry 17,400 → don't exercise → loss ₹95
Expiry 17,595 → payoff 95 - premium 95 = 0 → BEP = X + P
= 17,595
Expiry 18,000 → payoff 500 → profit 405
Lot 50: contract value ₹8,75,000 ; max loss 95 × 50 =
₹4,750
```

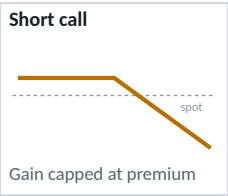


Index options cash-settled: ITM holder receives (closing index − strike)

Call buyer pays no margin

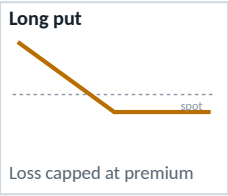
Short call (worked) ★★★★★

- Mirror image of long call; max gain = premium ₹95; loss unlimited as index rises
- Index 17,000 → +95; 17,700 → −105
- BEP = X + P = 17,595 for short call too — **BEP depends on the instrument, not the position**
- Writer pays margin (obligation, potentially unlimited loss — risk to system)



Long put (worked) ★★★★★

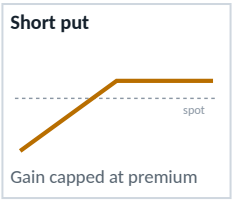
```
Buy put 17,500 at ₹150
Expiry ≥ 17,500 → lapse → loss ₹150
BEP = X - P = 17,350
Max profit when index = 0 → 17,500 - 150 = ₹17,350
Expiry 17,100 → payoff 400 → profit 250
Lot 50: max loss ₹7,500 ; contract value ₹8,75,000
```



Put buyer pays no margin — no obligation

Short put (worked) ★★★★★

- Opposite of long put; max gain = premium ₹150; losses below BEP 17,350; max loss when index hits zero
- Expiry 17,100 → −250; 17,400 → +50; ≥ 17,500 → +150
- Writer pays margin (obligation to buy, losses can be huge)



	Risk	Return
Long option	Premium paid	Unlimited
Short option	Unlimited	Premium received

Options vs futures; leverage ★★★★★

- Options: **asymmetric** payoff; futures: **symmetric**
- Leverage: small premium vs contract value (₹4,750 or ₹7,500 vs ₹8,75,000) → large % gains from small moves
- Leverage magnifies % loss too; worthless expiry loses whole premium
- Both used for trading, hedging, arbitrage

Futures	Options
Buyer and seller both obliged	Buyer has right, no obligation; seller has obligation

Futures vs options — full comparison ★★★★★

Futures	Options
Buyer and seller both obliged	Buyer has right, no obligation; seller obliged if exercised
Both pay initial margin	Buyer pays premium to seller; **seller deposits initial margin**
Both: unlimited gains or losses	Buyer: unlimited gain, loss limited to premium; seller: unlimited loss, gain limited to premium
Both pay/receive daily MTM	**Only option seller** subject to daily MTM margins

Five factors that determine option price ★★★★★

- Prices are never fixed by exchanges or SEBI — price discovery by market
- Unexpired option always has time value; intrinsic value only if ITM
- Time value depends on time to expiry and volatility

Factor	Change	Call	Put
Spot price	↑	↑	↓
Spot price	↓	↓	↑
Strike price	↑	↓	↑
Strike price	↓	↑	↓
Volatility	↑	↑	↑
Volatility	↓	↓	↓
Time to expiry	Longer	↑	↑
Time to expiry	Shorter	↓	↓
Interest rates	↑	↑	↓
Interest rates	↓	↓	↑

Index 17,562: 17,600 call > 17,700 call; 17,700 put > 17,600 put

Volatility = magnitude of moves either way — same effect on calls and puts

3-month option > 1-month (same strike)

Time decay — options are 'wasting assets'; sellers have a structural advantage

Interest rates matter more for stock/index options than options on futures

Option Greeks ★★★★★

Greek	Formula	Key points
Delta (δ, Δ)	Δ premium ÷ unit Δ underlying	Most important; long call +, short call −; long put −, short put +; 0.60 → ₹1 move = 60 paise; used in margining; = **hedge ratio**: n shares ÷ delta = calls to write for a **delta-neutral** hedge
Gamma (γ)	Δ delta ÷ unit Δ underlying	Second derivative; 'acceleration' of delta; positive; call δ 0.50, γ 0.08 → +₹1 → δ 0.58
Theta (θ)	Δ premium ÷ Δ time to expiry	Time decay per day; call with 5 days, θ 1.2 → loses ₹1.20/day; usually negative for long call & put
Vega (v)	Δ premium ÷ Δ volatility	Per 1% change in implied volatility; vega 0.80 → 0.80% per 1% IV change; positive for long call & long put
Rho (ρ)	Δ premium ÷ Δ cost of funding	Per 1 percentage point in risk-free rate; calls +, puts −; put ₹0.50, ρ −0.05, rates 5% → 4% → ₹0.55; call ₹0.25 → ₹0.20

```
ATM put δ -0.50, γ 0.004
Underlying +10 → Δδ = 0.004 × 10 = +0.04 → new δ -0.46 (put loses delta)
Underlying -10 → Δδ = -0.04 → new δ -0.54 (put gains delta)
```

Option pricing models ★★★★★

Model	Points
Binomial	William Sharpe, **1978**; binomial tree of up/down moves at equal time steps with probabilities; flexible, intuitive, very accurate (iterative) but complex and slow
Black & Scholes	Fisher Black & Myron Scholes, **1973**; popular, simple, fast — no iteration

B-S assumptions: European options only · lognormal prices (random walk / geometric Brownian motion, never negative) · constant known volatility · constant known risk-free rate · no dividends (adjustable) · no transaction costs or taxes · continuous trading, perfect liquidity · short selling with full use of proceeds · no arbitrage

```
C = S·N(d1) - X·e^(-rt)·N(d2)
P = X·e^(-rt)·N(-d2) - S·N(-d1)
d1 = [ln(S/X) + (r + v²/2)t] ÷ (v√t)     d2 = d1 - v√t
S stock, X strike, t years, r continuous risk-free rate,
v annual volatility, N(x) standard normal CDF
```

Implied volatility ★★★★★

- **Historical volatility**: SD of past % price changes (e.g. weekly Nifty changes over a year)
- Traders want expected volatility over the option's remaining life
- Pharma example: historical 15%, court verdict next week → expected 70–80%
- **Implied volatility** = volatility backed out by running Black-Scholes in reverse from the market premium (index 17,562, 17,500 call, 9 days, ₹95, rate 6%)
- High IV → traders sell options (premiums rich); low IV → buy options
- Writers demand higher premiums as safety margin for asymmetric payoff; if risk priced in, even adverse verdict may not hurt writers

Choosing ITM / ATM / OTM ★★★★☆

Deep ITM: high intrinsic value → highest premium

ATM/OTM: only time value → cheaper

ATM: highest uncertainty — small move tips it ITM or OTM

Call buyer & seller analysis (index 17,562) ★★★★★

Strike	17,300	17,400	17,500	17,600
Premium	327	250	185	130
BEP	17,627	17,650	17,685	17,730
Buyer P&L at 17,000	−327	−250	−185	−130
Buyer P&L at 17,800	+173	+150	+115	+70
Buyer ROI at 17,800	53%	60%	62%	54%
Seller P&L at 18,000	−373	−350	−315	−270

17,300 call: IV 262, TV 65; 17,600 call: all time value

Below 17,300 all calls worthless — biggest loss on 17,300

Call seller: neutral-to-bearish; gains premium below strike

Put buyer & seller analysis (index 17,562) ★★★★★

Strike	17,300	17,400	17,500	17,600
Premium	65	91	121	167
BEP	17,235	17,309	17,379	17,433
Buyer P&L at 17,000	+235	+309	+379	+433
Buyer ROI at 17,000	362%	340%	313%	259%
Seller P&L at 18,000	+65	+91	+121	+167

17,600 put ITM: IV 38, TV 129; others pure time value

Deep OTM put buyer gets highest ROI if index falls sharply

Put seller: neutral-to-bullish; selling deep ITM puts = higher premium but higher risk; deep OTM = less risk, less premium

Workbook sample answers ★★★★☆

```
Short put strike 245, premium 49 → max loss = 245 - 49 = ₹196
Stock 286, 260 call at 45 → intrinsic value = 26
Option buyer can't lose more than premium → true for ALL options
European call 'on or before' expiry → FALSE (only on expiry, at strike not market price)
ITM option = positive intrinsic value
```

17 Strategies using Equity Futures & Equity Options

≈9 marks · 26 subtopics · chapter priority ★★★★★

Long hedge with stock futures ★★★★★

Purpose	Lock price of a planned future purchase
Example	Buy 1,500 ABC shares on 20 Jun (FD matures); spot ₹455 on 10 May; June futures (expiry 29 Jun) ₹457.30, lot 1,500 → long 1 lot

Scenario on 20 Jun	Cash cost	Futures P&L	Net cost / share
Price ↑ to 520 (futures 521.20)	₹7,80,000	+₹95,850	₹6,84,150 → **₹456.10**
Price ↓ to 390 (futures 391.10)	₹5,85,000	−₹99,300	₹6,84,300 → **₹456.20**

Either way effective price ≈ initial futures price ₹457.30

Short hedge with stock futures ★★★★★

Purpose	Lock price of a planned future sale
Example	Sell 1,200 PQR shares on 10 Jul for loan; spot ₹1,690 (10 May), July futures ₹1,706, lot 600 → short 2 lots

Scenario on 10 Jul	Cash received	Futures P&L	Net per share
Price ↓ to 1,440 (futures 1,441)	₹17,28,000	+₹3,18,000	₹20,46,000 → **₹1,705**
Price ↑ to 1,800 (futures 1,802)	₹21,60,000	−₹1,15,200	₹20,44,800 → **₹1,704**

Hedging a portfolio with index futures ★★★★★

- Diversified 15-stock portfolio → specific risk ≈ 0 but market risk remains
- Options: sell entire portfolio and hold cash, or short index futures

```
Hedge ratio = (β × portfolio value) ÷ (futures price × lot)
= (1.3 × 90,00,000) ÷ (17,700 × 50) = 13.22 → short 13 contracts
```

Futures for trading: bullish & bearish ★★★★★

	Bullish: long XYZ futures	Bearish: short RSB Bank futures
View	Stock 1,298 → 1,350	Stock 1,595 → 1,520
Position	Futures 1,300, lot 850 → contract ₹11,05,000; margin 20% = ₹2,21,000	Futures 1,602, lot 550 → contract ₹8,81,100; margin 20% = ₹1,76,220
Outcome	Futures 1,346 → profit (1,346 − 1,300) × 850 = ₹39,100 → ROI **17.69%** vs buying shares: ₹39,950 on ₹11,03,300 = 3.62%	Futures 1,527 → (1,602 − 1,527) × 550 = ₹41,250 → ROI **23.41%**

Leverage explains popularity

Shorting futures expresses a bearish view without owning shares (cash sale needs shares in demat)

Cash-and-carry arbitrage (worked) ★★★★★

```
Stock A spot ₹1,500; 3-month futures ₹1,550; multiplier 100; carry 9% p.a. (0.75%/month)
Fair = 1,500 × e^(0.09 × 3/12) = 1,534.13 → futures overvalued by 15.87
Buy 100 shares, sell 1 futures → arbitrage profit = 100 × 15.87 = ₹1,587
Expiry 1,580: +8,000 - 3,000 = 5,000 - financing 3,413 = ₹1,587
Expiry 1,480: -2,000 + 7,000 = 5,000 - 3,413 = ₹1,587
Expiry 1,500: 0 + 5,000 - 3,413 = ₹1,587   (same in every case)
```

Reverse cash-and-carry arbitrage (worked) ★★★★★

```
Stock B spot ₹100; 1-month futures ₹90 (negative carry); multiplier 200
Buy 1 futures at 90, sell 200 shares at 100 → locked ₹2,000
Expiry 110: -2,000 + 4,000 = 2,000 ; Expiry 85: +3,000 - 1,000 = 2,000 ; Expiry 100: 0 + 2,000
Interest on ₹20,000 sale proceeds at 9% for 1 month = 20,000 × e^(0.09/12) - 20,000 = ₹150.56 → total ₹2,150.56
Early exit on 15th: spot 130, futures 135 → -6,000 + 9,000 = ₹3,000 + interest 15 days ₹74.11 = ₹3,074.11
```

Assumes arbitrageur owns the stock; else borrow stock and include borrowing cost

Positions can be squared off before expiry when favourable

Calendar spread arbitrage (futures) ★★★★★

```
Spot 120; near futures 121.30; mid futures 121.50; rate 8%
Fair near = 120 × e^(0.08 × 1/12) = 120.80 ; fair mid = 120 × e^(0.08 × 2/12) = 121.61 → fair gap ≈ 0.81
Actual gap 0.20 → near is overpriced vs mid → SHORT near at 121.30, LONG mid at 121.50
Case 1: near expires 122, mid 122.82 → -0.70 + 1.32 = +0.62
Case 2: near expires 120, mid 120.80 → +1.30 - 0.70 = +0.60
```

Buy underpriced, sell overpriced

Low risk, low return; no directional bet

Rare in practice — mispricing vanishes quickly; both legs must be executed and squared simultaneously

Option spreads: vertical, horizontal, diagonal ★★★★★

Spread	Construction
Vertical	Same expiry, different strikes (calls or puts); bullish or bearish
Horizontal (time / calendar)	Same type, same strike, different expiries; bet on time-value gap; no payoff chart possible
Diagonal	Different expiries and strikes; complex; better suited to OTC

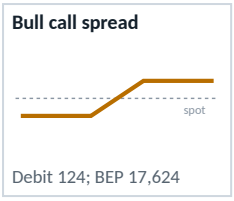
Spreads = same underlying, same type, different strikes/maturities

Limited profit, limited loss

Bull call spread (worked) ★★★★★

```
Long 17,500 call @ 185 + short 17,800 call @ 61 → net debit 124
Max loss 124 (≤ 17,500) ; max profit = 300 - 124 = 176 (≥ 17,800)
BEP = lower strike + net premium = 17,500 + 124 = 17,624
```

Bullish but wants lower cost; gains above short strike given up; higher short strike → bigger profit cap but smaller premium



Bull put spread (worked) ★★★★★

```
Short 17,500 put @ 125 + long 17,000 put @ 34 → net credit 91
Max profit 91 (≥ 17,500) ; max loss = 500 - 91 = 409 (≤ 17,000)
BEP = higher strike - net credit = 17,500 - 91 = 17,409
```

Bullish; long lower put = insurance against unlimited loss on short put

Net premium receipt strategy

Bear call spread (worked) ★★★★★

```
Short 17,500 call @ 185 + long 17,800 call @ 61 → net credit 124
Max profit 124 (≤ 17,500) ; max loss 176 (≥ 17,800) ; BEP 17,624
```

Bearish; long higher call caps the unlimited loss of the short call; starts with net inflow

Bear put spread (worked) ★★★★★

```
Long 17,500 put @ 125 + short 17,000 put @ 34 → net debit 91
Max profit = 500 - 91 = 409 (≤ 17,000) ; max loss 91 (≥ 17,500) ; BEP 17,409
```

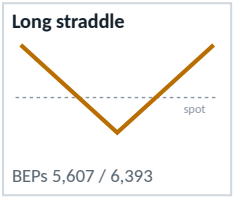


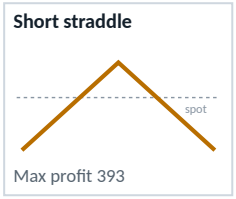
Long & short straddle (worked) ★★★★★

```
Stock 6,000; ATM call 257, put 136 → total 393
Long straddle: max loss 393 at expiry = 6,000
BEPs = strike ± total premium = 5,607 and 6,393
Expiry 5,300: put +564, call -257 → +307 ; expiry 6,700: call +443, put -136 → +307
Short straddle: max profit 393 at 6,000; losses beyond 5,607 / 6,393 — unlimited
```

Long straddle: market-neutral; expects big move either way (major announcements) — a volatility view

Short straddle: expects stability; limited profit, unlimited loss — use with care

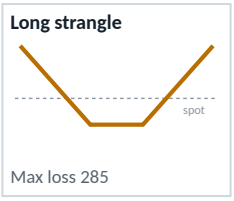




Long strangle (worked) ★★★★★

```
Stock 6,100; buy OTM 6,200 call @ 145 + OTM 6,000 put @ 140 → total 285 (max loss, between strikes)
Expiry 5,700: put +160, call -145 → +15 ; expiry 6,800: call +455, put -140 → +315
BEPs = 6,000 - 285 = 5,715 and 6,200 + 285 = 6,485
```

Same outlook as straddle but different strikes, both OTM → cheaper, needs bigger move



Short strangle ★★★★☆

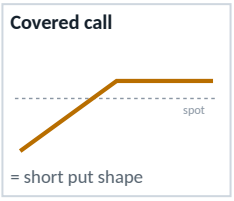
```
Sell OTM 6,200 call @ 145 + OTM 6,000 put @ 140 → max profit 285 between 6,000 and 6,200
BEPs 5,715 and 6,485; loss unlimited beyond either
```

Outlook: price stays stable (like short straddle)

Covered call ★★★★★

```
Buy stock 1,590 + sell 1,600 call @ 10
Expiry 1,640: stock +50, call -30 → +20 (max profit = 1,600 - 1,590 + 10 = 20)
Expiry 1,520: stock -70, call +10 → -60 ; BEP 1,580
```

- Generates income on existing holdings, lowers acquisition cost
- Payoff = **synthetic short put** (capped upside, large downside)
- Arbitrage: if 1,600 put trades at ₹17 (≠ ₹20 implied) → covered call + buy put = ₹3 risk-free (before frictions)
- Strike choice: near spot → more premium but gains capped early; far → less premium, longer ride
- Sell calls at your target exit price; when reached, sell stock and cover the call



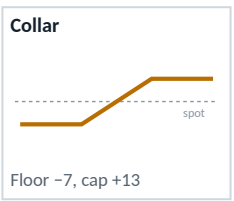
Collar ★★★★★

```
Long stock 1,590 + short 1,600 call @ 10 + long 1,580 put @ 7
Expiry 1,490: -100 + 10 + 83 = -7 (covered call alone: -90)
Expiry 1,690: +100 - 80 - 7 = +13 (covered call alone: +20)
Max loss 7, max profit 13
```

Extension of covered call: own stock + buy OTM put below price + sell OTM call above price

Call premium offsets put cost

Put lowers downside but also cuts max profit; BEP rises by put premium



Butterfly spread ★★★★★

```
Long call butterfly: buy 6,000 call @ 230, sell 2 × 6,100
calls @ 150, buy 6,200 call @ 100
Net cost = -230 + 150 + 150 - 100 = -30 (max loss, ≤
6,000 or ≥ 6,200)
Max profit at 6,100 = 100 - 30 = 70
Lower BEP = 6,000 + 30 = 6,030 ; upper BEP = 6,200 - 30 =
6,170
```

- Extension of **short straddle** — buy OTM call and OTM put to cap losses
- Long call butterfly: neutral, expects very low volatility; bull spread + bear spread; buy 1 ITM, sell 2 ATM, buy 1 OTM, strikes equidistant
- Risk = net premium paid; profit = adjacent strike gap − net debit
- With puts: buy highest, sell 2 middle, buy lowest; with calls + puts: buy low call, sell middle call, buy high put, sell middle put — limited profit & loss

Strategy	Construction	View
Long put butterfly	1 long ITM put, 2 short ATM puts, 1 long OTM put	Neutral/stable
Short call butterfly	1 short ITM call, 2 long ATM calls, 1 short OTM call	Significant volatility
Short put butterfly	1 short ITM put, 2 long ATM puts, 1 short OTM put	Significant volatility
Iron butterfly	Short ATM call + short ATM put + long OTM call + long OTM put	Neutral/stable

Long call butterfly



Max profit 70 at 6,100

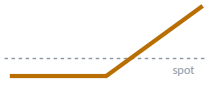
Hedging with options; protective put ★★★★★

- Buying a call = insurance against a surprise rally in a stock you plan to buy (stock 463, buy 460 call @ 19; if 520, call gain offsets; if 430, buy in market)
- Buying a put protects a holding against unexpected falls
- Fund manager fearing a fall: selling portfolio or shorting futures gives up upside → buy puts instead (insurance)

```
Protective put: buy stock 1,600 + buy 1,600 put @ 20
Expiry 1,530: -70 + 50 = -20 (loss capped at premium)
Expiry 1,660: +60 - 20 = +40 ; BEP 1,620
```

Payoff = **synthetic long call** (limited loss, unlimited gain)

Protective put



= long call shape

Put-call parity arbitrage ★★★★★

```
c + X·e^(-rt) = p + S0      (European options, same
strike and expiry)
Stock 1,251; 1,240 call (1 month) 47.50; r 8% → fair put
= 28.26
Put trades 23.15 (underpriced) → buy put + buy stock,
short call → outflow 1,226.65 borrowed at 8%
Repay 1,226.65 × e^(0.08/12) = 1,234.86
Expiry 1,275: 1,275 - 35 (call) - 1,234.86 = +5.14 ;
Expiry 1,200: 1,200 + 40 (put) - 1,234.86 = +5.14
Gain = fair - traded put price
```

Hard to execute: three simultaneous legs; one leg failing → naked exposure

Synthetic long futures = buy ATM call + sell ATM put (same expiry) → compare with actual futures

Synthetic short put = covered call → compare with actual put

Costs (transaction, impact, margin, borrowing) ignored in examples

Delta hedging ★★★★★

```
Stock 100; short 10 ATM calls, lot 50, delta 0.50
₹1 rise → loss 0.50 × 50 × 10 = ₹250
Futures delta ≈ 1 → long 5 futures lots (1 × 5 × 50 =
+250) vs calls -0.5 × 10 × 50 = -250 → delta-neutral
Stock → 110, call delta → 0.60 → need 6 lots
```

Delta-neutral only for small moves

Delta changes with price → keep adjusting = 'delta hedging'

Open interest & delta-adjusted OI ★★★★★

- OI = positions opened and not closed (May index futures OI 2,56,000)
- **New OI (SEBI 29 May 2025)**: portfolio-level net **delta-adjusted** open positions across futures and options — Future Equivalent OI (FutEq OI)

Position	Delta
Long futures	+1
Short futures	−1
Long call	0 to +1
Short call	0 to −1
Long put	0 to −1
Short put	0 to +1

```
Client 2 long 500 calls, δ 0.5 → FutEq OI 250 (notional
500)
Market-wide example: notional OI 2,800 → FutEq OI 900
```

Futures price	OI	Signal
↑	↑	Bullish — go long
↑	↓	Short covering
↓	↑	Short build-up, bearish — go short
↓	↓	Long positions squared up

Put-call ratio (PCR) ★★★★★

```
PCR = put volume (or OI) ÷ call volume (or OI)
Put OI 8,000, call OI 12,000 → PCR = 0.67
```

- Generally a **contrarian** indicator
- PCR < 1: more calls written → sellers don't expect rise → bearish signal; PCR > 1 (e.g. 1.25): bullish signal
- Very high PCR → oversold → possible rebound (consider buying calls); very low PCR → overbought → possible fall (consider buying puts)
- Limitations: weigh other factors; use with other indicators; read the chart correctly

Rollover ★★★★☆

```
Roll cost = (Next series price - Current series price) ÷
Current series price × 100
```

- Carry position to next expiry: close current, open next; expressed as % of positions; compared with trailing 3-month average
- Lower-than-average rollover → uncertainty; higher → strong sentiment
- May→June index rollover 70% vs 65% average → conviction — but check OI base
- High rollover + rising cost of carry (premium over spot) → bullish; high rollover + falling carry → bearish
- Price ↑ + higher rollover → strength continues; price ↓ + higher rollover → weakness continues; price ↑ + lower rollover → strength fading; price ↓ + lower rollover → weakness easing / bottoming

Volatility index (India VIX) ★★★★★

- Market's expectation of near-term volatility ('rate and magnitude of price changes')
- 'VIX' is a trademark of **CBOE**, first volatility index in **1993**
- India VIX by NSE from best bid-ask of near and next-month **NIFTY 50 options**; shows volatility over next **30 calendar days**
- Higher VIX = more expected volatility; options more valuable for buyers when volatility rises
- Trade volatility via straddles/strangles (expensive); no India VIX futures currently
- India VIX and NIFTY 50: **negative correlation**
- VIX peaks → managers raise high-beta exposure; low VIX → low-beta stocks (as stated in workbook)
- VIX, rollover, PCR, OI = sentiment indicators, no guarantee

Workbook sample answers ★★★☆☆

Buy lower-strike call + sell higher-strike call = bullish spread

Hedged position = protective put

Put-call parity: call & put on same stock, same strike, same maturity

Bullish: rising futures price + rising OI

18 Interest Rate, Interest Rate Instruments & Fixed Income Market

≈6 marks · 48 subtopics · chapter priority ★★★★☆

Interest rate concept: loan vs debt instrument ★★★★☆

```
A borrows X from B for 3 months, repays X + Δ ; Δ = X × R% × 3/12
```

- Signed agreement = **loan** (generally non-tradable); written 'note' = **debt instrument** (tradable) — A issuer, B investor, R yield
- Interest rate = charge for use of an asset; cost for borrower, return for lender; quoted annually = nominal annual rate
- 'Interest rate is subjective': varies with tenor, amount, credit risk (home loan 15 yrs 6%, personal loan 7%, home 20 yrs 6.5%, > ₹50 lakh 6.25%, 5-yr G-sec 5%, 5-yr AAA 5.50%)
- Low-risk borrower → lower rate; high-risk → higher

Macro & micro factors affecting interest rates ★★★★★

Macro factor	Effect
Demand for money	Strong economy → more demand → rates ↑; recession → rates ↓
Supply of money	Central bank tightens in inflation → policy rate ↑
Fiscal deficit & govt borrowing	High borrowing → higher rates, corporates crowded out
Inflation	Savers need compensation → nominal rates ↑; falls as inflation falls
Global rates & FX	Higher rates abroad → capital moves → domestic rates must rise
Central bank actions	Repo ↑ → money market rates ↑ → bond yields ↑

Micro factor	Effect
Maturity / tenor	Rates usually rise with tenor
Credit risk	Higher risk → higher rate
Seniority	Order of payment on default; junior → higher rate
Security	Unsecured pays more than secured
Liquidity	Easy-to-trade bonds command a premium (lower yield)
Investor sentiment	Negative mood → rush to G-secs

Effective rate; nominal vs real rate ★★★★★

```
Effective rate = (1 + annual rate ÷ n)^n - 1
6% compounded semi-annually: ₹1,000 → ₹30 + ₹30.90 = ₹60.90 → (1.03)² - 1 = 6.09%
(1 + r)(1 + i) = (1 + R)  ; approx r = R - i
Nominal 9%, inflation 6% → real ≈ 3% ; nominal 7%, inflation 9% → real ≈ -2%
```

Interest rate risk = uncertainty in rate movement — both directions hurt someone

Fixed income securities & key components ★★★★★

- Pay fixed interest (coupons) periodically, principal at maturity; bonds/debentures most common
- 'Fixed' because life is fixed (exception: Dutch perpetual bond of **1648**; Indian perpetuals have issuer call option) and cash flows mostly fixed (floaters: timing known, size not)

Component	Meaning
Issue price	Usually = face value for coupon bonds; below FV for zero-coupon
Face value (par, principal)	Coupon calculated on it; repaid at maturity (sometimes with premium)
Coupon / coupon rate	Cash flow at fixed dates; % of FV; usually stated annually
Coupon frequency	Monthly, quarterly, semi-annual, annual
Interest payment dates	Dates coupons are paid
Maturity date	Principal repaid; security ceases
Call / put date	Issuer / investor may redeem early
Redemption value	Above FV = premium; below = discount; equal = par

```
Example: FV ₹1,000 issued 1 Apr 2021 at ₹1,000, 10 yrs, 7% half-yearly (1 Apr & 1 Oct), maturity 1 Apr 2031, no call/put, redeemed ₹1,000
```

Bond vs loan; classification criteria ★★★☆☆

Bond is tradable, can change hands; loan usually not

Bond value changes continuously with rates and credit; loan value doesn't

Classification by: issuer · maturity · coupon · currency · embedded options · security · priority of claims · purpose · underlying · taxation

By issuer ★★★★★

Issuer type	Points
Government / sovereign / gilt-edged	Domestic currency; issued by auction, traded in secondary market; risk-free (can print money) — but defaults possible (European debt crisis 2008–2012); lowest rates; largest Indian segment incl. **State Development Loans (SDLs)** and special securities (oil, fertiliser, bank recapitalisation subsidies)
Municipal ('muni')	Local authorities fund infrastructure, libraries, parks; few in India; SEBI regulation for issuance & listing
Corporate bonds / NCDs	Depend on company's revenues; cheaper than equity, no dilution; may be collateralised; higher risk → higher rates; CPs for short-term/working capital; public issue or private placement; CRAs rate (AAA safest; 'junk'/'high-yield'); credit spread = corporate − govt yield
Securitised debt	Illiquid loans → SPV → tradable asset-backed securities

By maturity ★★★★☆

Type	Tenor
Overnight	Banks borrow from money market & RBI; collateralised cheaper than clean; RBI supplies/absorbs liquidity via banks & Primary Dealers
Ultra-short (money market)	Up to 1 yr: CP, CD, T-bills, Cash Management Bills (CMB)
Short term	1–5 yrs
Medium term (intermediate)	5–12 yrs — bulk of issuance
Long term	> 12 yrs (GoI bonds mostly)
Staggered / serial	Portions mature every few years — orderly retirement
Perpetual (consol)	No maturity; banks issue for Tier-I capital (Basel III)

No fixed rule for short/medium/long; MFs classify debt funds by **duration**, not maturity

Long-term bonds usually costlier (lock-in, illiquidity)

By coupon ★★★★★

Type	Feature
Plain vanilla / straight / bullet	Fixed coupon, defined maturity, issued & redeemed at FV
Zero-coupon (ZCB)	No interest; issued at discount, redeemed at FV; single cash flow; e.g. T-bills, CMBs, **STRIPS**; highly rate-sensitive; used by pension funds/insurers
Floating rate (FRB)	Coupon linked to benchmark (e.g. 182-day T-bill), reset each coupon date; trades near FV; largely immune to interest rate risk
Cap / floor / collar	Cap protects issuer; floor protects investor; both = collar
Inverse floater	Coupon moves opposite to benchmark (fixed coupon minus spread over MIBOR); NBFC issuers, MF investors
Inflation-indexed	Par (or par and coupons) indexed to WPI/CPI
Step-up / step-down	Low then high coupons (start-ups, risky) / high then low (declining revenues, e.g. leasing)
Deferred coupon	Mix of coupon bond and ZCB — nothing early, high coupons later (long gestation)
Deep discount	ZCB at big discount (≥ 20%, longer maturity); infrastructure cos; high risk

By embedded options ★★★★☆

Bond	Feature
Straight	Regular coupons, principal at maturity
Callable	Issuer may repay early on call date — used when market rates fall; usually pays a premium when called
Puttable	Investor may seek early redemption — used when rates rise; usually at a discount
Call & put	Both rights

Embedded options raise risk and are valued using option premia

By security & seniority ★★★★☆

Type	Feature
Secured	Backed by specific assets; paid first from collateral; higher recovery
Unsecured	No collateral; issuer's name
Credit-enhanced	Extra collateral, insurance, or third-party guarantee → better rating, lower cost
Senior	Paid before junior debt and shareholders; lower rate
Subordinated (junior)	Paid just before equity; higher coupon; banks issue for Tier II capital

Other instruments ★★★★☆

Instrument	Feature
Annuities	Regular payments over time; fixed income but not tradable; most consumer & housing loans structured as annuities
AT1 bonds	Basel III Additional Tier 1: unsecured, perpetual, non-convertible bonds meeting RBI conditions (Tier 1 = CET1 + AT1)
Tier 2 bonds	Subordinated, unsecured; min **5 years**; no put; call only after ≥ 5 yrs with RBI approval
Convertible bonds	Convert to equity at pre-fixed price; issuer can't refuse
REITs / InvITs	SEBI-registered trusts in commercial real estate / infrastructure
Green bonds	For climate & environmental projects; usually same rating as issuer's other debt
Tax-free bonds	Govt enterprises; interest exempt under Section 10
Tax-saving bonds	Tax benefit on investment, lock-in; interest taxable
Asset-linked bonds	Linked to asset performance; credit enhancement
ESG debt securities	SEBI framework 5 Jun 2025: green, social, sustainability, sustainability-linked bonds per international frameworks
Masala bonds	Rupee-denominated bonds issued outside India by Indian entities

Equity vs debt; pros & cons of debt ★★★★☆

	Equity	Debt
Represents	Ownership	Loan
Maturity	None	Yes
Returns	Variable (dividend, gains)	Predefined interest
Voting	Yes	No
Liquidation	Last	Before equity
Risk	Higher	Lower

Debt pros: no loss of control · lender priority · known obligations · interest tax-deductible (lowers WACC) · private placement simpler · no shareholder meetings/votes · committed lenders · cheaper long-run

Debt cons: covenants/exclusivity · must be repaid · refinancing/cash flow risk · leverage raises default risk · fixed regardless of conditions · poor planning hurts operations · high D/E = riskier · liens / sinking funds

Risk-free rate & risks in bonds ★★★★★

- Fixed income ≠ fixed return — timing (and often size) of cash flows known, return not guaranteed
- 7-yr 9% corporate bond: **credit risk** (may not pay) · **price / interest rate / market risk** (sell before maturity at unknown price) · **reinvestment risk** (coupons reinvested at unknown rates)
- Interest rate risk hits both sides: future borrower fears rise, future lender fears fall; banks/NBFCs on both sides
- Also liquidity, call, inflation risk
- Sovereign in home currency → no credit risk → **risk-free rate**; others pay a **credit spread** (risk-free 8.25% + AAA 0.10% = 8.35%; AA 0.25% → 8.50%)
- Risk-free rate = benchmark/opportunity cost for all valuations
- Market risk disappears if held to maturity; reinvestment risk disappears for zero-coupon
- **Sovereign, home-currency, zero-coupon, no call, held to maturity → no risk; fixed income = fixed return**

Rates	Price risk	Reinvestment risk
Rise	Prices fall (loss)	Reinvest at higher rate (gain)
Fall	Prices rise (gain)	Reinvest at lower rate (loss)

Term structure of interest rates ★★★★★

- Rate depends on **term** and on **borrower credit**
- Standard tenors — money market: ON, 1W, 2W, 1M to 1Y monthly; bond market: 2Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y, 40Y
- Sovereign rate quoted directly; others as credit spread by rating

Term	Risk-free	AAA spread	A spread	BBB spread
1M	5.00%	0.15%	0.25%	0.35%
3M	5.25%	0.25%	0.50%	0.75%
1Y	5.75%	0.40%	0.75%	1.10%
5Y	6.50%	0.85%	1.50%	2.25%

Rate vs term plotted = term structure = **yield curve** (risk-free curve, AAA curve, BBB curve…)

Risk-free term structure = most important valuation tool (ultimate opportunity cost)

What sets short-term & long-term rates ★★★★★

Segment	Main driver	Central bank tools
Short-term rates	Liquidity in the system	Repo / reverse repo (policy rates)
Long-term rates	Inflation outlook, capex demand	Bank rate, CRR, SLR, OMO

SLR forces banks to hold G-secs → distorts G-sec yields (captive demand)

Global rates matter through capital flows (uncovered interest parity)

Yield curve shapes ★★★★★

Shape	Meaning
Normal (upward)	Long rates > short rates — usual case
Inverted (downward)	Short rates > long rates — often signals slowdown
Flat	Short ≈ long rates
Humped	Medium rates highest

Yield curve shifts ★★★★★

Shift	Description
Steepening	Gap long – short widens; curve turns **anti-clockwise**
Flattening	Gap narrows; curve turns **clockwise**
Parallel	All rates move by the same amount
Twist	Short and long move in opposite directions
Convexity change	Curvature changes (middle moves vs ends)

Base SR 7%, LR 8% → new	Result
6.90 / 8.10	Steepening twist
7.10 / 7.90	Flattening twist
7.10 / 8.10	Parallel shift up

Simple vs compound interest; FV & PV ★★★★★

```
₹100 at 8% for 5 yrs: simple → 140 ; compound → 146.93
FV single period: ₹2,000 at 9% → 2,180
Multi-period (5 yrs): 2,000 × 1.09^5 = 3,077.25 (interest
1,077.25 vs simple 900)
₹1,00,000 at 7.3% for 4 yrs → 1,32,555.85 (annual) ;
1,33,215.41 (semi-annual)
FV = PV × (1 + r/m)^(m×n)     PV = FV ÷ (1 + r/m)^(m×n)
```

12% nominal compounded	Effective rate
Annually	12.00%
Semi-annually	12.36%
Quarterly	12.55%
Monthly	12.68%

More frequent compounding → higher effective rate

Day count conventions ★★★★★

Count first day, exclude last day

Convention	Used for (India)
Actual/Actual	Corporate bonds
30/360 (European)	**G-secs**
Actual/365	Money market instruments
Actual/360	Swaps

```
30/360: days = 360(Y2-Y1) + 30(M2-M1) + (D2-D1), day 31 →
30
31-Dec-2020 to 25-Jan-2021 = 25/360
```

Accrued interest; clean & dirty price ★★★★★

- **Clean price** = quoted / negotiated price
- **Dirty (invoice / full) price** = clean + accrued interest = settlement price
- AI grows from 0 after each coupon, drops at coupon → **saw-tooth** pattern
- AI change is deterministic; clean price change is stochastic
- Equities have no 'accrued dividend'

```
6.90% semi-annual bond, coupons 13 Jan & 13 Jul, trade 5
Mar
Actual/Actual: AI = 3.45 × 51/181 = 0.972099
30/360: AI = 3.45 × 52/180 = 0.996667
Clean 101.50 → dirty 101.50 + 0.9967 = 102.4967
```

Spot (zero) rate & holding period return ★★★★☆

```
₹100 → ₹150 in 3 yrs: annual 14.4714% ; semi-annual
13.9826% ; quarterly 13.7464% ; monthly 13.5919%
```

Spot/zero rate = yield on a single cash flow (zero-coupon)

HPR known only after the event (ex post)

Sources of bond return ★★★★★

Source	Point
Coupon	7% GS 2027 → ₹3.50 each half-year; 7.17% GS 2028 → ₹3.585 on 8 Jan & 8 Jul
Capital appreciation	Price rises when yields fall
Term premium	Extra yield for longer tenor
Reinvestment income	YTM assumes coupons reinvested at YTM

Price-yield relation; current yield ★★★★★

- Price and yield move **inversely**
- Coupon > yield → premium; coupon < yield → discount; coupon = yield → par

```
All yield 7.5%: 10% coupon → 117.37 ; 8% → 103.47 ; 6% →
89.58
Current yield = annual coupon ÷ clean price = 7.17 ÷
96.2290 = 7.45%
```

Current yield ignores capital gain/loss and time value

YTM, BEY, discount yield, effective yield ★★★★★

Measure	Formula / point
YTM	IRR of bond cash flows at current price; Excel YIELD(); assumes held to maturity and reinvestment at YTM
Bond equivalent yield (BEY)	(FV − P) ÷ P × 365 ÷ days ; T-bill 99.6898, 34 days → 3.34%
Discount yield	(FV − P) × 360 ÷ (FV × days) ; used for 91-day T-bill futures
Effective yield	(1 + r/m)^m − 1 ; 4.20% paid monthly → 4.28%

TRAP Doubling a semi-annual yield **UNDERSTATES** the effective annual yield

Bond pricing (worked) ★★★★★

```
10% coupon, 5 yrs, yield 8%, annual
Discount factors: 0.9259 / 0.8573 / 0.7938 / 0.7350 /
0.6806 ; PVIF(annuity) 3.9927
Price = 10 × 3.9927 + 100 × 0.6806 ≈ 107.99
Semi-annual: 5 × PVIF 8.1109 + 100 × DF → 108.11
P = Σ C/(1+y)^t + FV/(1+y)^n ; continuous: P = Σ
C·e^(-yt)
Under 1 yr: V = FV ÷ (1 + R × n) ; zero-coupon: V = FV ÷
(1 + R/m)^(m×t)
```

Valuing bonds between coupon dates (worked) ★★★★☆

- Dirty price = PV of the bond = clean price + accrued interest
- Step 1: value the bond at the previous coupon date (coupon, yield, residual maturity)
- Step 2: carry that value forward to the settlement date with a future value factor
- Step 3: deduct accrued interest → clean (trade) price

```
Settle 12-Aug-2020; maturity 11-May-2030; coupon 5.79%
semi-annual; yield 5.90%; 30/360E
Last coupon 11-May-2020 → 91 days = 0.252778 yr
AI = 5.79 × 91/360 = ₹1.4636
Dirty ₹100.6464 (sum of 20 discounted cash flows) → clean
= 100.6464 - 1.4636 = ₹99.1828
```

Spot rates vs YTM; YTM drawbacks ★★★★★

- Spot (zero) rate = true return: counts premium/discount, redemption gain/loss and reinvestment; read directly from zero-coupon prices
- Bond price = each cash flow discounted at its own zero rate
- Price set by demand-supply for **money** (via zero rates), not for the bond itself
- A coupon bond has many returns, one per cash flow (10Y semi-annual → 20); YTM averages them into one number
- YTM is just another way to quote price — **not a true return** and not a test of mispricing

```
3Y 8% bond, zero rates 7.75% / 8.00% / 8.25%
8/1.0775 + 8/1.08² + 108/1.0825³ = 7.4246 + 6.8587 +
85.1413 = 99.4246 → YTM ≈ 8.29%
Bond B (9% coupon, same issuer & maturity) → 101.9983,
YTM 8.2215%
```

Both bonds fairly priced on the same zero curve — mispricing only if market price differs from 99.4246 / 101.9983

Prefer B if you expect rates to rise (more to reinvest); prefer A if capital gains taxed lower than income

YTM drawback	Why
Assumes flat zero curve	Same rate for every cash flow
Coupon effect	Same maturity, different coupons → different YTMs
Inconsistent reinvestment rate	Assumes each bond's own YTM; actual rates unknown

Price-yield relationship rules ★★★★★

- Inverse; uses **clean** price; curve is non-linear (convex)
- Longer maturity → more price volatility (same coupon and price)
- Higher coupon → less volatility; lower coupon → more
- Higher YTM → less volatility than lower YTM (same coupon and maturity)

Bond selling at	Coupon vs yield	CR, CY, YTM
Par	CR = R → V = F	CR = CY = YTM
Discount	CR < R → V < F	CR < CY < YTM
Premium	CR > R → V > F	CR > CY > YTM

Required rate (10Y 9% bond)	5%	7%	9%	11%	13%
Value ₹	130.89	114.05	100.00	88.22	78.30

Each 1% rise cuts price by a smaller amount (8.81 → 4.75)

Price time path; pull to par ★★★★☆

Years left	Discount (11%)	Par (9%)	Premium (7%)
10	88.22	100	114.05
5	92.61	100	108.20
1	98.20	100	101.87
0	100	100	100

Pull to par: price moves to face value as maturity nears — premium bonds fall, discount bonds rise

Interest rate risk; price volatility properties ★★★★★

- Sovereign → no credit risk; held to maturity → no price/liquidity risk (pull to par); zero-coupon → no reinvestment risk
- Price risk and reinvestment risk act in opposite ways; price effect is instant, reinvestment effect slow
- Longer bonds more sensitive: 25 bp move → 2030 bond ≈ ₹1.6–2.1, 2060 bond ≈ ₹2.4–4.9
- 5.79% GS 2030: +25 bp → −1.89%; −25 bp → +1.93%; SD of price changes for 50 bp shocks ≈ 4× that for 25 bp

Property (option-free bond)	Point
1	% price change differs across bonds with different coupon, maturity, yield (8.50% GS 2030 falls 1.76% vs 1.89% for 5.79% GS 2030)
2	Very small yield change (1–2 bp): % change ≈ same up or down
3	Large change (100–200 bp): up and down moves differ
4	Price gain when yields fall > price loss when yields rise (5.79% GS 2030: +7.88% vs −7.18% for 100 bp) — due to **convexity**

Macaulay duration ★★★★★

- Time-weighted average of PV of cash flows; weighted average maturity / payback period; in **years**
- Zero-coupon: duration = maturity; coupon bond: duration < maturity
- Falls as time passes; does not rise endlessly with maturity — it levels off
- 5.77% GS 2030 at 5.85% → 7.50 yrs; at 6.50% → 7.43 yrs

```
Mac D = Σ [PV(CFt) × t] ÷ market price
Spreadsheet: 5.79% GS 2030 on 07-Jul-20 → 761.4442 ÷
100.5953 = 7.5694 yrs
```

Bond (5.77%)	Yield	Duration
2030	5.85%	7.50
2060	6.50%	14.67
2090	6.80%	14.93

Factor ↑	Duration
Maturity	↑
Coupon	↓
Yield (YTM)	↓

Portfolio duration ★★★★☆

```
D portfolio = Σ wi × Di
30% in 3-yr + 70% in 5-yr → 0.3×3 + 0.7×5 = 4.40 yrs
```

Modified duration ★★★★★

```
MD = Mac D ÷ (1 + YTM/n)     n = coupons per year
Mac D 5.34, yield 4.5%: annual → 5.11 ; semi-annual →
5.34/1.0225 = 5.22
% ΔPrice ≈ -MD × ΔYield
```

MD 5.11 → 100 bp yield move ≈ 5.11% price move, opposite direction

Applies to the **dirty** price

Linear approximation — good for small moves, poor for large ones; convexity fixes this

PV01 (price value of a basis point) ★★★★★

```
PV01 = Dirty price × MD ÷ 10,000
Price 100 → 99.95 for 1 bp → PV01 = ₹0.05 (5 paise)
```

Convexity ★★★★★

- Duration = first derivative (tangent line); convexity = second derivative — curvature of price-yield curve
- MD **overestimates** the fall when yields rise and **underestimates** the gain when yields fall; error grows with shock size (+2% shock: actual −11.85 vs MD −13.22)

```
Approx convexity = (P+ + P- - 2P0) ÷ [2 × P0 × (Δy)²]
Convexity adjustment = Convexity × 100 × (Δy)²
3-yr 10% semi-annual bond at 9% → convexity 8.2135 ;
divide semi-annual convexity by 4 to annualise
% ΔP = -MD × Δy + Convexity × 100 × (Δy)²
MD 10.66, convexity 81.96, yield 5% → 3%: +21.32% + 3.28%
= +24.60%
Yield 5% → 7%: -21.32% + 3.28% = -18.04%
```

Factor ↑	Convexity
Coupon	↓
Yield	↓
Maturity	↑

Portfolio convexity = weighted sum

Positive convexity → gain on fall > loss on rise

Role of the debt market ★★★★☆

- Super asset classes: underlying, derivative, structured finance (+ collective schemes, physical, alternative assets)
- Underlying classes: money, bond, equity, forex — independent, 'traditional'
- Money + bond = fixed income market; debt + equity = capital market (money market = short-term < 1 yr)
- Underlying markets → **financing**; derivatives → **risk management**; structured finance → both
- Debt = charge on income; equity return = appropriation of profit; debt avoids dilution but too much → bankruptcy risk

Importance: govt raises debt cheaply · liquid market lowers cost for all, better pricing · less reliance on banks (risk spread) · better bank ALM · channels MFs and retail · lets pension funds/insurers match liabilities

Primary vs secondary debt market; participants ★★★★☆

- Primary: govt issues by **auction**; corporates mostly by **private placement**
- Secondary: exit route + price discovery (credit appetite, spreads, default probability)
- Issuance needs underwriting, rating, listing, issue managers, legal and bankruptcy framework; SMEs prefer bank loans (regulatory cost)

Role	Who
Issuers	Govts, banks, PSUs, private companies
Intermediaries	Investment / merchant banks, brokers
Investors	Corporate treasuries, MFs, insurers, banks, pension funds, HNIs; domestic & foreign
Regulators	**RBI** and **SEBI**

Segments: G-secs (GoI dated, T-bills; states' SDLs) · PSU & bank bonds · private corporate debt

GoI also issues FRBs, inflation-indexed bonds, special securities, CMBs; states issue UDAY bonds

Banks issue CDs + long bonds; corporates issue bonds, debentures, CPs, FRNs, ZCBs

G-sec primary market: auctions ★★★★★

- RBI issues on behalf of govt — aim: fund fiscal deficit at low cost
- Centre: T-bills + dated securities; states: dated securities only
- Yield-based or price-based auctions, generally **weekly**; min bid **₹10,000** and multiples

Auction / bid	Rule
Uniform price	All winners pay the cut-off rate
Multiple price	Each winner pays own bid
Competitive	Institutions (banks, PDs, insurers, FIs, MFs) bid price/yield; multiple bids allowed; allotted if within cut-off
Non-competitive (NCB)	Retail without current/SGL account; via aggregators (banks, PDs, exchanges) or **RBI Retail Direct**; no price quoted; allotted at **weighted average** of accepted competitive bids

G-sec secondary market ★★★★★

Venue	Points
NDS-OM	Anonymous screen-based order matching, **Aug 2005**; run by Clearcorp Dealing Systems (CCIL subsidiary) for RBI; direct for banks/PDs, indirect via custodians / primary members; retail via primary members who are DPs
OTC / telephone	Via bank / PD / FI or SEBI broker; report on NDS-OM within **15 minutes**
NDS-OM-Web	**29 Jun 2012**; gilt account holders see live quotes, primary members act as risk controllers
Stock exchanges	Retail platform; demat settlement
RBI Retail Direct	Retail Direct Gilt (RDG) account with RBI; rbiretaildirect.org.in

- RBI 7 Feb 2025: SEBI-registered non-bank brokers may access NDS-OM ('Stock Broker Connect' for demat clients)
- SEBI 11 Feb 2025: via a **Separate Business Unit (SBU)** — ring-fenced, NDS-OM only, separate accounts, separate net worth; exchange grievance mechanism, IPF and SCORES **not** available to SBU clients

G-sec settlement & market operations ★★★★★

Item	Rule
Settlement	Net basis — **DvP-III**; CCIL is CCP by **novation**; outright trades **T+1** (FPIs OTC: T+1 or T+2, reported on trade date)
Switch / conversion auction	Sell source security, buy destination from GoI; bids on **e-Kuber**; source price = previous day FBIL close
OMO	RBI sells G-secs to absorb excess liquidity, buys to inject — durable liquidity tool
Buyback	Govt redeems early: cut cost (high coupons), fewer securities, liquidity, cash management; auction (large) or NDS-OM (small)
Securities lending (GSL)	RBI Directions Dec 2023; lend G-secs for a fee against other G-secs; borrowers = short-sale eligible; lenders = repo eligible

Holding G-secs ★★★★☆

- RBI **Public Debt Office (PDO)** = registry and central depository
- RBI-regulated entities must hold G-secs in demat (SGL) form from **20 May 2002**
- Physical stock certificate: transfer by transfer form registered with PDO, not by endorsement

Demat route	Who
SGL account with RBI	Banks, PDs, select UCBs & NBFCs
Gilt account	With bank/PD holding a **CSGL (SGL II)** account at RBI
Depository demat (NSDL/CDSL)	Enables trading on exchanges

Corporate bond primary market ★★★★★

- Mostly private placement to institutions; small share by public issue
- Public issue: prospectus; must list; offer document **not** approved by SEBI — in-principle approval from exchange; Companies Act 2013 + SEBI NCS Regulations 2021
- Private placement: Private Placement Memorandum (PPM)
- Face value of privately placed listed debt / NCRPS: **₹1 lakh** (reduced from ₹10 lakh, Oct 2022); ₹10,000 allowed with conditions

EBP rule	Detail
Introduced	SEBI 21 Apr 2016; revised 10 Oct 2022 & 16 May 2025
Mandatory	Single issue ≥ **₹20 crore** (incl. green shoe); shelf ≥ ₹20 crore in a FY; subsequent issue taking FY total ≥ ₹20 crore
Optional	Securitised debt, security receipts, CPs, CDs, REIT/SM REIT/InvIT units, issues < ₹20 crore
Documents	PM + term sheet ≥ 2 working days before (first-time issuer: 3); bidding announcement ≥ 1 working day before
Disclose	Open/closed bidding · uniform/multiple yield allotment · settlement via clearing corp or escrow · T+1 or T+2
Anchor portion	Up to 30% / 40% / 50% of base issue, by rating
Cut-off	Estimated cut-off yield to EBP ≥ 1 hour before bidding
Participants	Issuers, arrangers (merchant bankers, brokers, PDs), QIBs, non-QIBs, custodians (for FPIs)

Corporate bond secondary market ★★★★★

- Mainly OTC, institutional, negotiated; reported to exchange platform; settled DvP-I via clearing corporation
- Every trade reported within **15 minutes**; settlement T+0 to T+2
- **RFQ platform** launched early 2020 — pre-trade transparency, audit trail; exchanges also run order matching for bonds
- RFQ cash-flow dates not adjusted for day count — use scheduled due date

SEBI step	Detail
19 Oct 2022	Debt-segment brokers may bid on RFQ for clients
28 Oct 2022	Face value ₹10 lakh → ₹1 lakh
14 Nov 2022	Online Bond Platform orders must go through exchange RFQ, settled by clearing corporations
9 Jan 2023	RBI-authorised payment mechanisms allowed besides RTGS
2 Aug 2023	From 1 Apr 2024 brokers do ≥ **25%** of proprietary CB trades (value, monthly) via RFQ (OTO or OTM)
7 Aug 2023	FPIs ≥ **10%** of CB secondary trades via RFQ, quarterly

Similar RFQ mandates for MFs, AIFs, PMS

OBPP = online platform (not exchange, not EBP) offering listed debt — gives retail access

Money market instruments ★★★★★

Instrument	Key facts
Call money	Unsecured, overnight; only SCBs + PDs; NDS-CALL (CCIL)
Notice money	Unsecured, 2–14 days; NDS-CALL
Term money	Unsecured, 15 days – 1 yr; NDS-CALL
Market repo	Ready forward: sell security + repurchase at higher price (interest); reverse repo = collateralised lending; G-sec repo on **CROMS**, settled by CCIL
Triparty repo (TREPS)	Triparty Agent handles collateral, settlement, custody; CBLO (20 Jan 2003) → TREP **5 Nov 2018**; exempt from CRR/SLR; seller may substitute security
T-bills	Zero-coupon, discount; 91/182/364 days; weekly auction **Wednesday**
CMBs	< 91 days; temporary govt cash mismatches
Commercial paper	Corporates/NBFCs/AIFIs; ₹5 lakh multiples; 7 days – 1 yr; demat; min rating **A3**; IPA appointed; others need net worth ≥ ₹100 crore; bank facilities must be standard assets
Certificate of deposit	Banks; usance promissory note; ₹5 lakh multiples; 7 days – 1 yr; AIFIs 1–3 yrs
Repo in corporate bonds	RBI 2010; listed bonds (not own/related), CPs, CDs, debt ETF units
Tri-party repo on corporate debt	On exchanges; **AMC Repo Clearing Ltd (ARCL)** = LPCC and tri-party agent

Call money market, MIBOR & repo utility ★★★★☆

- Call money = second most liquid asset after cash; day-end liquidity, no collateral, same-day settlement
- Pure interbank (incl. PDs) since **Aug 2005**
- **Overnight MIBOR**: administered by **FBIL**, CCIL calculation agent; volume-weighted average of NDS-CALL trades in the **first hour**; RBI 'significant benchmark'
- Repo market funds positions and lets traders short G-secs (borrow security to deliver); short sale allowed up to **90 days**

Role of regulators ★★★★☆

Regulator	Scope
RBI	Govt borrowing (Centre, states, UTs); money market and G-sec market; bank instruments; laws: RBI Act 1934, Govt Securities Act 2006, PSS Act 2007, FEMA 1999, Banking Regulation Act
SEBI	Corporate bonds incl. bank/PSU issues > 1 yr; risk disclosures; FPIs, MFs; debenture trustees, CRAs, merchant bankers

Credit rating agencies ★★★★★

- Sovereign home-currency debt = credit risk free; others priced on issuer creditworthiness
- CRAs regulated by SEBI — **SEBI (CRA) Regulations 1999**; at least one rating mandatory for debt issue
- Rating = relative probability of default; used to set credit spread; agencies ask investors to use other information too; bonds often rated by more than one CRA

Rating	Meaning
AAA	Highest safety
AA	High safety
A	Adequate safety
BBB	Moderate safety
BB	Moderate risk
B	High risk
C	Very high risk
D	Default

AAA to BBB− = investment grade · BB+ to CCC− = junk · D = default · short-term scale A1+ to D

'+' / '−' show position within category

Workbook sample answers ★★★☆☆

BBB has higher credit risk than A or AAA

Credit spread = price of credit risk

LR 10%, SR 8% → normal/positive curve

Accrued interest applies to coupon bonds

Higher coupon → modified duration decreases

19 Interest Rate Derivatives

≈2 marks · 14 subtopics · chapter priority ★★★☆☆

Derivative definition & accounting criteria ★★★★☆

- Derivative derived from an independent underlying; cannot exist without it (wheat farmer selling harvest forward)
- Accounting: **FAS 133** (US), **IAS 39** (EU), **AS 30** (India)

Criterion	Meaning
1. Value linked to underlying	Price set by underlying, not by derivative demand-supply (raw material vs finished product)
2. Settles on a 'future' date	Later than underlying settlement (underlying T+2 → derivative after T+2)
3. No full cash outlay on trade date	Gives leverage
4. Net settlement (FAS 133 only)	Settles or can settle on net, not gross, basis

5 asset classes: rate, credit, equity, forex, commodity · 4 generic products: forward, futures, swap, option

Commodity derivatives alone for ~300 years; financial derivatives after 1970; ~two-thirds of transactions by 1990s

Legal definition (SCRA 1956) ★★★★☆

- Securities Laws (Second Amendment) Act **1999** added derivatives to 'securities'
- SCRA derivative: security derived from debt, share, loan, risk instrument or contract for differences · contract deriving value from prices/index of securities · commodity derivatives · others notified by Central Govt
- **Section 18A**: derivative contracts legal and valid if traded on a recognised exchange, settled on its clearing house per its rules, and between parties/terms notified by Central Govt

Approaches to risk ★★★★★

Approach	Meaning
Speculation (trading)	Taking risk — profit or loss possible
Hedging	Eliminates existing risk; locks future return; no explicit cost
Insurance	Removes downside, keeps upside; explicit upfront cost; needs **options**
Diversification	Reduces risk more than return — minimises risk per unit return

Farmer–miller forward: ₹10,000 for 100 kg wheat in 2 months; buyer long, seller short; agreed price = forward price, settlement date = expiration date

Each side cuts one risk and takes another (farmer loses upside; miller loses benefit of a fall); still counterparty / no-wheat risk

Economic functions of derivatives ★★★★★

Primary role: **risk management** | Hedging

Price discovery (futures/forwards show expected spot)

Market efficiency (replication keeps prices in line)

Access to unavailable assets/markets (e.g. swaps give cheaper rate)

Price stability (RBI uses derivatives for INR) | More volume in underlying

Speculation — calculated risk for knowledgeable traders

Shifts speculation from unorganised to organised market

Risk of derivatives	Meaning
Counterparty	Default by other side
Price	Loss from price move
Leverage	Magnifies gains and losses
Liquidity	Can't exit
Legal / regulatory	Enforceability
Operational	Fraud, poor documentation, bad execution

Read the broker's **Risk Disclosure Document**; not for those with limited resources, experience or risk tolerance

Forwards & FRA ★★★★★

- Forward: OTC, customised, both obliged; mostly delivery-settled; generally no margin/MTM
- **FRA**: exchange of interest on notional principal for a future period at agreed rate; cash-settled; borrower = long, lender = short
- Benchmark > FRA rate → long receives difference; payment discounted to settlement date

```
₹100 cr, borrow after 60 days for 91 days at 5%; 91-day T-bill at settlement 5.5%
Saving = 100 cr × 0.5% × 91/365 = ₹12.46 lakh (before discounting)
3×6 FRA: ₹10,00,000, FRA 5%, floating 6% → diff = 10,00,000 × 1% × 3/12 = ₹2,500
Settlement = 2,500 ÷ (1 + 6% × 3/12) = ₹2,463.05
```

Bond forwards (RBI 2025) ★★★★☆

- RBI (Forward Contracts in Government Securities) Directions, **21 Feb 2025**; OTC
- Buyer agrees to buy a specific **G-sec only** on a future date at a price fixed now
- Eligible: residents; non-residents eligible under FEM (Debt Instruments) Regs 2019
- Market-makers: banks, PDs etc.; users: non-retail users under Rupee IRD Directions 2019
- Physical or cash settlement; cash → bilateral or RBI-approved clearing; physical → **CCIL** or approved agency

Futures, options, caps, floors, collars ★★★★★

- Futures = exchange-traded forwards; mostly cash-settled; margins + MTM; clearing corporation guarantees
- **Interest Rate Futures (IRF)**: standardised; on notional security, interest-bearing instrument, index or rate; includes money market futures
- **Interest Rate Option (IRO)**: OTC and exchange-traded

Product	Structure
Cap	Series of interest rate **call** options (caplets); pays when rate > strike
Floor	Series of interest rate **put** options; pays when rate < strike
Collar	Buy cap + sell floor (same rate, maturity, notional)
Reverse collar	Buy floor + sell cap

Swaps & swaptions ★★★★★

- Swap = series of forwards; exchange cash flows by formula on notional principal; usually one fixed, one floating
- IRS: fixed-to-floating or floating-to-floating

```
A pays overnight MIBOR, receives 3.75% fixed; ₹100 cr; 32 days
A pays (daily compounded MIBOR) ₹31,58,169 ; receives 100 cr × 3.75% × 32/365 = ₹32,87,671
Net to A = +₹1,29,503 ; B = -₹1,29,503
```

Swaption	Right to enter swap as
Call swaption	**Receiver** — receive fixed, pay floating
Put swaption	**Payer** — pay fixed, receive floating

CDS: protection buyer pays periodic fee; seller pays loss on reference obligation if credit event occurs

Four generic derivatives compared ★★★★☆

Derivative	Feature	Market
Forward	Buy/sell underlying for future settlement; customised	OTC
Futures	Same, standardised	Exchange
Swap	Exchange returns of one asset for another / cash over a period	Mainly OTC
Option	Right to buy/sell for future settlement	OTC & exchange

Underlying	Forward	Futures	Swap	Option
Interest rate / instrument	FRA, bond forward	IR & bond futures	IRS	IR & bond options, swaptions
Equity	Equity forward	Equity futures	Equity swap	Equity option
Currency	FX forward	FX futures	FX swap	FX option
Commodity	Commodity forward	Commodity futures	Commodity swap	Commodity option

Bond swap = sell one bond, buy another — two cash trades, not really a derivative

Growth drivers; IRD market size ★★★★☆

Drivers: higher volatility · global market integration · technology · better risk tools · innovation (better risk-return, lower cost)

BIS data	Interest rate	Currency
ETD notional, Mar 2025 — futures / options (USD bn)	44,861 / 55,387	353 / 149
ETD daily turnover 2024 — futures / options	10,364 / 2,262	156 / 12
OTC notional 2024-S2 (USD bn)	5,48,341 (of 6,99,477 total)	1,30,093 FX

IRDs are the largest derivatives globally — every business faces interest rate risk; equity, FX, commodity risk only for some

Participants ★★★★☆

Type	Role
Hedgers	Reduce exposure — banks, MFs, insurers, corporates, home-loan borrowers, debt fund investors
Speculators / traders	Take views; derivatives offer leverage, lower cost, speed in size
Arbitrageurs	Exploit price gaps — cash vs derivative, OTC vs exchange, futures vs options

Participation rules from RBI, SEBI, IRDAI for banks, PDs, MFs, insurers, FPIs

Underlyings of IRDs ★★★★★

Feature	Interest rate	Notional bond	Single bond	Bond index
Underlying	MIBOR, repo, 91-day T-bill yield	Theoretical bond, fixed coupon & maturity (10-yr GoI)	Specific security (6.10% GOI 2031)	Index (8–13 yr GoI index)
Quote	Rate	Yield or price	Price	Index value
Settlement	Mainly cash	Cash or physical	Cash or physical	Cash
Settlement price	From benchmark rates	Basket of deliverable bonds	Spot price of bond	Index value

Rate derivatives: borrower = long the rate (expects rise); lender = short

Bond/index derivatives: buyer is **short the rate** (expects yields ↓, price ↑); seller is long the rate

OTC vs exchange-traded derivatives ★★★★★

	OTC	Exchange-traded
Negotiation	Private, bilateral	Screen-based order matching
Terms	Customised	Standardised lot & expiry
Counterparty / settlement risk	Yes	No — CCP by **novation**, margins & MTM
Transparency	Lower	Higher
Small sizes	Hard / costly	Possible

ETD limits: imperfect hedge (fixed size/date), cash settlement may not suit hedgers, daily MTM is operationally heavy

Gap closing: exchanges offer RFQ, Exchange-for-Physical; OTC has vanilla vs exotic, e-trading, CCP clearing — **CCIL** in India

Workbook sample answers ★★★☆☆

Role of derivatives = risk management

Largest market = interest rate derivatives

Hedgers reduce interest rate risk

FRA, IRS, IRF all are IRDs

Theoretical bond underlying = notional bond futures

20 Exchange Traded Interest Rate Futures

≈10 marks · 19 subtopics · chapter priority ★★★★★

Futures basics ★★★★☆

- Futures fix the limits of forwards: standardised, exchange-traded, clearing corporation guarantees
- Buyer = long, seller = short; 'buy/sell' figurative — no money or asset changes hands at trade
- Exchange sets every term except price

Two parties via exchange · centralised platform · price discovery · **margins from both sides** · expiry fixed · lot size fixed

History of IRF in India ★★★★★

Year	Attempt	Result
Jun 2003	91-day T-bill, 6% 10Y notional bond, zero-coupon 10Y notional bond futures	Failed, withdrawn
Aug 2009	Notional 7% 10Y GOI bond futures — **cheapest-to-deliver**, physical	Not adopted
Mar 2011	91-day T-bill futures; then 2Y and 5Y bond futures	Limited
Dec 2013	**Cash-settled single-bond futures on on-the-run 10-yr GOI** (SEBI + RBI)	Success
Jun 2015	6-year and 13-year GOI futures permitted	
10 Jan 2023	SEBI: futures on **corporate bond indices**	

Pre-Dec 2013 failures: cheapest-to-deliver, physical settlement, zero-coupon yield curve

IRDs took off only on the **third attempt, 2014**; OTC OIS very successful

Exchange derivatives more successful in equity & currency than rates

Futures terminology ★★★★★

Term	Meaning
Underlying	Rate (overnight MIBOR), G-sec, index (8–13 yr G-sec, corporate bond index) or notional security (10Y)
Quantity freeze	Max lots per single order — prevents fat-finger errors
Contract cycle	Monthly (near, mid, far), quarterly, some weekly; up to 1 yr; single-bond G-sec: 3 serial monthly + 3 quarterly (Mar/Jun/Sep/Dec)
Expiry	Last trading day; single-bond G-sec: **last Thursday**; holiday → previous trading day
Tick size	₹0.0025 for single-bond futures (99.9975, 100.0000, 100.0025)
Contract value	Price × lot size
Base price	Day 1 = theoretical futures price; later = daily settlement price
Price band	± % of base; where none, dummy operating range relaxed in direction of move
MTM	Daily vs settlement price; losers pay, gainers receive
Daily settlement price	VWAP of last **30 minutes**; else theoretical
Final settlement price	From underlying market; cash (P&L only) or physical delivery
Open interest	Contracts outstanding; count one side only (longs = shorts); shows depth

Positions & rollover ★★★★☆

- Long = outstanding buy; short = outstanding sell; open position = all unsettled longs/shorts
- Opening increases open position; closing reduces it (sell what you bought / buy what you sold)
- Example: short 5 bond futures + long 3 T-bill futures; next day sell 2 T-bill → short 5 + long 1
- Rollover = close current month + open next month; % of positions; compared with trailing 3-month average; low → uncertainty, high → strong sentiment

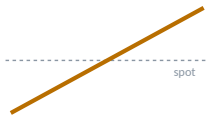
All open positions MTM'd · square off by opposite trade · trader alone responsible for margins · **zero-sum** · unlimited profit and loss

Futures payoffs ★★★★★

```
Long at ₹100: expiry 101 → +1 ; 99 → -1 ; 98 → -2 ; 102 → +2
Short at ₹100: expiry 99 → +1 ; 101 → -1
Cash-settled → only P&L paid/received
```

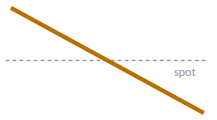
Linear, symmetrical payoff; X-axis = price at expiry (never below 0), Y-axis = P&L

Long futures



Gains as price rises

Short futures



Gains as price falls

Cash-settled GOI bond futures (6/10/13-yr) specs ★★★★★

Spec	Rule
Underlying	Single GOI dated security, residual maturity **4–8 / 8–11 / 11–15 years**; chosen by exchanges with FIMMDA (e.g. 6.33% GS 2035)
Lot	FV **₹2 lakh** = 2,000 bonds
Quote	Clean price per ₹100 FV
Contract value	Price × 2,000
Tick	₹0.0025
Price band	±3% of base; may widen by 0.5% after 30 minutes, max 2 times a day
Hours	Mon–Fri **9:00 am – 5:00 pm** (aligned with NDS-OM)
Cycle	3 serial monthly + 3 quarterly (Mar/Jun/Sep/Dec)
Expiry	Last Thursday (holiday → previous day)
Settlement	Cash
Final settlement price	Weighted average price on NDS-OM in **last 2 hours**; < 5 trades → FIMMDA/FBIL price
Daily settlement price	VWAP of last half hour across exchanges, or theoretical
MTM & final settlement	T+1

FBIL = independent benchmark administrator for interest rates and FX

91-day T-bill futures specs ★★★★★

Spec	Rule
Underlying	91-day GOI T-bill
Lot	2,000 units (FV ₹2 lakh)
Quote	**100 − futures discount yield** (5% → 95.0000)
Contract value	2,000 × (100 − 0.25 × y)
Tick	₹0.0025
Price band	±1% of base
Hours	9:00 am – 5:00 pm
Cycle	3 serial monthly + 3 quarterly
Expiry	**Last Wednesday, 1:00 pm** (holiday → previous working day)
Daily settlement	2,000 × (100 − 0.25 × Yw), Yw = weighted futures yield last half hour
Final settlement	2,000 × (100 − 0.25 × y), y = weighted average discount yield of RBI's 91-day T-bill auction on expiry day
Settlement	Cash; T+1

Overnight MIBOR futures specs ★★★★★

Spec	Rule
Underlying	Daily FBIL overnight MIBOR for contract month
Unit	Interest on ₹5 crore notional for one month, 30/365 basis, at average daily MIBOR
Quote	Interest rate
Base rate	Day 1: MIBOR OIS rate of matching tenor; later: daily settlement rate
Contract value	Rate × 100 × 411 (1 bp = 5 cr × 0.01% × 30/365 = **₹411**)
Tick	0.25 bp (0.0025) = **₹102.75**
Hours	9:00 am – 5:00 pm; expiry day 9:00 – 10:00 am
Cycle	3 serial monthly + 3 quarterly
Expiry	**Last working day** of month
Daily settlement	VWAP last 30 min (min 5 trades), else last 60 min (min 5 trades), else theoretical
Final settlement	Simple average of daily FBIL overnight MIBOR for the month (4 decimals)
Settlement	Cash in INR; T+1

Other / earlier IRF designs ★★★★☆

Contract	Key features
10Y notional 7% GOI — physical	Lot ₹2 lakh; 4 quarterly contracts; expiry **7th business day before last business day** of delivery month; deliverable GOI 7.5–15 yrs with ≥ ₹10,000 cr outstanding; conversion factor = price to yield 7%; last delivery = last business day
10Y notional GOI — cash	Basket of GOI with 8–11 yrs residual; coupon set by exchange; 3 serial monthly + quarterly; last Thursday; FSP = weighted average yield of basket (NDS-OM last 2 hrs; < 5 trades → FIMMDA/FBIL); 6-yr and 13-yr versions possible
2Y / 5Y notional 7% GOI	Semi-annual coupon, FV ₹100; lot ₹2 lakh; 3 serial monthly; last Thursday; cash; FSP from yields polled by **FIMMDA**; basket 1.5–2.5 yrs / 4.5–5.5 yrs

Corporate bond index futures (CBIF) ★★★★★

- SEBI **10 Jan 2023**: derivatives on indices of corporate debt rated **AA+ and above**

Index rule	Limit
Single issuer weight	≤ **15%**
Minimum issuers	**8**
Group weight	≤ 25% (excl. PSU, PFI, PSB)
Sector weight	≤ 25% (excl. PSU, PFI, PSB)
Review	At least half-yearly
Track record	≥ 1 year
Exposure	Aggregated at issuer level
Duration buckets	Set by exchanges

Contract rule	Detail
Contract value	≥ **₹2 lakh** at launch; reviewed half-yearly
Hours	9:00 am – 5:00 pm
Tenure	Up to **3 years**
Cycle	Weekly, 3 serial monthly, 1 quarterly or 1 half-yearly (Jun/Dec)
Quote / tick	INR; tick set by exchange
Expiry	Last Thursday of cycle
Daily settlement	Last half-hour VWAP, else theoretical (method on website)
Final settlement	Closing index value on expiry
Settlement	Cash; MTM and final next working day

Lot size, tick value & contract value ★★★★★

Contract	Contract value	Tick value / lot
GOI bond futures	2,000 × price; 5 lots at 101 → ₹10,10,000	2,000 × 0.0025 = **₹5**
91-day T-bill futures	2,000 × (100 − 0.25y) × lots; 10 lots at 95 (y = 5%) → ₹19,75,000	2,000 × 0.0025 × 0.25 = **₹1.25**
Overnight MIBOR futures	Rate × 100 × 411 × lots; 10 lots at 4.50% → ₹18,49,500	411 ÷ 4 = **₹102.75**

Bond futures FV always a multiple of ₹2 lakh (no ₹3 or ₹5 lakh)

Wholesale cash market lot ₹5 crore = 250 futures contracts

Market value = face value × price ÷ 100

Contract value drives margins, transaction and regulatory charges

```
Sample: 10 lots of bond futures at ₹99 → 10 × 2,000 × 99
= ₹19,80,000
```

Why exchange-traded IRDs in India ★★★★☆

- Banks, insurers, PDs, provident funds hold > **80%** of G-sec rate risk — need a credible 'true hedge'
- Households: savings + housing loans → rate risk; **Fisher effect** (expected inflation ↑ → nominal rate ↑) makes IRDs the main inflation hedge
- Policy aim: **market completion** — wider hedging tools
- Exchange benefits: near-zero counterparty/settlement risk, transparency, wider reach — complementary to OTC
- OIS is successful but not a true hedge for the stock of G-secs
- G-sec curve = risk-free proxy for time value of money; all assets priced at a spread over it

FRA vs interest rate futures ★★★★★

Parameter	FRA	IRF
Mechanism	Bilateral OTC (may use e-platform)	Exchange platform
Terms	Custom	Standardised
Underlying	Usually interest rate	Rate, instrument, index
Price discovery	Negotiation — not efficient	Free interaction on exchange
Liquidity	Low	High
Market maker	Scheduled banks, PDs, AIFIs	Currently not applicable
Settlement	Bilateral, usually cash	Clearing corporation, guaranteed; cash or physical
Information	Report to trade repository within **30 minutes**	Online nationwide

	Advantages	Limitations
FRA	Perfect hedge (custom); fewer margin/MTM issues	Liquidity risk, counterparty risk, not open to all
IRF	Transparency; no counterparty risk; open to all; credit agnostic; lower liquidity risk; lower impact cost	Imperfect hedge (fixed size/date); MTM and margin operations

FRA: borrower long the rate; benchmark > FRA rate → long receives; below → long pays

Bond/index futures buyer is long bond = **short the rate**

Forward rates ★★★★★

```
₹1,000 for 2 yrs: 2-yr spot 6% → 1,123.60 ; 1-yr spot 5%
then F
1,000 × 1.06² = 1,000 × 1.05 × (1 + F) → F = 7.0095%
F = [(1 + S1)^n1 ÷ (1 + S2)^n2]^(1/(n1 - n2)) - 1
6-m spot 5.60%, 1-yr spot 5.85% (semi-annual) → 6-m rate
6 m from now = 6.10%
```

No arbitrage: one long investment = rolled short investments

Cost of carry, income & basis ★★★★★

- Financing cost = interest to carry the asset to expiry
- Income on cash position = accrued interest to expiry + coupons received + interest on reinvested coupons
- **Basis = spot − futures**
- Futures > spot → **negative basis** (contango); futures < spot → **positive basis** (backwardation)
- Bond futures: basis usually **positive** — income on bond > financing cost, so futures trade at a discount

Bond futures pricing (worked) ★★★★★

```
Futures price = cash price + financing cost - income on
cash position
5.75% GS; last coupon 29 Apr; trade 3 Aug, settle 4 Aug
(T+1); clean 101.5000
AI (30/360, 29 Apr-4 Aug) = 1.5174 → dirty (A) = 103.0174
Financing 4% on dirty, 4-27 Aug, Actual/365 → (B) =
0.2597
Future dirty = 103.2771 ; AI 29 Apr-27 Aug (C) = 1.8847
Theoretical futures clean = A + B - C = 101.3924
```

Each participant has its own funding cost → many 'fair values' — that keeps the market trading

Assumptions: underlying plentiful · easy to hold · can be shorted · no transaction costs · no taxes · no margins

Not valid for non-storable or non-shortable assets

Expectancy model & convergence ★★★★★

- Futures price = **expected future spot price**; matters most when asset can't be stored or shorted
- Futures may trade at premium or discount; show expected direction
- Futures > spot bond price → market expects rates ↓ (prices ↑); futures < spot → expects rates ↑
- Oct-2021 6.10% GS 2031 futures at ₹98.50 in Sep → market expects spot ₹98.50 at close on last Thursday (28 Oct 2021)
- Futures and spot **converge at expiry** — all futures settle at underlying cash price

91-day T-bill futures pricing ★★★★☆

- Use forward rate: trade 1 Oct 2021, expiry 27 Oct 2021 → need 26-day and 117-day yields from FBIL T-bill curve (interpolate/extrapolate if missing)
- Forward formula gives the 91-day rate on 27 Oct
- All models theoretical, based on assumptions

Workbook sample answers ★★★☆☆

91-day T-bill futures = cash settled

Cash-settled 10-yr bond futures last trading day = last Thursday

Total contracts outstanding = open interest

Short at 100, expiry 101 → loss ₹1

10 lots at 99 → ₹19,80,000

21 Exchange Traded Interest Rate Options

≈6 marks · 15 subtopics · chapter priority ★★★★☆

Option basics (land example) ★★★★☆

- Futures = linear, unlimited P&L; options let buyer ride upside with loss capped
- Mr X pays Mr Y ₹50,000 non-refundable (1 Aug 2021) for the right to buy land at ₹10,00,000 in 3 months → **call option**
- Strike ₹10 lakh · expiry 31 Oct 2021 · time to maturity 3 months · premium ₹50,000 · X buyer, Y writer · land = underlying
- Land at ₹15 lakh on expiry → X exercises, pays ₹10 lakh
- Car insurance = buying a **put**; insurance premium = option premium

Similarities with futures: buyer & seller · set price · set settlement date; used for hedging, views, arbitrage

Difference: option buyer has right only (risk = premium); seller obliged (unlimited risk, gain capped at premium)

Option styles ★★★★★

Style	Exercise
European	Only on expiry (exchange options can still be sold before expiry)
American	Any time on or before expiry
Bermudan	On specified days, usually near expiry — between European and American

Moneyness; intrinsic & time value ★★★★★

	Call	Put
ITM (positive cash flow)	Strike < spot	Strike > spot
ATM (zero)	Strike = spot	Strike = spot
OTM (negative)	Strike > spot	Strike < spot

```
Call IV = max(0, S - X) ; put IV = max(0, X - S) ; never negative
Time value = premium - intrinsic value
```

Only ITM has intrinsic value

Longer to expiry → more time value (2-month call > 1-month)

At expiry option value = intrinsic value; time value = 0

Exercise decision may include transaction/regulatory charges

Five option price factors ★★★★★

Factor	Change	Call	Put
Spot	↑	↑	↓
Strike	↑	↓	↑
Volatility	↑	↑	↑
Time to expiry	Longer	↑	↑
Interest rate	↑	↑	↓

Prices never fixed by exchange or SEBI

Time decay → 'wasting assets'; sellers have a built-in advantage

Rate = risk-free rate

Option Greeks (bond examples) ★★★★★

Greek	Measures	Example
Delta	Δ premium per unit Δ underlying; long call/short put 0 to 1; long put/short call 0 to −1	6.10% GS 2031 at 98.80, 98.50 call ₹0.45, δ 0.55 → bond 99.00: +0.11 → **0.56**; bond 98.60: → **0.34**
Gamma	Δ delta per unit Δ underlying; 'acceleration'	ATM put δ −0.50, γ 0.004: +10 → −0.46; −10 → −0.54
Theta	Premium lost per day	₹2.75, θ −0.05 → ₹2.70 next day; long option θ negative, short positive — 'friendly Greek to the seller'
Vega	Δ premium per 1% Δ volatility; + for long calls and puts; biggest with more days left	Bond 99.50, Nov 100 call ₹0.10, vega 0.02: vol 10→11% → 0.12; 10→8% → 0.06
Rho	Δ premium per 1 pp Δ risk-free rate; calls +, puts −	Put ₹0.50, ρ −0.05, rate 5→4% → 0.55; call 0.25 → 0.20

Put-call parity ★★★★★

```
C + PV(X) = P + S
Holds only for European options with same underlying, strike and expiry
```

Option pricing models ★★★★★

Model	Points
Binomial	William Sharpe **1978**; up/down tree at equal steps; flexible, intuitive, accurate (iterative) but slow
Black-Scholes	Black & Scholes **1973**; fast, no iteration; European, no dividends; uses **spot**
Black (1976)	For options on futures/forwards; replaces S with F·e^(−rt) — solves negative carry; used for futures options, bond options, caps & floors, swaptions, commodity options; **used in India for exchange-traded IROs on G-secs**

```
B-S: C = S·N(d1) - X·e^(-rt)·N(d2) ; P = X·e^(-rt)·N(-d2)
- S·N(-d1)
d1 = [ln(S/X) + (r + v²/2)t] ÷ (v√t) ; d2 = d1 - v√t
Black: C = e^(-rt)[F·N(d1) - X·N(d2)] ; P = e^(-rt)
[X·N(-d2) - F·N(-d1)]
d1 = [ln(F/X) + (σ²/2)t] ÷ (σ√t) ; d2 = d1 - σ√t
```

Dividends cut stock price → calls worth less, puts more

Short-term options: S − PV(expected dividends); long-term: adjust with constant dividend yield

Implied volatility ★★★★★

Volatility	Meaning
Historical	From past closing prices; easy; used in calculators
Forecasted	Predicted with statistical models
Implied (σ)	Market consensus expectation, backed out from premium; only model input not directly observable

IV shows size of move, not direction

IV usually rises in bearish markets, falls in bullish ones

No traded options → no IV

IV ↑ after trade → good for buyer, bad for seller; IV ↓ → good for seller

Long & short call on G-sec (worked) ★★★★★

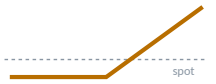
```
6.10% GS 2031 at 98.40; buy 98.50 call @ 0.20, expiry 28
Oct 2021
Expiry 98.25 → lapse, loss 0.20 ; BEP = X + P = 98.70 ;
98.90 → +0.40 - 0.20 = +0.20
Max loss per lot = 2,000 × 0.20 = ₹400
Short call: mirror — max gain 0.20, unlimited loss; BEP
also 98.70
```

Long call/short call: right vs obligation; seller can be assigned

BEP is instrument-specific, not position-specific

Cash-settled: buyer receives only the gain (0.40)

Long call



BEP 98.70

Short call



Gain capped 0.20

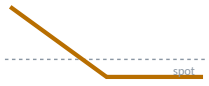
Long & short put on G-sec (worked) ★★★★★

```
Buy 98.50 put @ 0.30
Expiry 98.75 → lapse, loss 0.30 ; BEP = X - P = 98.20 ;
98.00 → +0.50 - 0.30 = +0.20
Max loss per lot = 2,000 × 0.30 = ₹600
Short put: max gain 0.30; max loss when price → 0; BEP
98.20
```

Long call = long underlying; **long put = short underlying**

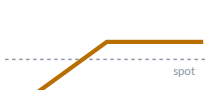
Options asymmetric; futures symmetric

Long put



BEP 98.20

Short put



Gain capped 0.30

Squaring off an option (worked) ★★★★☆

```
1 Oct 2021: hedge ₹100 cr G-secs — buy 5,000 lots Oct
6.10% GS 2031 put, strike 101.00, at 0.20/0.21 (pays
offer 0.21)
15 Oct: sells bonds, closes put at 0.10/0.11 (receives
bid 0.10)
Workbook: paid 5,000 × 0.21 = 1,050 ; received 5,000 ×
0.10 = 500 ; net -550
```

TRAP The workbook figures leave out the 2,000-unit lot. Per ₹100 of face value the net loss is 0.11; on the whole position it is 5,000 × 2,000 × 0.11 = ₹11 lakh. Buy at the offer, sell at the bid.

Exchange-traded IRO specs ★★★★★

- RBI **29 Dec 2016** introduced IROs; FIMMDA to list eligible underlyings; superseded by Rupee IRD Directions **26 Jun 2019**
- First exchange IROs on single GOI securities: **Aug 2019**

Spec	Rule
Underlying	GOI dated security chosen with FIMMDA (e.g. 6.33% GS 2035)
Lot	FV ₹2 lakh = 2,000 bonds
Quote	Premium in ₹
Contract value	Price × 2,000
Tick	₹0.0025
Price band	Delta-based, from previous underlying close and volatility; minimum operating range; computed daily, flexed intraday
Hours	9:00 am – 5:00 pm
Type	Premium-style **European** call & put
Strike interval	**₹0.25** (99.50, 99.75, 100.00…)
Strikes	Min **8 ITM, 8 OTM, 1 near-the-money**
Cycle	3 serial monthly + 3 quarterly
Expiry	Last Thursday
Settlement	Cash via clearing corporation (guaranteed)
Final settlement price	NDS-OM weighted average, last 2 hrs; < 5 trades → FIMMDA/FBIL
Daily settlement price	Last half-hour VWAP across exchanges or theoretical
Final settlement	T+1

Exchange-traded vs OTC IRO ★★★★☆

	Exchange IRO	OTC IRO
Products	Mainly European IROs	European IROs, caps, floors, collars, reverse collars, swaptions, structured products
Terms	Standardised	Custom
Price discovery	Exchange	Negotiation
Liquidity	High	Low
Market maker	Not applicable	Scheduled banks, PDs, AIFIs
Settlement	Clearing corporation, guaranteed	Bilateral
Reporting	Online	Trade repository within 30 minutes
Limits	Imperfect hedge; not every product listed	Liquidity, counterparty risk, not open to all

Open interest (worked) & PCR ★★★★★

Date	Trade	OI	Volume
1 Jun	A buys 1, B sells 1	1	1
2 Jun	C buys 5, D sells 5	6	6
3 Jun	A sells 1, D buys 1 (both close)	5	7
4 Jun	E buys 5 from C (C closes)	5	12

Underlying	OI	Reading (Ch 21)
↑	↑	Long build-up — bullish
↑	↓	End of bullish trend
↓	↑	Short build-up — bearish
↓	↓	End of bearish trend

- PCR = put OI ÷ call OI (or volumes)
- Ch 21 reading: PCR > 1 → bearish sentiment; < 1 → bullish; equity options average ≈ **0.7**; falling to ≈ 0.5 → bullish
- As contrarian: very high PCR → oversold, rebound (buy calls); very low → overbought, fall (buy puts)
- Limits: weigh other factors, use with other indicators, read the chart correctly

> **TRAP** Ch 17 gives the contrarian reading (PCR < 1 = bearish signal, > 1 = bullish). Ch 21 gives the direct sentiment reading (high PCR = bearish). Match the answer to how the question is worded.

Workbook sample answers ★★★☆☆

Price paid by buyer = premium · Buyer limited risk, seller unlimited · Negative cash flow on exercise = OTM · Premium − intrinsic = time value · 98.50 put @ 0.20, expiry 98.50 → loss ₹0.20

22 Strategies using Exchange Traded Interest Rate Derivatives

≈6 marks · 14 subtopics · chapter priority ★★★★☆

Hedgers, speculators, arbitrageurs ★★★★☆

Type	Role	Example
Hedger	Real rate exposure; removes it	₹10 cr in 6.10% GOI pays ₹61 lakh/yr; rates → 6.50% → new bonds pay ₹65 lakh; selling old bond means lower price
Speculator	No exposure; takes view; supplies other side, depth and liquidity	Trader buys wheat futures from farmer expecting higher price
Arbitrageur	No exposure, no risk taken; locks profit across markets; brings prices to equilibrium	Spot 100, March futures 110, lot 50 → buy cash, sell futures → ₹10 × 50 = **₹500** whether expiry is 108 or 95

Arbitrage risks: execution gap, illiquid leg (naked exposure), unwinding when not cash-settled

Costs (transaction, impact, carry) ignored in examples

Traders and arbitrageurs bring liquidity → better price discovery, lower costs, less manipulation

Choosing instrument, side, contract month ★★★★★

Parameter	Rule
Instrument	Match tenor: 3-month rate → T-bill futures; 10-yr rate → 10Y GOI bond futures/options
Market side	Match expected rate direction
Contract month	Match timing of expected rate change

Expect rates	Position	Trades
↑ (rise)	Long the rate	Sell GOI bond futures · buy GOI bond put · sell T-bill futures · **buy** overnight MIBOR futures
↓ (fall)	Short the rate	Buy GOI bond futures · buy GOI bond call · buy T-bill futures · **sell** overnight MIBOR futures

Future borrower fears a rise → sells bond futures OR buys MIBOR futures

Short hedge (worked) ★★★★★

```
20 Sep 2021: X holds ₹5 cr 6.10% GS 2031 at 100.00 (yield 6.10%); fears rise after Oct policy
Sell 250 lots (5 cr ÷ 2 lakh) Oct futures at 99.95
Expiry 28 Oct: bond 98.36 (yield 6.32%)
```

Scenario	Amount on expiry	Plus	Minus
I. Sell now, T-bills 3.20% for 38 days	₹5,07,53,106	No rate risk	Loses accrued interest
II. Hold unhedged	₹5,00,86,528	Accrued interest; gains if price rises	Full rate risk
III. Hold + short futures	**₹5,08,81,528** (futures gain (99.95 − 98.36) × 250 × 2,000 = ₹7,95,000)	Accrued interest; no rate risk	Impact cost; expiry vs trade price mismatch; gives up gains if price rises

Bondholder = long bond, short the rate → hedge by **selling** bond futures

Fund manager liquidating bonds in 45 days: sell bond futures to lock price

Portfolio (duration-based) hedge ★★★★★

```
Max short IRF = (portfolio MD × portfolio market value) ÷ (futures MD × futures price ÷ par)
₹26 cr, portfolio duration 6.1, futures duration 4.7, futures 98.50
Lots = 26,00,00,000 × 6.1 ÷ (98.50 × 4.7 × 2,000) ≈ 1,713
```

Makes position duration zero

Limits: duration accurate only for small moves → large moves cause **under-hedging** (convexity); **basis risk** if prices don't move together

Hedging future borrowing & long hedge ★★★★★

```
Future borrowing: corporate to issue 10-yr NCDs in 1 month at G-sec + 50 bp
14 Aug: sell 7.59% GOI futures 95.60 (yield 7.93%) ; 24 Sep FSP 94.80 (8.08%) → gain ₹0.80 per bond offsets higher cost (basis risk remains)
Long hedge: insurer expects ₹200 cr premiums in March; fears fall in rates
12 Feb: 6.45% GS 2029 spot 99.79 (6.48%), March futures 99.60 (6.51%) → buy 10,000 lots
Expiry 101.58 (6.23%): gain (101.58 - 99.60) × 10,000 × 2,000 = ₹3.96 cr
If expiry 99.00 (6.59%): loss ₹1.20 cr — hedging also cuts potential profit
```

Long hedge = buy futures to lock purchase price/yield (insurers with inflows, bonds maturing soon)

Options alternative: long hedge → buy bond **call**; short hedge / future borrowing → buy bond **put**; max loss = premium

Vertical spreads on G-sec options ★★★★★

Spread	Legs	Net	Max profit	Max loss	BEP
Bull call	Long 98.75 call @ 0.45, short 99.50 call @ 0.20	Debit 0.25	0.50 (≥ 99.50)	0.25 (≤ 98.75)	99.00
Bull put	Short 99.50 put @ 0.40, long 99.00 put @ 0.10	Credit 0.30	0.30	0.20	99.20
Bear call	Short 98.50 call @ 0.60, long 99.25 call @ 0.20	Credit 0.40	0.40	0.35	98.90
Bear put	Long 99.50 put @ 0.40, short 99.00 put @ 0.25	Debit 0.15	0.35	0.15	99.35

Spreads: same underlying & type, different strikes/expiries; limited profit and loss

Bull = view rates ↓ (bond prices ↑); bear = rates ↑

Bull put = **buy lower-strike put, sell higher-strike put**; net premium receipt

Horizontal (calendar): same strike, different expiry — play on time-value gap; no payoff chart

Diagonal: different strikes and expiries; complex

Straddles & strangles on G-sec options ★★★★★

Strategy	Legs	Max loss / profit	BEPs
Long straddle	Buy 99 call @ 0.30 + 99 put @ 0.20	Max loss 0.50 at 99; unlimited profit	98.50 / 99.50
Short straddle	Sell both	Max profit 0.50; unlimited loss	98.50 / 99.50
Long strangle	Buy OTM 99.25 call @ 0.20 + 98.75 put @ 0.15	Max loss 0.35 between strikes	98.40 / 99.60
Short strangle	Sell both	Max profit 0.35; unlimited loss	98.40 / 99.60

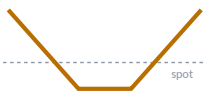
Long = expects big move either way (rate uncertainty); short = expects stable yields — use with care

Long straddle



BEPs 98.50 / 99.50

Long strangle



BEPs 98.40 / 99.60

Covered call & protective put on bonds ★★★★★

```
Covered call: buy bond 99.50 + sell 100 call @ 0.10 → = synthetic short put, strike 100, premium 0.60
If 100 put trades at 0.50 → covered call + buy put = 0.10 risk-free (before costs)
Protective put: buy bond 99.50 + buy 99.50 put @ 0.20 → loss capped at 0.20, upside open
```

Covered call 'sells volatility' for income; strike near price → more premium, gains capped early

Sell calls at your target exit price

Protective put keeps upside, unlike short futures

Coupons/accrued interest ignored in these examples

Long call butterfly on bonds ★★★★☆

```
Buy 98.50 call @ 0.60, sell 2 × 99.00 calls @ 0.30, buy
99.50 call @ 0.20
Net cost = 0.60 - 0.60 + 0.20 = 0.20 debit
Lower BEP 98.70 ; upper BEP 99.30 ; max profit 0.50 -
0.20 = 0.30 at 99.00
```

Neutral, low-volatility view; bull + bear spread; 4 options, 3 strikes, equal gaps

Variants: short call, long/short put, iron butterfly

Speculation with IRF ★★★★★

Long view (rates ↓)	Buy bond (₹5 lakh)	Long IRF, 5% margin, ₹1 cr	IRF ₹5 lakh + CD for rest
Buy price (2 Mar 2020)	100.70	100.62	100.62
Exit (26 Mar)	101.59	101.59	101.59
Price gain	₹4,450	₹97,000	₹4,850
Accrued interest	₹2,150	Nil	Nil
CD return	—	—	₹2,300
Total	₹6,600	**₹97,000**	₹7,150

- Leverage magnifies losses too; futures earn no accrued interest; buy calls to cap loss
- Short view (rates ↑, prices ↓): not everyone may short G-secs → sell bond futures or buy bond puts
- Change portfolio duration fast and cheaply: buying bond futures raises duration, selling lowers it

TRAP The workbook says managers 'lengthen duration when rates are expected to rise'. That is backwards. Expect rates to rise → **SHORTEN** duration (sell futures). Expect rates to fall → lengthen (buy futures).

Arbitrage with single-bond IRF (worked) ★★★★★

	Cash-and-carry	Reverse
When	Futures > theoretical	Futures < theoretical
Trades	Buy bond 102.60 (repo funded 5%), sell futures 102.62; 500 lots, FV ₹10 cr	Sell bond 102.60, lend in repo 5%, buy futures 102.40
Expiry 102.30	AI 4,42,750 + IRF 3,20,000 − bond loss 3,00,000 − funding 2,95,151	Bond gain 3,00,000 + lending 2,95,151 − AI 4,42,750 − IRF loss 1,00,000
Expiry 102.80	AI + bond gain 2,00,000 − IRF loss 1,80,000 − funding	IRF gain 4,00,000 + lending − AI − bond loss 2,00,000
Net (both cases)	**₹1,67,599**	**₹52,401**

Synthetic T-bill: long 15-yr bond + short 3-month futures = riskless 3-month security; if its yield > cash T-bill yield → yield enhancement

Risks: execution gap, illiquid leg, settlement type, short-sale rules, costs (brokerage, stamp duty, margin, impact)

Spread trading with IRF ★★★★★

```
Calendar: sell Sep 6.10% GS 2031 futures 99.00, buy Oct
99.05 (spread 0.05)
After 10 days: Sep 98.80, Oct 98.95 → spread 0.15 → gain
0.10 × 2,000 = ₹200 per contract
```

Calendar spread: only basis risk → lower margins; use exchange spread order to avoid execution risk

Inter-bond spread: long one GOI bond future, short another, same expiry (6.10% 2031 vs 5.85% 2030; or 10-yr vs 5-yr 7.27% 2026) — a play on steepening/flattening

Limits of IRDs for hedgers ★★★★★

Standardised, cash-settled → imperfect hedge, basis risk

Expiry fixed (last Thursday) vs exposure date

Not available on every instrument/tenor (corporate bonds, floating home loans poorly correlated)

Cash settlement: timing/price gap between closing futures and real trade

Offsets: transparency, small lots, easy execution

Workbook sample answers ★★★☆☆

Multi-bond portfolio → duration-based hedge ratio

Bull put spread → buy lower-strike put, sell higher-strike put

Expect rates ↑ → sell GOI bond futures

Buy one month, sell another = calendar spread

Hedger limits → standardised contracts AND not available on all underlyings